School of Computing and Mathematical Sciences

CO7201 Individual Project

Final Report

AI Symptom Tracker - CogniCare

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Glossary

Term	Definition
CTE	Chronic Traumatic Encephalopathy; a progressive neurodegenerative disease associated with repetitive head impacts.
TES	Traumatic Encephalopathy Syndrome; the clinical disorder and collection of symptoms used to categorize symptomatic living patients.
RHI	Repetitive Head Impacts; the primary risk factor for CTE, including both symptomatic concussions and asymptomatic sub-concussive hits.
p-tau	Hyperphosphorylated tau protein; the primary biomarker that aggregates in the brain to cause neurodegeneration in CTE.
Mind Dump	The core feature of CogniCare that allows users to record unstructured qualitative data via voice or text.
Pillars	The five functional domains (Physical, Mood, Cognitive, Sleep, Social) used by CogniCare to categorize patient symptoms.
Capacitor Secure Storage	A hardware-level encryption layer used to protect sensitive session tokens and identifiers on mobile devices.
RBAC	Role-Based Access Control; a security method that restricts system access to authorized users (e.g, distinguishing “Patient” vs. “Carer” permissions).
LLM	Large Language Model; the AI engine (e.g, Gemini) used to normalize unstructured qualitative data into structured clinical insights.

Edge Functions	Serverless **Deno**-based scripts (Supabase) used for high-priority background transport, such as real-time push notifications.
NFTs	**Neurofibrillary Tangles**; abnormal accumulations of tau protein within neurons that characterize the pathology of CTE.
FCM	**Firebase Cloud Messaging**; the cross-platform messaging solution used to deliver real-time background alerts to mobile devices.
Bcryptjs	A password-hashing function used to implement secure, 12-round salted authentication for all account roles.
PII	Personally Identifiable Information; sensitive data that is masked by CogniCare before external AI processing to ensure privacy.
SSR	Server-Side Rendering; a web development technique used in Next.js to improve dashboard performance and SEO.
BFF Model (Backend-for-Frontend)	An architectural pattern used by CogniCare where a dedicated backend layer serves the specific needs of the mobile frontend to optimize performance and security.
Framer Motion	A production-ready motion library for React used in the CogniCare UI to provide visual feedback, such as pulses during voice capture or loading animations for predictive analysis.
Prisma ORM	The Object-Relational Mapper used to manage the PostgreSQL database, facilitating multi-role interactions and complex clinical data tracking.
Socket.io	The library used to implement stateful real-time communication, ensuring that updates like new logs or chat messages sync instantly across patient and carer dashboards.

TES Consensus Criteria	The 2021 clinical diagnostic framework for Traumatic Encephalopathy Syndrome used in CogniCare’s specialist assessment tools to categorize symptomatic patients during life.
Zero-Trace Security	A security design principle implemented during logout where all encryption keys, sensitive data, and application state are explicitly deleted or zeroed out from the device.
Deno	The secure-by-default runtime for JavaScript and TypeScript used to execute your Supabase Edge Functions.

1. Abstract

Chronic Traumatic Encephalopathy (CTE) is a progressive neurodegenerative disease caused by repetitive head impacts, currently only diagnosable post-mortem. Traumatic Encephalopathy Syndrome (TES) refers to the clinical collection of symptoms used to categorise affected individuals during their lifetime. **CogniCare** is a cognitive care coordination platform designed to bridge the gap between daily symptomatic reality and formal clinical assessment for individuals with suspected CTE/TES. Using a **dual-model** AI pipeline (Gemini 2.5 Flash and Pro) and hardware-level **Capacitor Secure Storage**, the application transforms unstructured voice and text entries into structured clinical insights across five functional pillars: Physical, Mood, Cognitive, Sleep, and Social. The platform addresses the fragmented healthcare journey of CTE patients through privacy-first design, collaborative care coordination, and automated clinical reporting details of which are outlined in the sections that follow.

1.1 Problem Statement: The Diagnostic Void

The primary challenge in managing CTE is the "clinical invisibility" of the disease during a patient's life. Currently, patients and their families navigate a fragmented healthcare journey where:

- **Narrative Attrition:** Crucial symptomatic data is lost between doctor visits because patients often struggle with memory and executive function.
- **Subjective Bias:** Caregivers lack a standardized way to quantify changes in mood or behaviour, leading to late-stage interventions.
- **Security Concerns:** Highly sensitive neurodegenerative data is often shared over insecure channels, risking patient privacy.
- **Diagnostic Lag:** Without a longitudinal record of "sub-concussive" exposure and symptomatic trends, clinicians lack the data required to apply the official TES consensus criteria effectively.

1.2 The Solution: Bridging the Gap with CogniCare

CogniCare solves these challenges by creating a **living clinical archive** that serves as a bridge between the home and the clinic. Unlike static journals, the platform provides:

- **Data Normalization:** The "Mind Dump" feature uses AI to distil raw voice/text into five functional pillars (**Physical, Mood, Cognitive, Sleep and Social**), providing a 360-degree view of health.
- **Collaborative Care Circles:** A dual-role authentication flow allows patients to securely invite "Carers" (family or doctors), ensuring that those on the front lines of care have real-time access to stability trends.
- **Clinical Automation:** By synchronizing daily narratives with clinician and TES frameworks during the reporting phase, the system automates the pre-filling of assessment forms, reducing the administrative burden on clinicians without medicalizing the daily patient experience.
- **Privacy-First Architecture:** Utilizing **bcryptjs** hashing and hardware-level **Capacitor Secure Storage**, CogniCare ensures that the patient's sensitive data are protected by industry-standard encryption.

2. Introduction

2.1 Background on CTE

2.1.1 Definition and Origins

Chronic Traumatic Encephalopathy (CTE) is a progressive neurodegenerative disease associated with a history of **repetitive head impacts (RHI)**, including symptomatic concussions and asymptomatic sub-concussive hits [3], [5]. The condition was first described in the medical literature by Harrison Martland in 1928 as **"punch-drunk"** syndrome (dementia pugilistica) in boxers who exhibited psychiatric symptoms and cognitive decline. Modern understanding of the disease expanded significantly following the 2005 neuropathological identification of CTE in a retired National Football League player, which established the condition as a broader public health concern beyond the boxing community [1], [2].

2.1.2 Pathophysiology

The defining pathological feature of CTE is the abnormal accumulation of hyperphosphorylated **tau protein (p-tau)** in the form of neurofibrillary tangles (NFTs) and astrocytic tangles [5], [6]. What distinguishes this process from Alzheimer's disease is not merely the presence of tau, but its precise anatomical signature: rather than dispersing diffusely across the cortex as in AD, the pathognomonic p-tau aggregates of **CTE cluster** around small blood vessels at the depths of the cortical sulci [5], [6]. This spatial specificity is further reinforced at the molecular level, where cryo-electron microscopy has revealed that the tau filament fold in CTE is structurally unique, enclosing a hydrophobic molecule entirely absent from Alzheimer's filaments [7]. Left unchecked, this cascade of molecular and anatomical disruption ultimately drives neuronal loss, axonal disintegration, and gross structural deterioration of the brain, manifesting as cerebral atrophy and ventricular enlargement [3], [5].

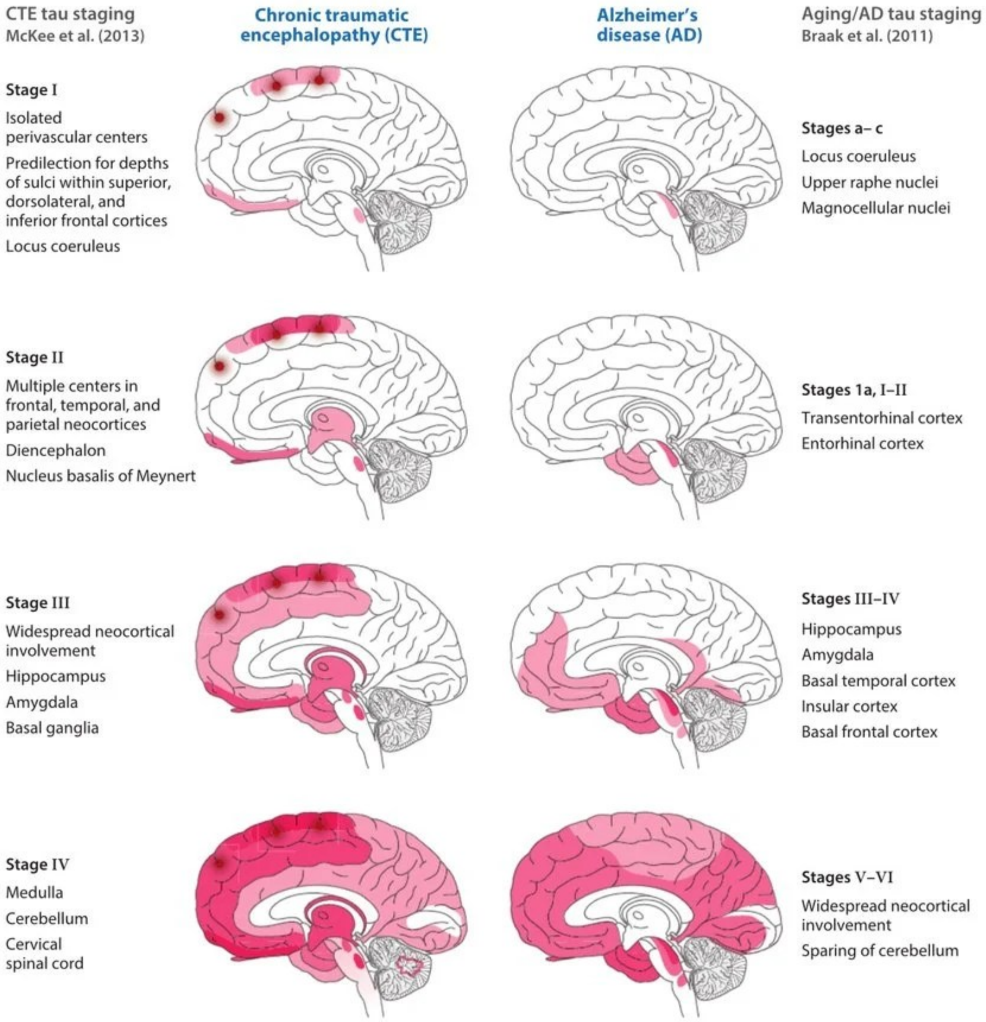


Figure 1: Pathological stages of CTE (McKee Staging) showing the unique perivascular spread of p-tau pathology compared to Alzheimer's Disease. Adapted from [3],[12] and [21].

2.1.3 General Impact on Daily Life

CTE fundamentally disrupts an individual's functional independence and social reintegration. Patients often experience a "living hell" characterized by a slow deterioration of self, leading to the disruption of relationships and an inability to perform routine activities such as managing finances or following complex instructions [8], [9]. This decline often imposes a significant burden on caregivers, who must manage the patient's emotional volatility and progressive cognitive deficits [8], [9].

2.2 Symptoms and Progression

2.2.1 Early vs. Late-Stage Symptoms

The clinical presentation of CTE is often divided into two distinct variants based on the age of onset and initial symptom profile [3], [4].

- **Early-Onset (20s - 30s):** This variant is typically dominated by behavioural and mood disturbances. Common early signs include irritability, impulsivity and depression, which may fluctuate in intensity [10], [11].
- **Late-Onset (60s):** This variant focuses on cognitive decline and memory impairment, which is significantly more likely to progress into full-blown dementia [3],[12].

The progression is categorized into four pathological stages (I - IV) based on the spread of tau pathology. As shown in the diagram below, there is a direct correlation between the severity of the pathological stage and the frequency of clinical dementia.

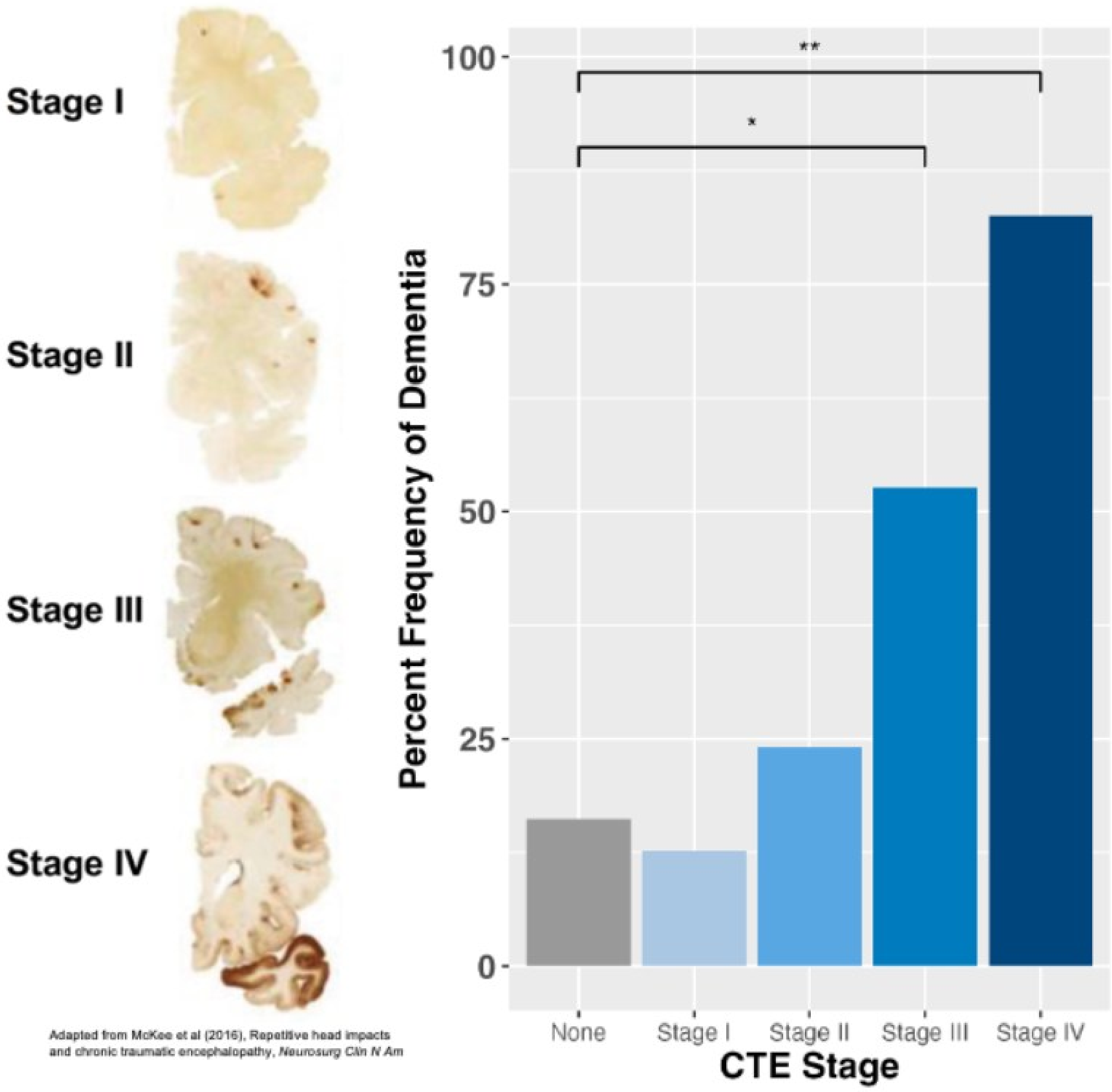


Figure 2: Pathological stages of CTE (McKee Staging) and the corresponding percentage frequency of dementia. Adapted from [3] and [12].

2.2.2 Categorization of Symptoms

The symptoms are generally classified into five functional domains:

- **Cognitive:** Significant impairments in episodic memory, attention and executive functions (e.g., planning and multitasking). Advanced cases are now recognized as a distinct cause of dementia [12].
- **Mood:** Manifests as severe depression, anxiety, apathy and emotional lability. Patients often experience significant hopelessness and a loss of interest in previously enjoyed activities [3].
- **Behavioural & Social:** Characterized by increased aggression, explosivity and a loss of social "filters". These changes often lead to social withdrawal and the breakdown of interpersonal relationships. Recent studies indicate that a family history of mental illness can moderate the severity of these aggressive behaviours [13].
- **Physical:** Motor abnormalities such as parkinsonism (tremors, rigidity), ataxia (balance issues) and dysarthria (slurred speech), particularly in the later stages [3], [5].
- **Sleep:** Frequent disturbances including insomnia, sleep apnea and REM sleep behaviour disorder. Poor sleep quality often exacerbates the cognitive and mood-related symptoms of the disease [4].

2.3 Risk Factors and Causes

2.3.1 Sports Contact

Participation in contact sports remains the most consistently associated risk factor for CTE. While the condition was historically associated with boxing where it was first described as "punch drunk" syndrome [1] recent clinical data highlights significant prevalence among retired professionals in **American Football, Rugby Union, and Association Football (soccer)**, where frequent sub-concussive impacts are a routine part of the game rather than exceptional events [3], [5]. Research has validated a robust dose-response relationship between the duration of play and disease severity; every additional 1,000 estimated head impacts increase the odds of a CTE diagnosis by approximately 21% [14]. This suggests that it is the cumulative burden of repeated sub-concussive hits, not just diagnosed concussions, that drives neurodegeneration [10], [11] a finding with significant implications for how contact sports are regulated and monitored at both amateur and professional levels.

2.3.2 Military Service

Military personnel are at risk due to exposure to explosive blasts. "Blast wind" and pressure waves cause rapid head acceleration and axonal stretching, triggering p-tau deposition and persistent memory deficits similar to those seen in athletes [15], [16].

2.3.3 Domestic Violence Context

Repetitive neurotrauma from intimate partner violence (IPV) is an under-researched but critical risk factor. Survivors often experience multiple blows to the head and non-fatal strangulation, the latter causing hypoxic-ischemic brain injury. Research indicates that survivors with high injury loads have significantly worse cognitive outcomes [17], [18].

2.4 Diagnosis and Challenges

2.4.1 Current Limitations

Currently, a definitive diagnosis of CTE can only be achieved through a post-mortem neuropathological examination of brain tissue [3], [5]. Clinical diagnosis during life is complicated because CTE symptoms significantly overlap with other neurodegenerative and psychiatric conditions, such as AD, PTSD and frontotemporal dementia [4], [19].

2.4.2 Ongoing Research into Live Diagnosis

Researchers are developing specialized tools to identify high-risk living individuals and identify biomarkers for in-vivo diagnosis:

- **TES Criteria:** The Traumatic Encephalopathy Syndrome (TES) consensus criteria (2021) allow clinicians to categorize symptomatic patients based on exposure history and progressive decline [4], [19].
- **Biomarkers:** Research is validating blood-based tests (e.g., p-tau217) and specialized PET imaging tracers to detect specific tau folds in the living brain [12], [20].

3. Literature Review & Research Framework

3.1 Research Questions (RQs)

To bridge the gap between clinical theory and technical implementation, this project is guided by three core Research Questions:

- **RQ1 (The AI Challenge):** To what extent can a multimodal AI pipeline utilizing Gemini 2.5 effectively normalize unstructured voice "Mind Dumps" into clinical data that aligns with established NHS diagnostic markers?
- **RQ2 (The Data Challenge):** How does the implementation of "longitudinal gating" - specifically the 7-day insights lock mitigate recall bias and improve the statistical validity of symptom analysis in patients with memory impairment?
- **RQ3 (The Collaborative Challenge):** In what ways does a "Patient-Owner, Carer-Collaborator" architecture with granular Role-Based Access Control (RBAC) permissions address privacy concerns while facilitating real-time care coordination?

3.2 The Research Gap: Acute vs. Longitudinal Monitoring

Current **Digital Health Interventions** (DHIs) in the neurological space are bifurcated into two extremes, leaving a significant "Diagnostic Void" for Chronic Traumatic Encephalopathy (CTE). Applications such as **Concussion Coach** are clinically validated for the 72-hour "return-to-play" window [22], yet research by Stern et al. highlights that CTE is a progressive tauopathy requiring monitoring over years or decades, not days [3], [21]. These existing tools lack the longitudinal architecture necessary to capture the subtle neurobehavioral shifts in the "Pillar" changes that occur between initial impact and late-stage neurodegeneration [4], [19].

A further limitation of existing mHealth trackers is their reliance on retrospective data entry. Research into the **"Seven Sins of Memory"** [24] demonstrates that patients with potential cognitive impairment suffer from significant Recall Bias, whereby current emotional states distort the accuracy of past memories. This is a particularly critical flaw in the context of CTE, where memory impairment is itself a core symptom of the condition being tracked.

While current literature focuses heavily on late-stage dementia management [23], there is a documented lack of Prospective Capture tools designed for the "middle years" of CTE progression. CogniCare directly addresses this gap by implementing a "Mind Dump" pipeline to record symptomatic data in real-time, effectively bypassing reliance on patient memory and providing the longitudinal continuity that existing tools fail to offer [25].

3.3 The Evolution of Clinical Information Extraction (CIE)

The extraction of structured clinical markers from unstructured patient speech has evolved through three distinct phases of **Natural Language Processing** (NLP). Early CIE systems relied on Rule-Based approaches using **Regular Expressions (Regex)** for keyword matching. While straightforward to implement, these systems proved clinically insufficient as they cannot interpret the linguistic nuance or "hedging" in how a patient describes complex symptoms such as cognitive fog, a particularly common presentation in CTE [26].

The introduction of **Transformer-Based Models** such as **BERT** and **BioBERT** marked a significant step forward by introducing context-awareness into clinical text analysis. However, these models often struggled with the high computational overhead required for real-time mobile deployment, making them poorly suited to the low-latency demands of an mHealth application used by cognitively fatigued patients.

The current research frontier centres on **Multimodal Large Language Models** (LLMs), which represent the state of the art in clinical information extraction. Benchmarking studies demonstrate that LLMs can achieve near-human accuracy exceeding 90% in transforming unstructured clinical narratives into structured data, particularly when using techniques such as **Chain-of-Thought prompting** [27], [28]. CogniCare leverages this capability through Google Gemini 2.5 Flash, selected not merely for its clinical reasoning ability but for its sub-second latency [29]. Research into mHealth user behaviour suggests that high latency leads to user attrition, which is an especially significant risk when the target population is experiencing cognitive fatigue. The "Flash" architecture therefore represents a deliberate, evidence-informed design choice to ensure the "Mind Dump" experience feels instantaneous and frictionless to the user.

3.4 Privacy-Preserving Architectures in mHealth

As mHealth applications move toward cloud-native processing, the **“Zero-Trust”** model has become the academic standard for protecting sensitive health data [30]. This paradigm assumes that no component of a system whether internal or external should be implicitly trusted, and that all data transactions must be verified, isolated, and minimised at every stage of processing.

Traditional architectures process sensitive data on a centralised server, creating a single point of **vulnerability**. Literature suggests that **Edge Isolation,** achieved through tools such as **Supabase Edge Functions**, reduces the attack surface by ensuring sensitive data is handled in isolated, serverless environments rather than passing through a shared processing layer [31]. CogniCare adopts this approach by offloading high-risk operations such as push notification delivery and authentication handshakes to decoupled Edge Functions, ensuring that the core application server is never directly exposed to sensitive cross-cloud communication.

At the device level, modern mobile security research advocates for **hardware-level encryption** as the gold standard for protecting session credentials [32]. By utilising **Capacitor Secure Storage**, CogniCare stores session tokens directly within the device’s Secure Enclave on iOS or the Keystore on Android, rather than in standard browser localStorage. This approach directly addresses the security vulnerabilities commonly cited in general health-tech audits [33], ensuring that even in the event of a physical device compromise, session credentials remain non-extractable and clinically sensitive data remains protected.

3.5 The “Diagnostic Void” & Pathological Justification

The technical architecture of CogniCare is directly mapped to the **McKee Staging Criteria** [3], [5]. Because Tau pathology begins in focal epicentres (Stage I) before becoming diffuse (Stage IV), a tool must be capable of tracking “Physical” and “Mood” pillars years before “Cognitive” decline is visible [6], [7]. CogniCare’s 5-pillar system provides the clinical granularity required to monitor this transition, which is currently invisible to standard concussion apps.

4. Aims, Objectives and Technical Requirements

This chapter defines the technical framework for the “CogniCare” platform, establishing the project’s engineering goals and the requirements necessary to address the clinical challenges of Chronic Traumatic Encephalopathy (CTE) management.

4.1 Project Aim

The primary goal of this project is to develop “**CogniCare”**, a supportive health platform designed to alleviate the cognitive burden of medical documentation for individuals with suspected CTE and their care networks. By transforming unstructured qualitative data (voice and text) into structured clinical insights across five key pillars, the system aims to bridge the “Diagnostic Void” between home-based symptoms and formal clinical assessments.

4.2 Objectives

To fulfill the project aim, the following technical and research objectives were established:

- **Develop an AI Normalization Pipeline:** Implement a **Large Language Model (LLM)** engine to automatically categorize raw narratives into five functional pillars: Mood, Behaviour, Sleep, Physical and Cognitive. This addresses the "Narrative Attrition" common in patients with executive dysfunction.
- **Implement Hands-Free Logging:** Integrate **voice-to-text transcription** to minimize typing fatigue and accommodate the motor abnormalities (e.g., tremors or ataxia) seen in late-stage CTE.
- **Ensure Data Safety:** Develop a privacy-first middleware layer designed to scrub **personally identifiable information (PII)** before external AI processing, maintaining compliance with digital health privacy standards.
- **Establish Granular Collaborative Care:** Utilize a relational database model to manage complex many-to-many "Care Circle" permissions, allowing patients to control the sharing of specific health pillars with authorized family members and clinicians.
- **Automate Clinical Workflow:** Implement a reporting engine to synthesize longitudinal data into weekly structured summaries optimized for clinical review and **Traumatic Encephalopathy Syndrome (TES)** assessment.

4.3 Functionality Requirements

Essential

• **AI-Driven "Mind Dump" Analysis:**
Patient Action: Ability to offload raw thoughts or symptoms into a simple, unstructured text interface.
System Action: Automatically parsing and tagging entries into the five health pillars using NLP.

• **Collaborative Symptom Tracking:**
Patient Action: View a chronological feed of their categorized symptoms to track personal trends.
Carer Action: Ability to view the shared timeline and add "Observation Notes" to provide a second perspective on the patient's status.

• **Granular Access Control:**
Patient Action: Use a "Care Circle" dashboard to invite carers and toggle permissions for carer access.
System Action: Enforcement of role-based permissions to ensure data privacy and patient-led sharing.

• **Clinical Summary Exports:**
User Action: Ability to trigger the generation of a structured flexible time-range PDF summary.
System Action: Compiling categorized data into a professional format suitable for medical consultations.

Desirable

• **Voice-to-Text Integration:**
Patient Action: Ability to record audio “brain dumps” to minimize typing fatigue.
System Action: Converting voice recordings into text for AI analysis.

• **Secure Care Circle Messaging:** A real-time, text-based chat system for patients and authorized carers to coordinate care using Socket.io.

Optional

• **Accessible Design (WCAG 2.1):** Rigorous implementation of simplified UI and navigation standards for users with cognitive impairments.
• **NHS Integration Framework:** Mapping internal data structures to FHIR standards to demonstrate potential for future NHS system integration.
• **Offline Entry:** PWA functionality to allow symptom logging without an active internet connection.

The requirements above reflect the initial classification established during the preliminary project phase, categorised into Essential, Desirable and Optional tiers. As development progressed, the full feature set expanded beyond these initial specifications both through planned elaboration and through features that emerged organically in response to clinical and architectural considerations. Figure 3 presents the complete evolved feature set using MoSCoW prioritisation (Must Have, Should Have, Could Have, Won't Have), which captures the broader scope of what was considered across the full development lifecycle. The mapping between the two frameworks is broadly consistent: Essential requirements correspond to Must Have, Desirable → Should Have, and Optional → Could Have, however the MoSCoW table additionally captures features that were identified and evaluated during development but deliberately deferred.

Figure 3: Evolved Feature Classification- MoSCoW Analysis (Post-Development)

4.4 Tech stack and Techniques used

Category	Technology	Technical Rationale
Frontend	React 19	The core library for building interactive, state-driven UI components. Used for complex Care Circle interactions and health log visualization.
Web & Mobile Shell	Next.Js 15+ (App Router)	Provides the routing engine and Server-Side Rendering (SSR) for the dashboard, ensuring SEO and performance optimizations for the web view and Capacitor shell.
Backend	Node.Js with Express	The primary server layer (found in /server) that handles API requests, clinical data normalization and manages the lifecycle of Care Circle.
Push Notifications	Supabase Edge Function (Deno)	The Node server triggers these serverless functions via secure REST calls. They resolve recipient tokens and broadcast to FCM/APNs, ensuring background delivery even if the app is closed.
Live Tracking	Socket.io	Powers active session synchronization. While Edge Functions handle *background* alerts, **Socket.io** ensures *instant* message delivery for users currently in the app.
Mobile Bridge	Capacitor 6.0	Bridges the web code to native iOS/Android, facilitating access to hardware-level features like **Secure Storage** for encrypted tokens and the **Voice Recorder**.
Styling	Tailwind CSS 4	The styling engine used for rapid, consistent UI development. Its utility-first approach ensures the application remains lightweight and highly customizable.

UI/UX	Shadcn ui & Framer Motion	Shadcn UI: Provides the high-fidelity component library. Framer Motion: Adds a layer of premium micro-interactions (e.g., smooth pillar transitions) that are vital for senior user accessibility.
Relational Database	PostgreSQL (via Prisma)	Ensures strict data integrity. The Prisma ORM provides type-safety for the complex relational logic between patients and multiple Carers.
AI Specification	Gemini 2.5 Flash/pro	Gives near-instant symptom extraction from raw logs. Used as a high-complexity fallback for diagnostic report synthesis and PDF automation.
Data Privacy	PII Masking (js-pii-mask)	Scans and redacts identifiable information (names, dates) at the server level before any AI processing or logging occurs.
Auth Security	Bcrypt (12-round salt)	Implements a robust multi-role security system (Patient vs. Carer) with industry-standard password hashing.
Testing Framework	Playwright	An end-to-end testing tool designed to automate and simulate user journeys across modern web applications.
Frontend Hosting	Vercel	Provides a seamless **CI/CD pipeline** for automated deployments and offers secure **environment variable encryption** and native support for Next.js Server Actions.
Backend Hosting	Render	Supports stateful **Socket.io connections**, bypassing the inherent **timeout constraints of serverless architecture** while enabling seamless GitHub-integrated CI/CD.

Table 1: List of tech-stack used and its rationale

4.4.1 Detailed Push Notification Lifecycle (Edge Functions)

Based on the current implementation and requirement, the **Real-time Pipeline** is optimized into two specialized **Deno** functions that maintain state between the database and the device:

1. **Patient Alerting (notify-patient-carer-note):**

- **Trigger**: Triggered by the Node server when a Carer logs a note.
- **Logic**: Fetches the patient's pushToken from patient_profiles, formats a priority alert including the carer's name and broadcasts via FCM.
- **Persistence**: Automatically saves a record to the notifications table for in-app history tracking.

2. **Care Circle Coordination (notify-care-circle-message):**

- **Trigger**: Triggered by the Express server during active chat sessions.
- **Logic**: Dynamically resolves the recipient (Patient or Carer) across profile tables, fetches their specific pushToken and delivers the message preview.
- **Priority**: Sets **FCM High Priority** and custom Android channel IDs (default) to ensure delivery even during doze mode or background state.

3. **Secure FCM Handshake (Google OAuth2):**

- The **Deno** runtime generates a custom **RS256 JWT** using **FIREBASE_PRIVATE_KEY** on-the-fly to exchange for a Google access token. This "Zero - Trust" approach ensures that long-lived keys are never sent to the client.

5. User Journey and Personas

The following detailed journey explores the two primary pathways: the **Patient (Owner/Controller)** and the **Carer (Collaborator/Contributor)**.

5.1 The Detailed User Journey Mapping

The user journey for a patient using **CogniCare** is designed to provide absolute control over personal data while ensuring that daily symptomatic "narratives" are captured which are often lost due to memory attrition are structured into high-confidence clinical data. Carer can add observation notes or symptoms on behalf of patient if provided access. Carer is more like an observing role.

5.1.1 Phase 1: Secure Onboarding

The onboarding process is to register both patient and carer into the system so that both can review their data through specialized role selection and secure authentication.

Patient Registration Journey

- **Step 1: Role Selection:** The user is presented with a dual-role entry screen. They select **"PATIENT"**, which triggers a specialized registration flow that includes family-linking features not available to carers.
- **Step 2: Basic Account Details:**

- **Full Name:** Collected to personalize clinical reports and the user interface.
- **Unique Identifier:** The user enters an **Email** or **Phone Number**.
- **Secure Password:** Managed by **bcryptjs** on the backend using 12 salt rounds to ensure hardware-level security.

- **Step 3: Family Details (Optional):** Unlike the Carer flow, the patient can add family contact by entering a **Family Member's Name, Email and Phone**. This is a potential 3rd pathway but due to vast application development, this pathway is in Future scope.
- **Step 4: Completion:** Upon clicking "Join CogniCare", the system hashes the password, creates a unique profile (prefixed with **PAT-**) and redirects the user to login page.

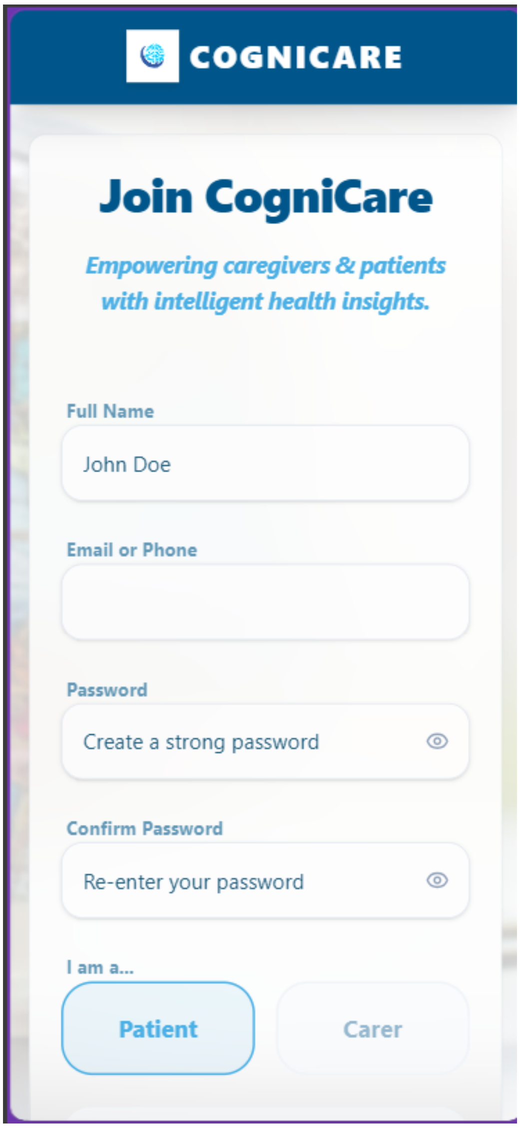


Figure 4: Registration Flow

Carer Registration Journey

- **Step 1: Role Selection**: The user is presented with a dual-role entry screen. They select **"CARER"**, which triggers a specialized registration flow designed to link their account to an existing patient for care management and monitoring.
- **Step 2: Basic Account Details**:
  - **Full Name**: Collected to personalize the carer dashboard and ensure proper attribution for clinical notes and messages.
  - **Unique Identifier**: The user enters an Email or Phone Number.
  - **Secure Password**: Managed by bcryptjs on the backend using 12 salt rounds to ensure hardware-level security.
- **Step 3: Patient Linking (Required)**: Unlike the Patient flow, the carer **must** enter a valid **Patient ID**. This unique identifier is verified against the database in real-time to link the carer to an active patient profile, granting them immediate access to health logs and AI-driven insights.
- **Step 4: Completion**: Upon clicking "Join CogniCare", the system validates the Patient ID, hashes the password, creates a unique profile (prefixed with **CAR-**) and redirects the user to the login page.

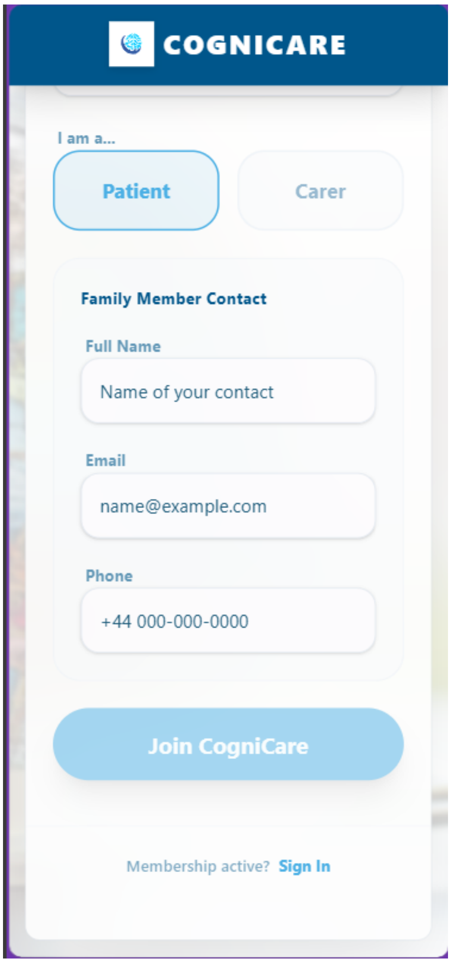


Figure 5: Registration flow (patient view)

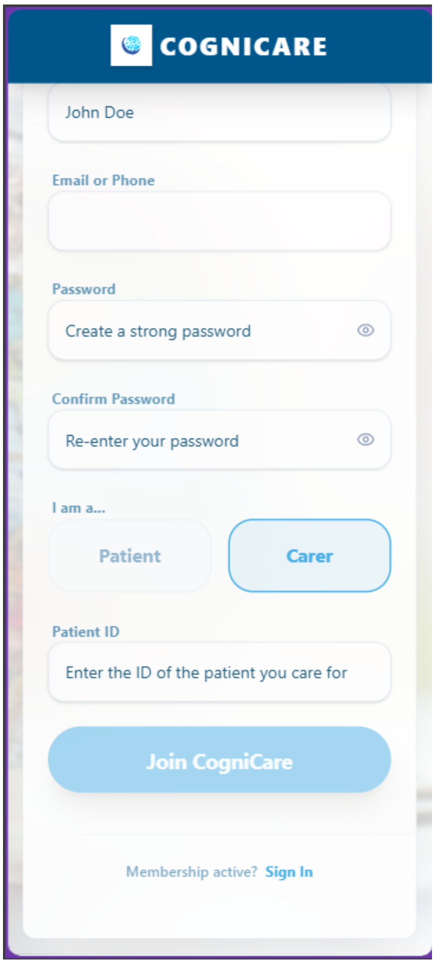


Figure 6: Registration flow (carer view)

Login Journey

- **Step 1: Authentication:** The patient/carer enters their credentials. The system uses bcrypt.compare via a secure API to grant access.
- **Step 2: Persona-Based Redirection:** The login engine resolves the user's role. Instead of a general dashboard, the patient is automatically redirected to the **"Mind Dump"** page to encourage immediate symptom capture before memory lapse occurs. Carer gets redirected to **"Carer Dashboard"** page where it can see list of patients assigned under him.

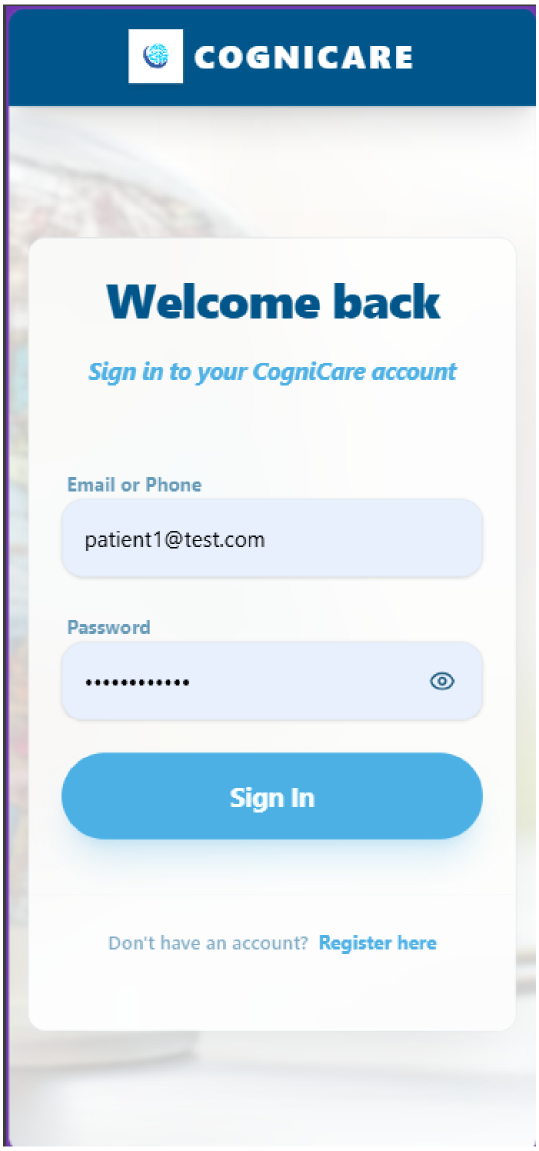


Figure 7: Login Flow

5.1.2 Phase 2: Dashboard Page

Patient View

The **Mind Dump** (internally referred to as “Brain Dump”) is the primary engine of the platform. It serves as a low friction “clinical bridge”, transforming raw human expression into longitudinal data.

Entry & Modality Selection

Immediately after the login redirect, the patient arrives at the exclusive **Mind Dump** interface. The UI is designed with large, high-contrast elements to minimize cognitive load.

- **Voice Capture (Primary):** A prominent microphone button is displayed for hands-free logging, addressing the motor or concentration barriers common in CTE.
- **Manual Text Entry:** A large, distraction-free text area at the bottom for those who prefer to type their state of mind.

The Capture Phase

The goal of this step is to capture the patient's subjective “inside-out” reality in their own natural language.

- **Voice Workflow:**
    - The patient taps the microphone.
    - **UI Feedback:** Visual pulses (implemented via Framer Motion) indicate the system is actively listening.
    - **Natural Speech:** The patient speaks freely (e.g., *“I’ve had a really bad headache since I woke up and I can't remember where I put my keys. I'm feeling a bit frustrated”*).
    - The patient taps the button again to terminate the recording.
- **Text Workflow:**
    - The patient types their observations and clicks “**Process Written Entry”.**

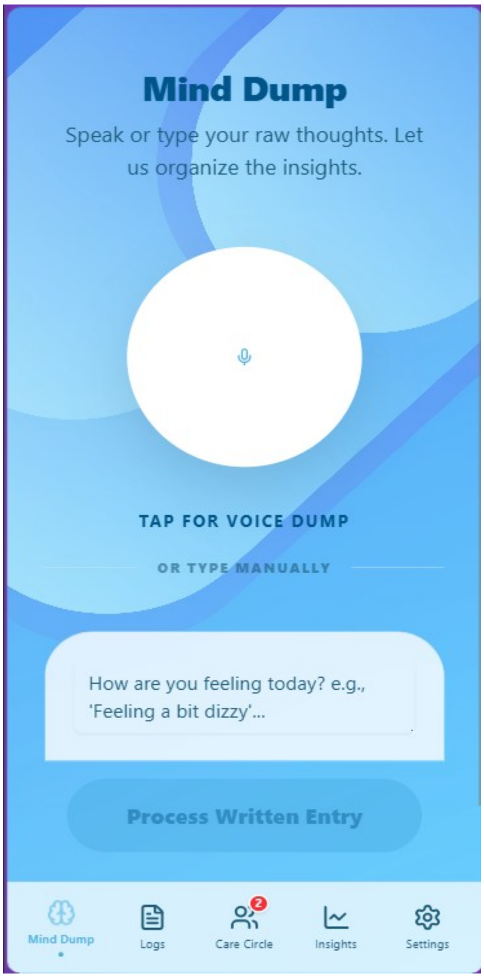


Figure 8: Patient Dashboard view

Real-Time AI Normalization Pipeline

The system initiates a dual-stage pipeline using **Gemini 2.5** to convert unstructured narrative into structured clinical metrics.

- **Transcription:** If voice was used, the **Google Gemini** engine transcribes or converts the audio to text.
- **De-identification:** PII masking redaction is performed server-side to ensure privacy before the text reaches the LLM.
- **The 5 Pillar Analysis:** The AI parses the text and categorizes symptoms into the **5 Pillars of CTE Management**: **Physical, Mood, Cognitive, Sleep and Social**.
- **Severity Mapping:** The AI assigns an initial severity score from **1 to 10** for each pillar detected. For example, "couldn't remember keys" triggers a **Cognitive** score of **4/10**.

Interactive Validation (The Symptom Analysis Card)

Recognizing that AI is a collaborative assistant rather than a standalone diagnostic tool, CogniCare implements a "Human-in-the-loop" validation step.

- **Visual Confirmation:** An interactive **Summary Card** slides into view at the top of the interface, showing the AI's "interpretation" of the log.
- **The "Human Override" Sliders:** If the patient feels the AI underestimated their pain (e.g., the AI gave **Physical** a **4**, but they feel an **8**), they can drag the slider to correct the metric manually.
- **Auto-Save:** Any adjustment to these sliders is saved to the database in real-time via the updateBrainDumpSeverity service, ensuring the database reflects the patient's lived experience over the AI's estimation.

Persistence & Data Density Tracking

- **Completion:** Once validated, the system clears the input area. The patient can select **"Clear"** to reset or enter another log immediately.
- **Progressing toward Insights:** Every log contributed is a timestamped entry in the SymptomLog table.
- **The 7-Day Threshold:** After this first log, the patient navigates to the **Insights Dashboard**. They see a persistent progress bar: **"1 / 7 days of data captured. Keep logging daily to unlock trend analysis"**.

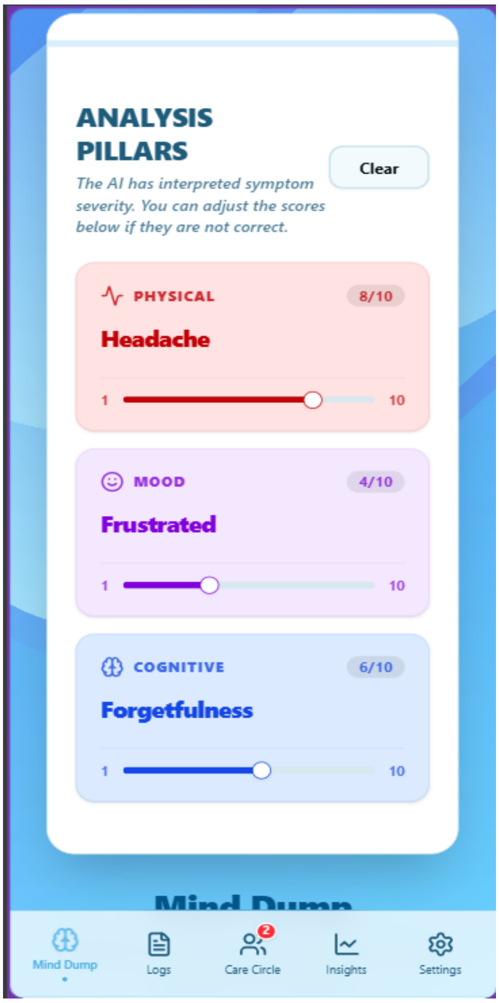


Figure 9: Symptom Analysis card

Technical and Clinical Summary of Phase 2

- **Data Integrity:** By enforcing the **7-day continuous logging gate**, CogniCare ensures that the “Improving” or “Worsening” signals on the dashboard are based on statistical trends rather than acute daily fluctuations.
- **Privacy-First:** All qualitative logs (the original text/voice) are only accessible to the patient if the patient restricts access to Carer for symptom log’s view. The Carer sees the resulting “normalized” logs with scores and can add any specific clinical notes separately.

Carer View

The Carer Dashboard is a high-fidelity monitoring station. It transforms the patient’s subjective “Mind Dumps” into actionable clinical intelligence, enabling longitudinal tracking and professional reporting while respecting the patient’s privacy.

Patient Selection & Real-Time Monitoring

- **The Global Overview:** Upon login, the carer is presented with a list of all linked patients. This serves as the primary “triage” screen for multiple-patient management.
- **Real-Time Alerts (The Pulsing Indicator):** Using **Socket.io** for live synchronization, a green pulsing dot appears next to a patient’s name the moment a new “Mind Dump” is processed.
- **Optimistic “Mark as Viewed”:** To minimize administrative fatigue, the system optimistically clears the notification dot when the carer clicks the patient, syncing this “read” state to the backend in the background.

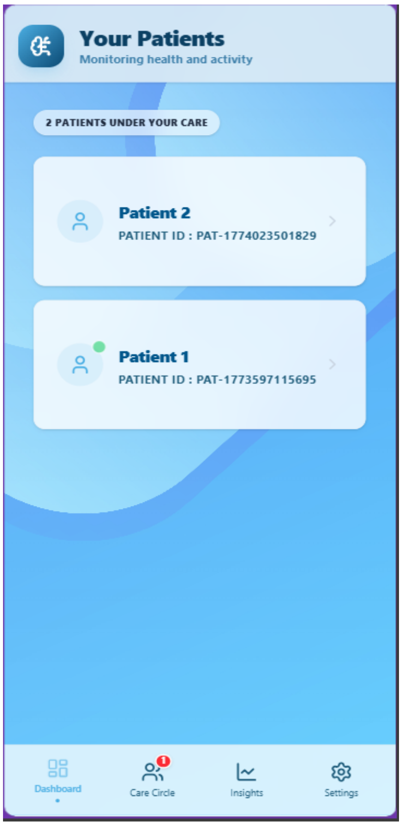


Figure 10: Carer Dashboard view

5.1.3 Phase 3: Symptom Log Page

The Symptom Log Page is used to track the record of the logs input by patient or carer on behalf of the patient. User can track the logs based on day, week and month wise.

Patient User Journey

- **Step 1: Historical Access**: The patient opens the Symptom Log page to view their symptom input history. The interface presents a unified timeline of all captured events, organized by date and time.
- **Step 2: Log Management (Full Control)**: The patient can **View, Edit, or Delete** any log entry they personally created (the "Mind Dumps").
- **Step 3: Human Override**: On their own logs, the patient can adjust the **Severity Sliders** to override AI categorizations, ensuring the data reflects their actual experience.
- **Step 4: Real-Time Alerts (Carer Awareness):** When a Carer adds a clinical observation or notes, the patient receives an Immediate **Push Notification (via Firebase)** and a **Real-Time Socket Event**. This ensures the patient is instantly aware of any clinical updates without needing to manually refresh the page.
- **Step 5: Notification Tab Integration:** Any new carer observation is automatically reflected on the Notification Tab with a red "unread" badge. The patient can click this notification to jump directly to the **specific log entry highlighted by the carer**.
- **Step 6: Initiating Discussions (Agency):** Upon reviewing a Carer's note, the patient alone has the power to click **"Start Discussion".** This anchors a specialized

chat thread in the Care Circle to that specific clinical log, allowing the patient to lead the conversation.

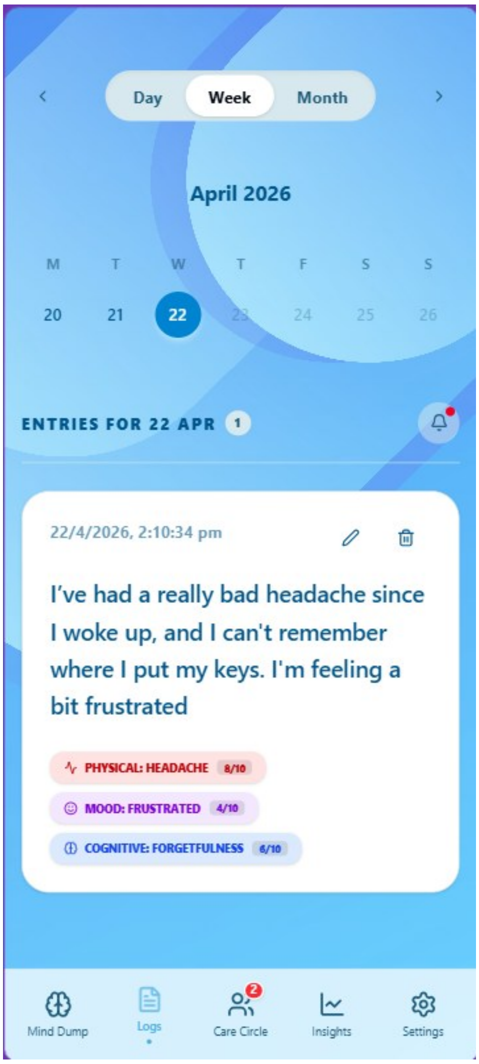


Figure 11: Log added by patient

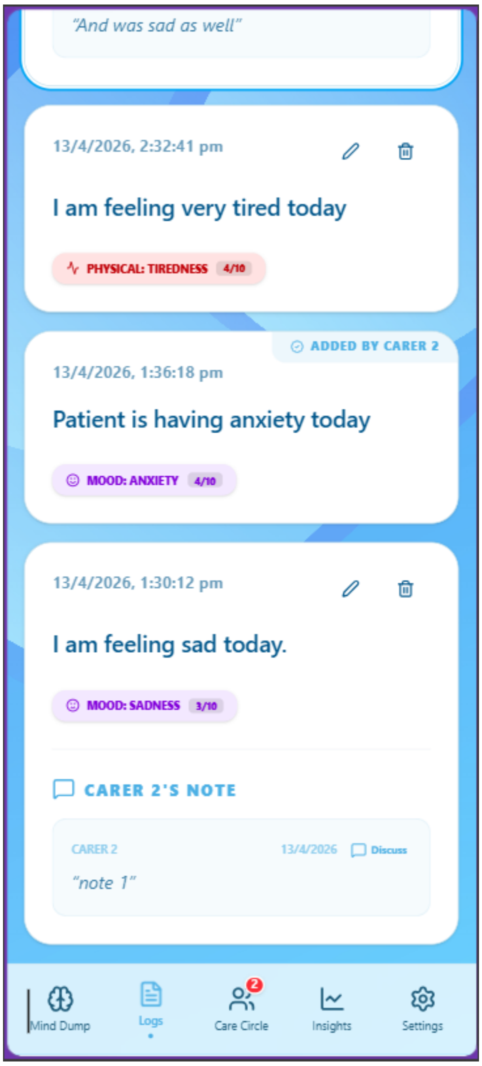


Figure 12: Carer note added on patient's log

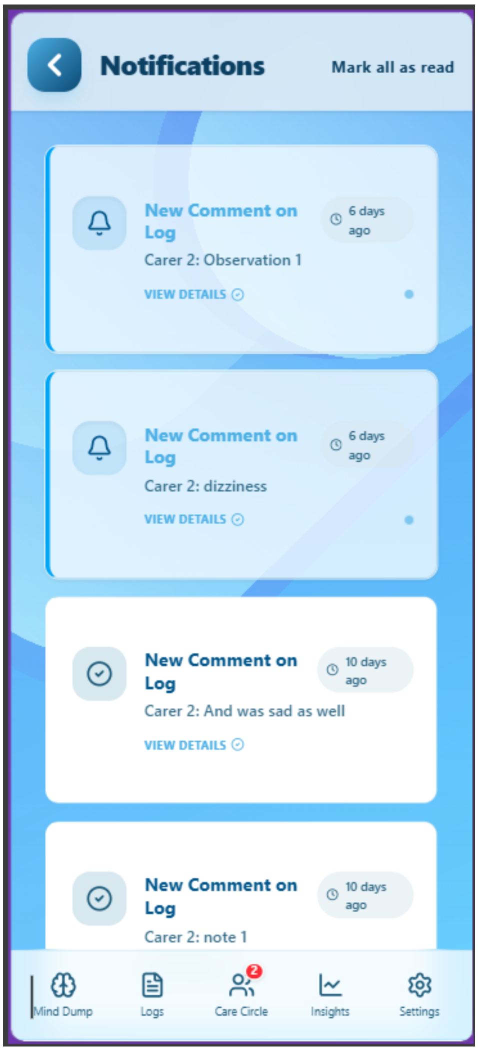


Figure 13: Notification's page

Carer User Journey

- **Step 1: Patient-Specific Surveillance**: The carer *can* enter a patient's timeline to review the health trajectory. They see all entries categorized into the 5 clinical pillars.
- **Step 2: Multi-Log Observation (Notes)**: The carer can add a **"Carer Note"** (observation) to any **Symptom Log** in the timeline. This includes:
- **Patient Logs**: Adding professional context to the patient's subjective "Mind Dump".
- **Carer Symptom Logs**: Adding further observations to a structured log previously created by themselves or another carer.
- **Step 3: Clinical Entry Creation**: The carer can click **"Add Log"** to record a new observation (e.g., "Noticed tremors during breakfast today"). This action triggers the **real-time notification pipeline** for the patient. *The logs added by Carer are marked as **"Added by carer_name".***
- **Step 4: Ownership Constraints**: The carer can **Edit or Delete only the entries they personally authored**. They have no permission to modify the patient's original narrative or notes created by other carers.

- **Step 5: Observational Role**: While the carer provides the data, they **cannot** initiate the *"Start Discussion"* chat threads from the logs page; this is reserved for the patient to ensure they lead the dialogue about their own health.

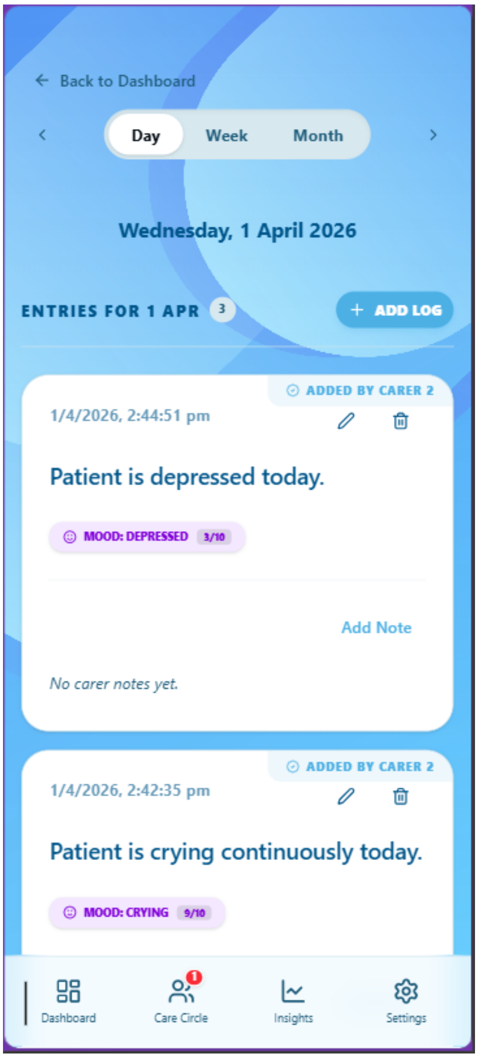


Figure 14: Carer log view

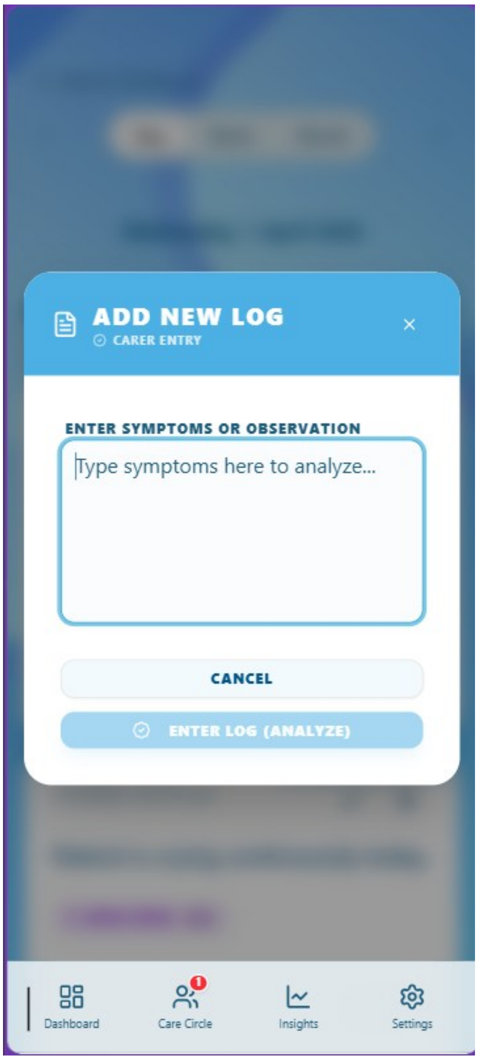


Figure 15: Carer Add Log view

5.1.4 Phase 4: Care Circle Page

Shared Collaborative Features (Patient & Carer)

- **Step 1: The Communication Hub**: Both users arrive at the Care Circle page, which serves as the central coordination point for the patient's care. The UI is split into two primary modes: **Clinical Threads** and **Direct Messaging**.
- **Step 2: Engaging in Clinical Threads**: Both roles participate in threads that are anchored to specific health logs. This ensures that every message is tied to clinical evidence, making discussions highly focused and data driven.
- **Step 3: Direct Messaging**: Both users can initiate private, one-on-one direct chats with other members of the circle. This is used for general check-ins or logistics that don't require a specific clinical anchor.

- **Step 4: Real-Time Sync**: Powered by **Socket.io**, the entire circle stays synchronized. When a message is sent, all participants receive an instant update, ensuring the care loop remains responsive and low latency.
- **Step 5: Issue Resolution**: Once a health concern or symptom query has been addressed, any member of the circle can mark a thread as **"Resolved".** This archives the discussion, keeping the dashboard clean and focused on active health issues.

Patient-Exclusive Capabilities

The Emergency Hotline: A prominent, red-gradient **Hotline Button** is exclusively available to the **Patient** in the page header. This provides one-click, immediate access to external crisis support, ensuring a safety net is always visible and accessible during acute events.

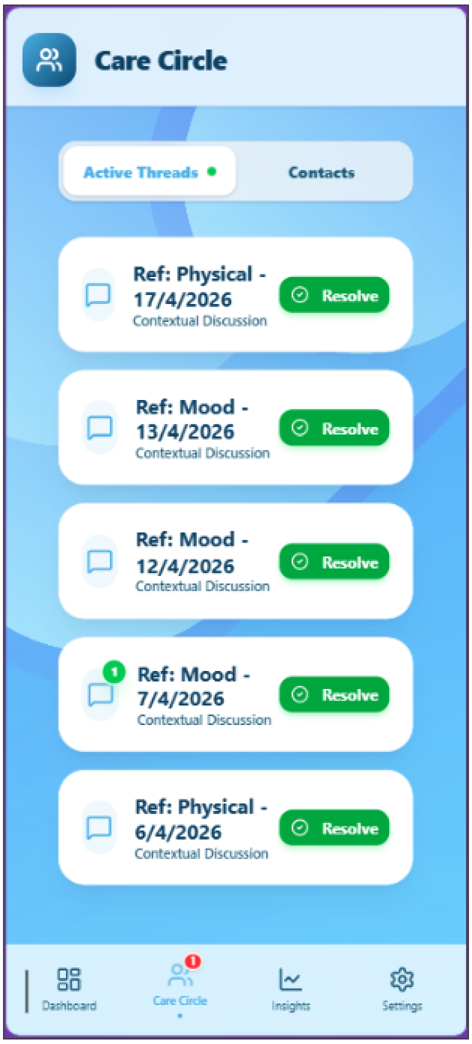


Figure 16: Chat Threads

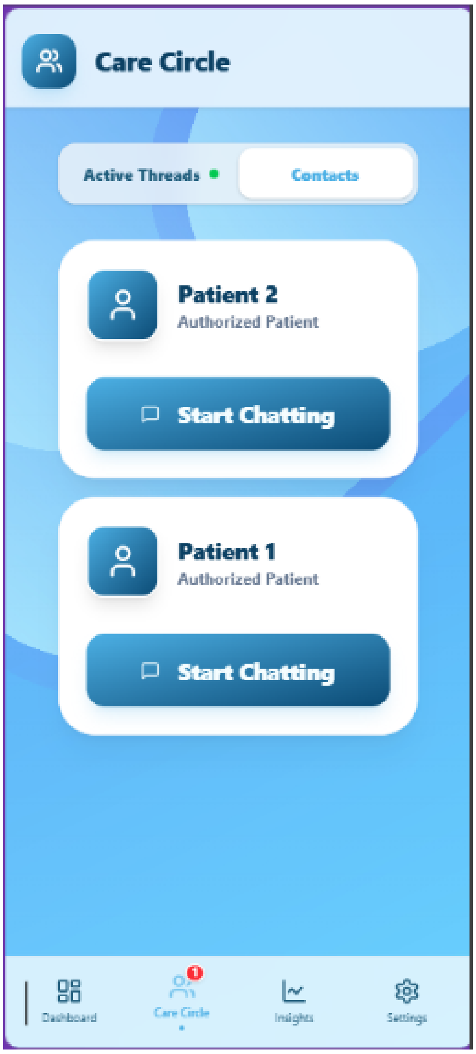


Figure 17: Direct chats

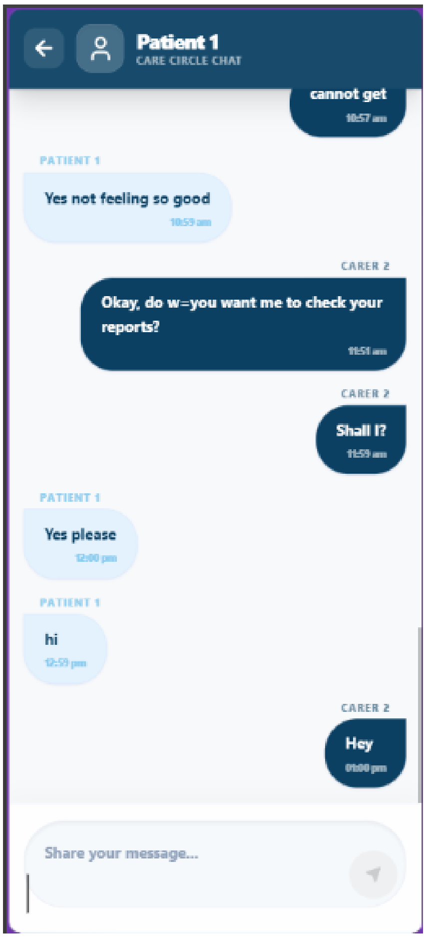


Figure 18: Care Circle direct chat

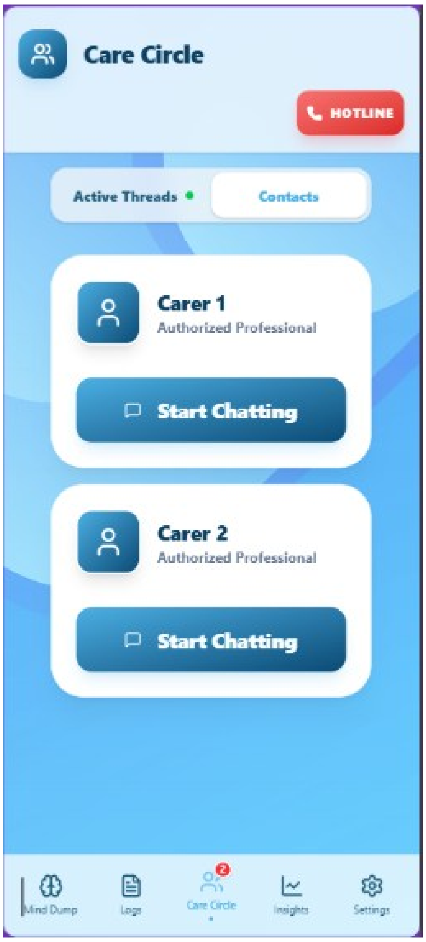


Figure 19: Direct chat (Patient view)

5.1.5 Phase 5: Clinical Insights & Predictive Intelligence (Unified User Journey)

The Data Sovereignty Gate (Eligibility)

- **The 7-Day Baseline**: Upon first entry, the system checks the getInsightsEligibility service. If the patient or carer on behalf of patient has fewer than 7 distinct days of symptom logs, the dashboard is **Locked**.
- **The Progress Visualizer**: Users see a high-fidelity progress tracker (e.g., "4 / 7 days") and a pulse-animated lock icon. This ensures all AI-driven insights are grounded in a statistically valid baseline of the patient's specific health patterns.
- **Unlocked State**: Once the 7-day threshold is met, the full analytical engine is automatically provisioned for both the patient and their carer.
- The Insights page is locked because, user cannot check the data until the system has at-least 7 days of data to give more relevant insights on the logged symptoms.

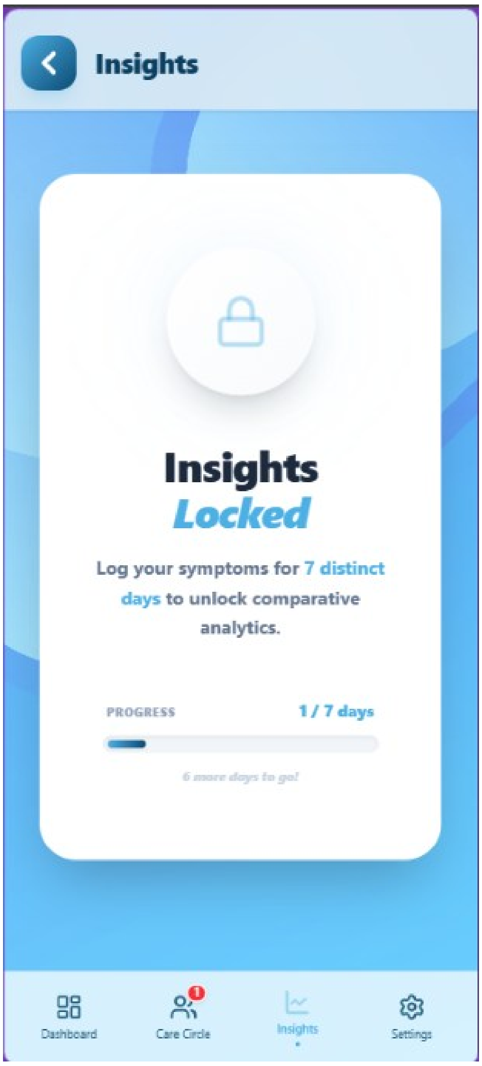


Figure 20: Insights Locked view

Immediate Health Snapshot (Major Symptoms & Alerts)

The **Major Symptoms Card** serves as the high-priority triage section of the Insights Dashboard. It is designed to immediately alert both the patient and carer to the most significant health concerns detected over the previous week.

- **The Red Alert Section**: If any **"Critical Risk"** keywords are detected in today's logs (e.g., "Self-harm", "Depression", "Suicidal thoughts"), a **prominent red alert** section appears at the top. This section pulses visually to demand immediate attention.
- **Top Symptoms Detection**: Below the alerts (if present), the system displays the **MajorSymptomsCard** with top 5 symptoms in a structured list across the patient's entire history (e.g., *"Persistent Cognitive Fatigue"*). Each symptom shows:
  - **Pillar Icon**: Color-coded icons representing the category (Physical, Mood, Cognitive, Sleep, Social).
  - **Source Label**: Indicates whether the symptom was primarily reported by the **Patient** or observed by a **Carer**.
  - **LVL (Severity)**: A numerical "Level" from 1 to 10.

Figure 21: Critical Attention Card

Figure 22: Major Symptom Card

How the "Level" is Calculated

The severity level shown on the card is the result of a multi-stage backend algorithm in insightsQueries.ts:

- **Stage 1: The 7-Day Window** The system aggregates every single log entry from the **last 168 hours (7 days)**. It ignores older data to ensure the card remains a "Real-Time" reflection of the patient's current state.
- **Stage 2: Peak Severity Logic** If a patient reports a symptom like "Headache" multiple times during the week, the system does not take an average. Instead, it records the **Peak Severity** score. *Example*: if a patient logs a Level 2 headache on Monday and a Level 8 headache on Thursday, the card will display **LVL 8**. This ensures that the most severe instance of a symptom is not "diluted" by milder days.
- **Stage 3: Risk Keyword Escalation** The system cross-references every log against a **Risk Keyword Library**. If a symptom name or the raw narrative contains a critical

keyword (e.g., *"Sudden dizziness"* or *"Suicidal ideation"*), the algorithm **automatically overrides the severity to Level 10**, regardless of what the AI initially suggested.

- **Stage 4: Triage Sorting (The Top 5)** The system then ranks all detected symptoms to decide which 5 make it onto the card. The sorting priority is:
  - **Risk Priority**: Any symptoms marked as **"Critical Risk"** always jump to the top.
  - **Severity Priority**: Symptoms with higher **"Peak Severity"** are ranked next.
  - **Frequency Priority**: If two symptoms have the same severity, the one mentioned more frequently across the 7 days takes precedence.

Longitudinal Analysis (Charting & Presets)

- **Multi-Scale Time Toggling:** Users can switch between predefined ranges **(Day, 7d, 15d, 1m, 3m, 6m) or a Custom Date Range.**
- **Pillar Comparison:** The Symptom Bar Chart visualizes the Combined Average of the 5 Clinical Pillars (Physical, Mood, Cognitive, Sleep, Social). Each bar represents the aggregate severity (1-10) for the selected period, allowing users to see briefly if, for example, "Sleep" is consistently more impacted than "Mood".
- **Interactive Drill-down:** Tapping a specific pillar on the chart highlights its individual score and triggers a refined view of the data points for that specific symptom.

Figure 23: Chart Visualization

Retrospective AI Insights (Clinical Synthesis)

When user clicks on "See AI Insights", it shows an overview of symptom logs. This phase transforms raw longitudinal data into actionable clinical intelligence, enabling patients and carers to identify patterns that are often invisible during daily monitoring.

Step 1: Deep Computational Analysis

- **The Intelligence Engine:** The system utilizes the **gemini-2.5-flash** model as the primary reasoning engine.
- **Resiliency Fallbacks:** If the primary model hits a quota or overload (429/503), the system automatically triggers a **multi-stage fallback:** 2.0-flash → 2.5-flash-lite → 2.5-pro.
- **Pattern Recognition:** The AI cross-references the 5 clinical pillars (Physical, Mood, Cognitive, Sleep, Social). It looks for "Perspective Gaps" (differences between patient-reported and carer-observed data) and "Causal Correlations" (e.g., identifying that a spike in Physical symptoms typically follows a drop in Sleep quality by 24 hours).

Step 2: Integrated Clinical Dashboard (The AI Reveal)

The AI output is rendered via the AI Insight Section using a high-fidelity, interactive layout:

- **The Status Badge:** A visual indicator at the top summarizes the overall trajectory as **Improving, Stable or Worsening,** with associated color-coded icons (TrendingUp/TrendingDown).
- **The Synthesis Narrative:** A concise clinical summary provides a holistic high-level view of the patient's state during the selected date range.
- **The Primary Concern:** A highlighted callout specifically identifies the "Pillar of Most Concern" (e.g., Cognitive) and provides the clinical rationale for why it was flagged.
- **Interactive Insights Carousel:**
  - **Critical Risks:** High-priority alerts (e.g., *"Increased risk of confusion during evening hours"*) are presented in a rose-tinted carousel.
  - **Key Findings:** Individual cards for each pillar allow users to swipe through detailed findings, such as specific cognitive sub-categories or social interaction trends.

Step 4: Role-Based Clinical Toning

- **Role Personalization**: The system automatically detects if the user is a Patient or Carer.
- **The Patient Perspective**: Insights are delivered in an **empathetic, supportive tone** focused on self-management and reassurance.
- **The Carer Perspective**: Insights are delivered in a **factual, observation-oriented tone,** providing the carer with "Triage-Ready" data to support their caregiving decisions or for communication with medical professionals.

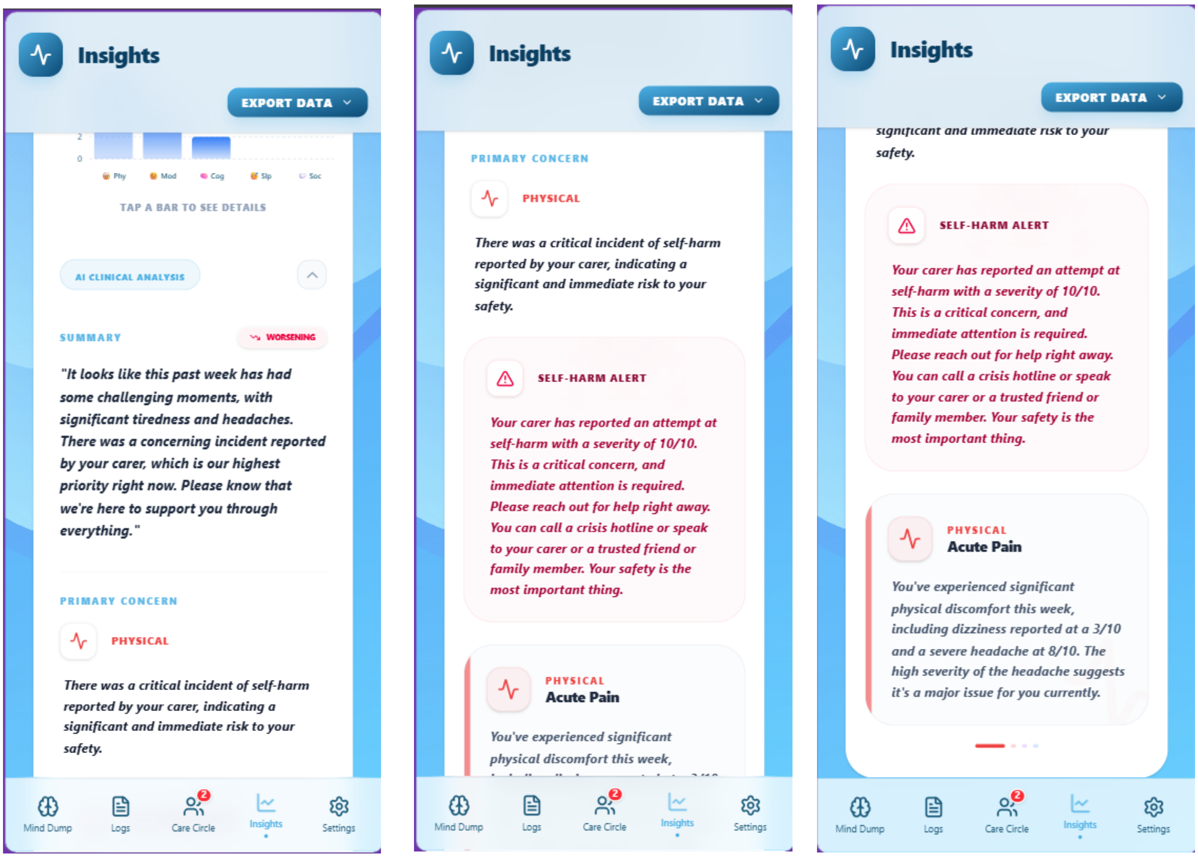


Figure 24: AI Insights View Snapshots

Predictive Analysis (The 7-Day Health Forecast)

It is a forward-looking feature designed to transform CogniCare from a reactive logging tool into a proactive early-warning system. By analyzing the "velocity" of symptoms over a 30-day window, the application provides users with a "clinical weather forecast" to help manage upcoming health fluctuations.

- **The Activation**: Upon clicking **"See Predictive Analysis"**, the app executes a high-level synthesis of the last 30 days of pillar data.
- **Visual Feedback**: A sleek loading animation powered by **Framer Motion** provides immediate feedback, signaling to the user that the AI is currently correlating complex historical patterns.

A. The Health Forecast Reveal

The results are presented through a "Future-Forward" UI that simplifies complex neurodegenerative data into actionable insights:

- **The Weekly Outlook**: At the top of the screen, the user is greeted with a high-level status (e.g., **Stable**, **Improving**, or **Potential Decline**). This summary provides an immediate "vibe check" of the coming week.
- **The Insights Carousel**: The app renders a series of cards identifying specific "Canary Symptoms" - minor changes in one area (like Sleep) that the AI predicts will lead to changes in another (like Mood).

B. Actionable Guidance & Prevention

The true power of the feature lies in the **Recommended Actions** section. Rather than just showing data, CogniCare tells the user exactly how to respond to the forecast:

- **Personalized Advice**: The AI generates a numbered list of proactive steps, such as *"Prioritize early evening rest"* or *"Reduce screen time to mitigate predicted cognitive fatigue"*.
- **The Proactive Loop**: This allows the patient to adjust their schedule *before* a flare-up occurs, effectively using the app to flatten the curve of their symptoms.

C. Collaborative Preparation (Role-Based Views)

The demonstration concludes by showing how this data is shared across the **Care Circle**:

- **The Patient's View**: Focuses on self-care and autonomy, empowering them to manage their own stability.
- **The Carer's View**: Provides specific "Clinical Vigilance" tips, telling the family member exactly what behaviours to watch for in the coming week.
- **Hardware Encryption**: The resulting forecast is never stored in local storage; it is either held in volatile state or encrypted via **Capacitor Secure Storage** if persisted.

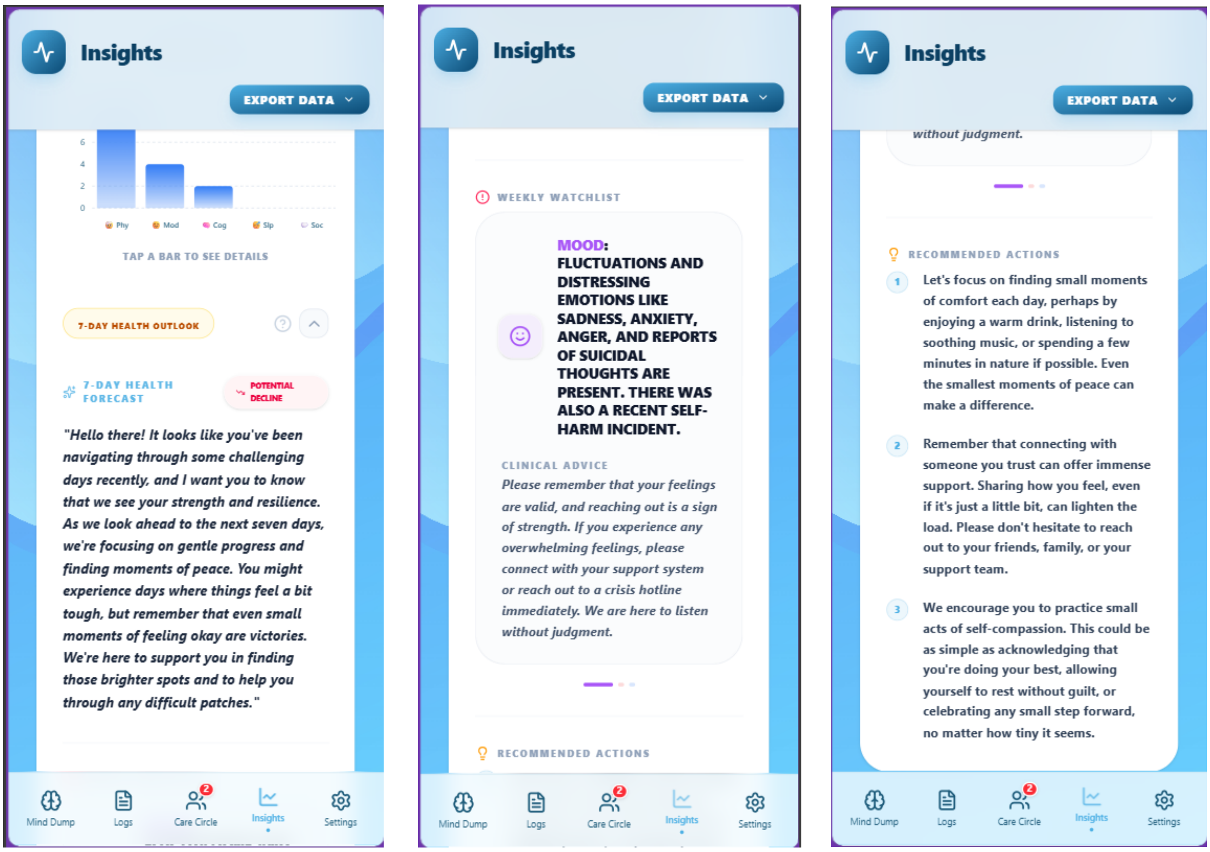


Figure 25: Predictive Analysis view Snapshots

5.1.6 Phase 6: Professional AI Health Report

This journey details the process of transforming thousands of data points into a single, clinical-grade document designed as a reference or symptom log overview for neurologists and GPs.

Step 1: The Export Intent & Eligibility

- **User Action**: From the Insights Dashboard, the user (Patient or Carer) selects a specific temporal window (e.g., *"Last Month"*) and opens the **Export** menu.
- **System Validation**: The system automatically checks <u>hasDataInRange</u> and <u>majorSymptoms</u>. If the patient has not logged enough data for the selected period, the **"Download AI PDF"** button remains disabled with a tooltip: *"No Logs in selected period"*.
- **Trigger**: The user clicks **"Download AI PDF".**

Figure 26: AI Clinical PDF

Step 2: Dual-Layer Data Synthesis (The Backend Engine)

- **Action**: This executes the generateProfessionalReportAction on the backend, which aggregates data across two distinct timelines to ensure the report is both precise and contextually accurate.
- **Primary Analytics (The Selected Range):** The engine calculates pillar-specific averages, log frequencies and a **"Snapshot Comparison"** (Start vs. End date) for the specific period selected by the user. This data provides the immediate metrics for the report's charts and metric bars.
- **Longitudinal Context (The 6-Month Baseline):** Simultaneously, the engine pulls the patient's all-time averages and a 6-month historical trajectory. While this data is not visually rendered as a chart in the final PDF, it serves as a critical **"Background Baseline"** for the AI.
- **AI Clinical Reasoning:** Gemini 2.5-Flash performs a final high-level synthesis, cross-referencing the **Selected Period** data against the Longitudinal Baseline. This allows the AI to provide expert-level insights (e.g., identifying if a current "Physical" score is a temporary flare-up or part of a worsening 6-month trend).
- **Risk Identification:** The system scans the raw text of all logs in the period for specific **"Clinical Safety Markers" (Risk Keywords)**, automatically escalating detected issues to the high priority **"Critical Risks"** section of the report.

Step 3: Multi-Page Document Assembly

The frontend renders a hidden ReportTemplate, constructed using a formal clinical architecture:

Page 1: Report Overview (Table of Contents)

Scope: Serves as a professional formal roadmap for the medical clinician.
Component 1: Clinical Roadmap: A structured list using high-fidelity icons that defines the four primary sections of the report:

- **Patient Health Summary**: Historical averages.
- **AI Clinical Insights**: Key findings and safety risks.
- **NHS Guidance Alignment**: Correlation with official markers.
- **Professional Clinician Memo**: The evaluation space.

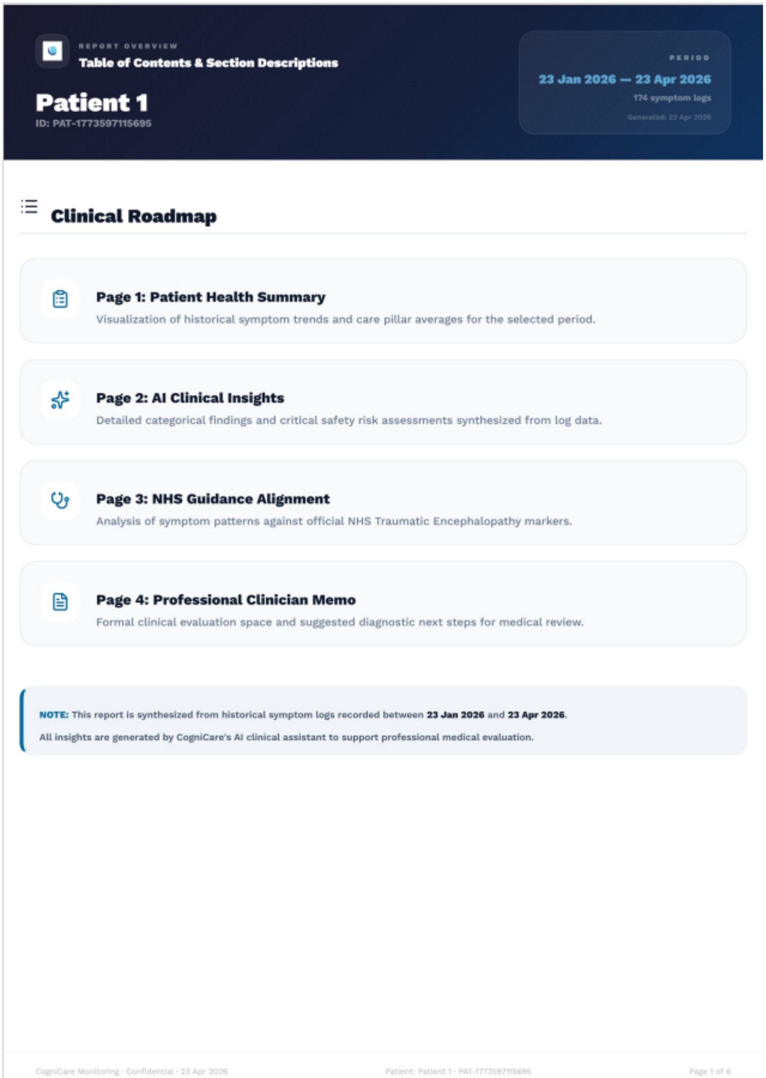


Figure 27: Report Overview (Table of Contents)

Page 2: Overall Symptom Analysis

Scope: This page is strictly focused on the **Selected Date Range** and the **Most Recent 7 Days** for major symptoms.

Component 1 (Clinical Metric Bar): At the top, the report displays a five-column summary bar showing:

- **Patient vs. Carer Log Count:** Validates data participation.
- **Total Logs:** The volume of data analysed.
- **Highest Burden:** Identifies which clinical pillar (e.g., Cognitive) was most severe during this period.
- **Most Managed:** Identifies which pillar remained most stable.

Component 2 (Symptom Burden Doughnut Chart): A large, color-coded pie chart (Doughnut Chart) that visualizes the percentage distribution of symptom severity across the 5 pillars for the **selected period**. It also includes small indicators showing how many logs contributed to each slice.

Component 3 (Major Symptoms & Alerts): A dedicated card (MajorSymptomsReportCard) listing the top symptoms and high-priority alerts triggered in the **last 7 days**. This ensures the doctor sees immediate "Red Flags" first.

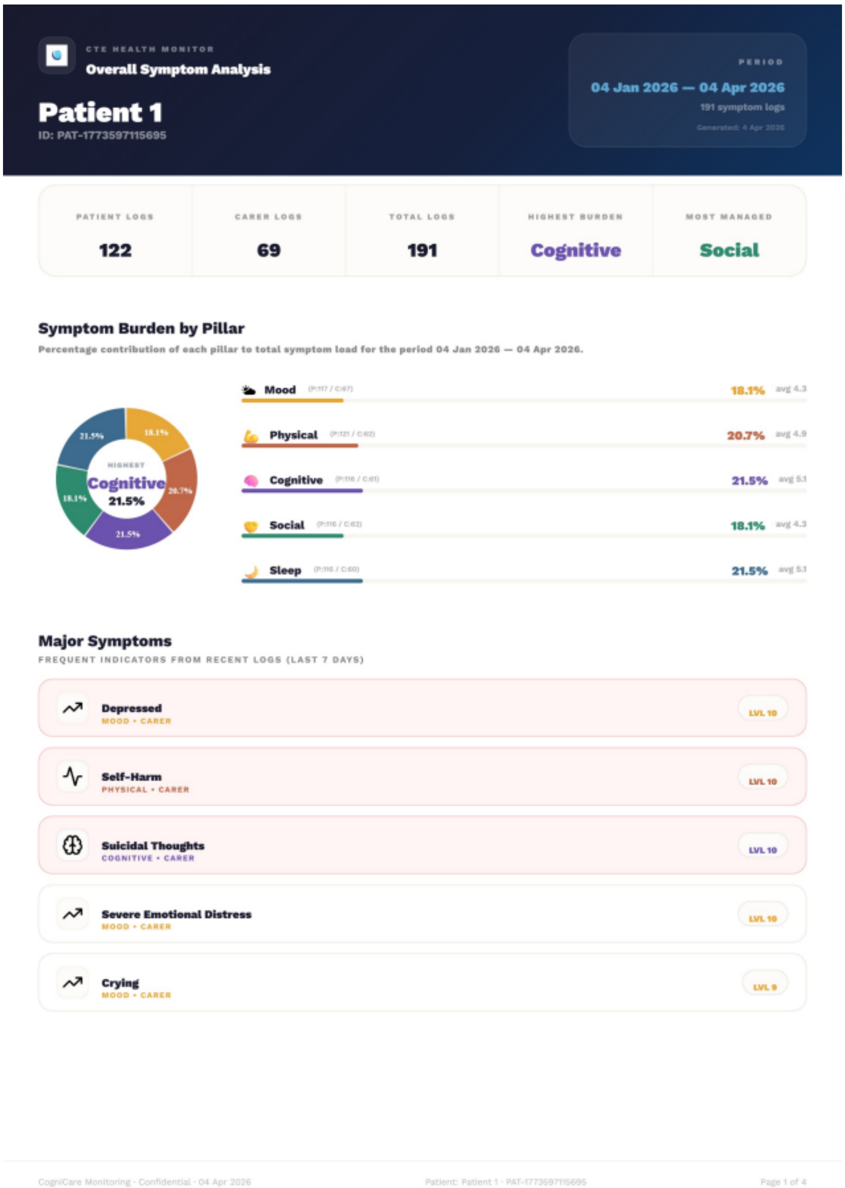


Figure 28: Symptom Analysis Overview

Page 3: AI Clinical Insights (The Primary Synthesis)

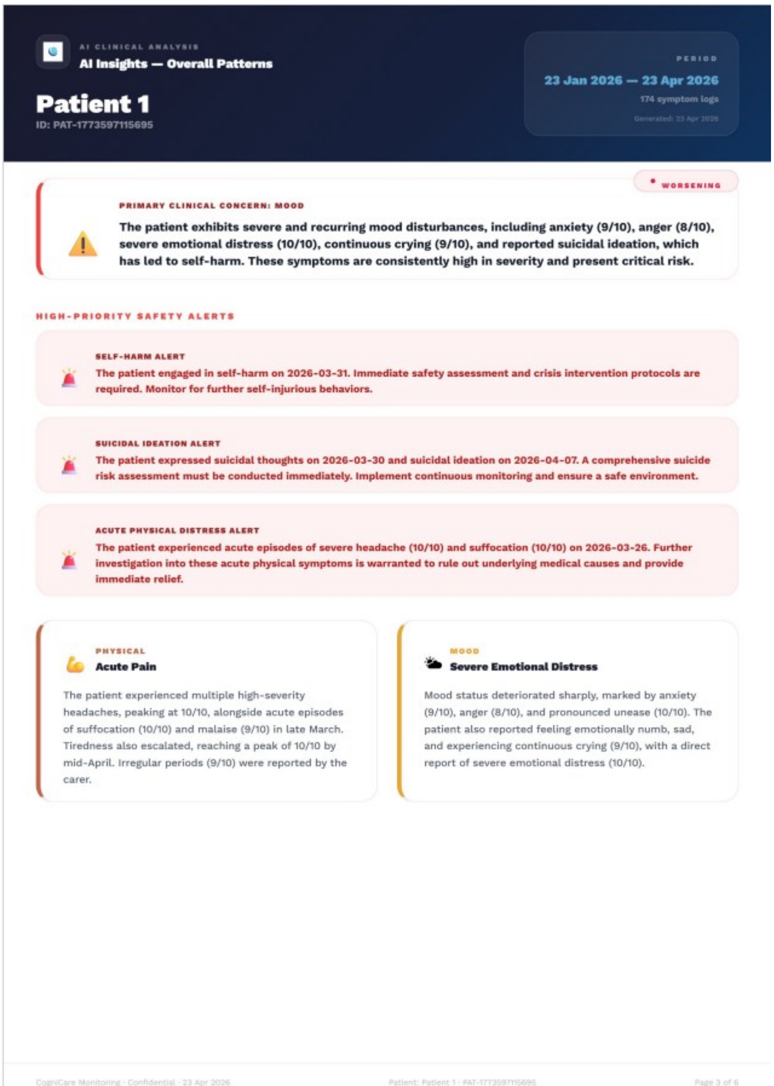
Scope: Translates raw logs into high-level diagnostic reasoning.

Component 1: Trajectory Status Badge: A dynamic badge in the top-right corner (Improving, Stable, or Worsening) provides an immediate "Clinical Direction" for the doctor.

Component 2: Primary Clinical Concern: A high-contrast red-bordered section that identifies the most critical pillar of concern and provides the specific AI-reasoned justification.

Component 3: Safety Alerts: A prioritized list of "Critical Risks" (e.g., *"Detected pattern of nighttime wandering"*) marked with emergency icons to ensure immediate triage.

Component 4: Categorized Findings: A two-column grid that breaks down specific findings by clinical pillar (Physical, Mood, Cognitive, etc.), providing a "Sub-Category" and a detailed narrative finding for each.



AI CLINICAL ANALYSIS

AI Insights — Overall Patterns

Patient 1

ID: PAT-1773597115695

PERIOD

23 Jan 2026 — 23 Apr 2026

174 symptom logs

Generated: 23 Apr 2026

WORSENING

PRIMARY CLINICAL CONCERN: MOOD

The patient exhibits severe and recurring mood disturbances, including anxiety (9/10), anger (8/10), severe emotional distress (10/10), continuous crying (9/10), and reported suicidal ideation, which has led to self-harm. These symptoms are consistently high in severity and present critical risk.

HIGH-PRIORITY SAFETY ALERTS

SELF-HARM ALERT

The patient engaged in self-harm on 2026-03-31. Immediate safety assessment and crisis intervention protocols are required. Monitor for further self-injurious behaviors.

SUICIDAL IDEATION ALERT

The patient expressed suicidal thoughts on 2026-03-30 and suicidal ideation on 2026-04-07. A comprehensive suicide risk assessment must be conducted immediately. Implement continuous monitoring and ensure a safe environment.

ACUTE PHYSICAL DISTRESS ALERT

The patient experienced acute episodes of severe headache (10/10) and suffocation (10/10) on 2026-03-26. Further investigation into these acute physical symptoms is warranted to rule out underlying medical causes and provide immediate relief.

PHYSICAL

Acute Pain

The patient experienced multiple high-severity headaches, peaking at 10/10, alongside acute episodes of suffocation (10/10) and malaise (9/10) in late March. Tiredness also escalated, reaching a peak of 10/10 by mid-April. Irregular periods (9/10) were reported by the carer.

MOOD

Severe Emotional Distress

Mood status deteriorated sharply, marked by anxiety (9/10), anger (8/10), and pronounced unease (10/10). The patient also reported feeling emotionally numb, sad, and experiencing continuous crying (9/10), with a direct report of severe emotional distress (10/10).

CogniCare Monitoring · Confidential · 23 Apr 2026

Patient: Patient 1 · PAT-1773597115695

Page 3 of 6

Figure 29: AI Clinical Insight PDF Overview - 1

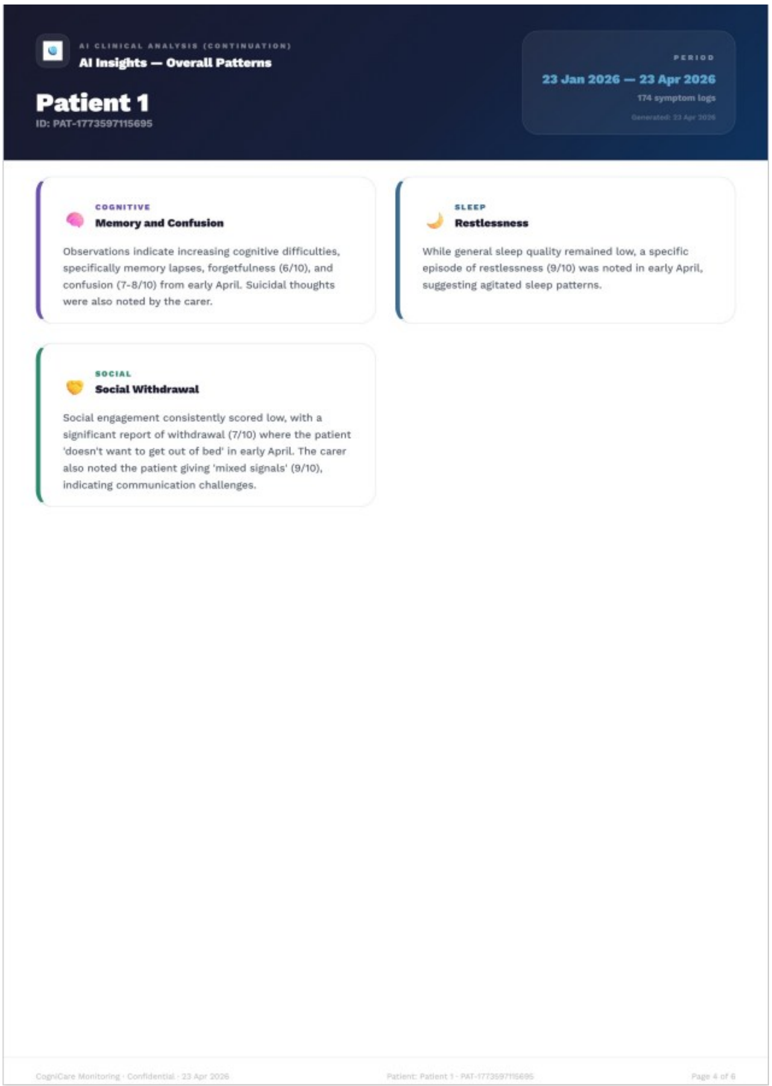


Figure 30: AI Clinical Insight PDF Overview - 2

Page 4: NHS Guidance Alignment (The Validation Layer)

Scope: Cross-references the patient's actual behaviour against official medical standards for CTE and Brain Injury.

Component 1: Clinical Alignment Narrative: A synthesized paragraph explaining how the patient's current trends correlate with known NHS health markers.

Component 2: Diagnostic Next Steps: A bulleted checklist of recommended clinical tests or screenings (e.g., *"Consult for MRI", "Memory Assessment"*) based on the detected symptom severity.

Component 3: Carer's Corner: A dedicated, purple-themed section that provides specific, observation-based tips for the carer to assist in managing the patient's current symptoms.

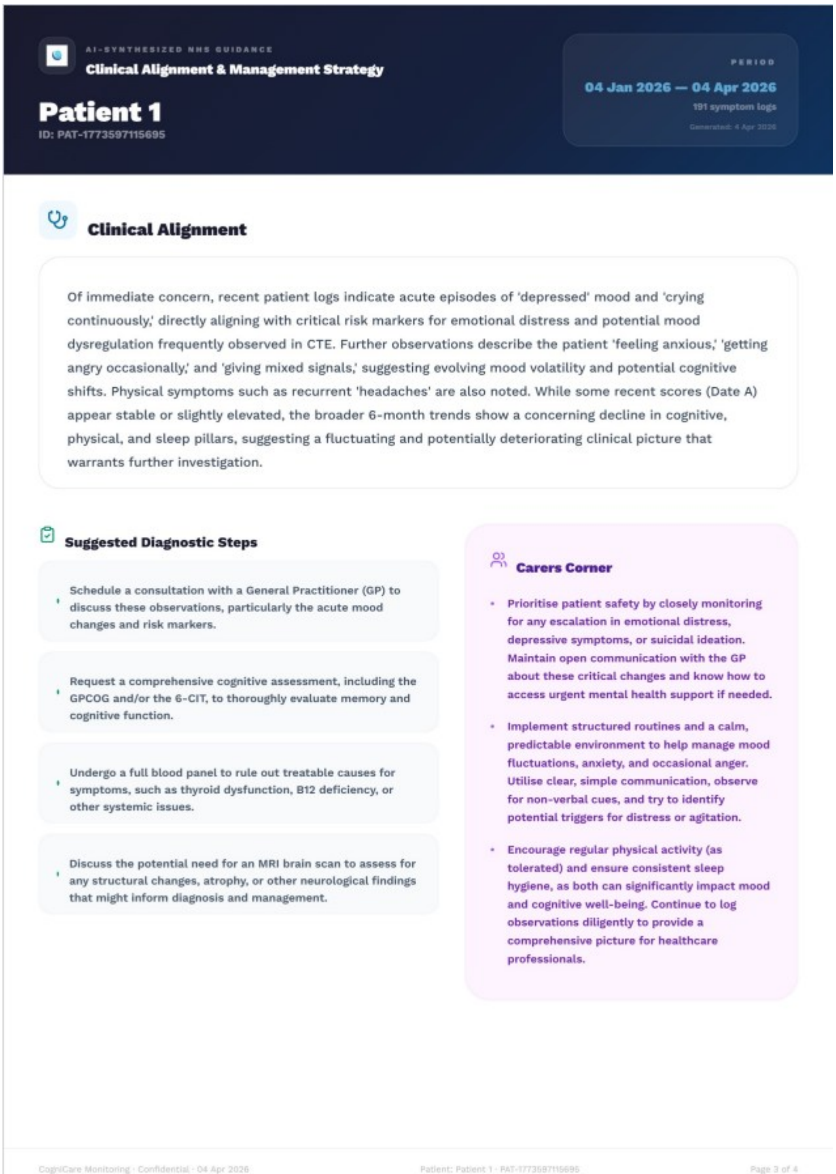


Figure 31: AI NHS Guidance PDF Overview

Page 5: Professional Clinician Memo (The Evaluation Space)

Scope: A formal concluding page designed to be hand-signed and used during an actual medical consultation.

Component 1: Professional Disclaimer: A clear legal and clinical notice stating that the report is a synthesis tool for professional medical evaluation, not a definitive diagnosis.

Component 2: Clinician Notes Space: A large, structured writing area with 16 dashed lines, allowing the doctor to record hand-written observations during the appointment.

Component 3: Formal Validation Block: Includes a professional signature line and a timestamped "Synthesized by CogniCare AI" version identifier to ensure the document's origin is transparent.

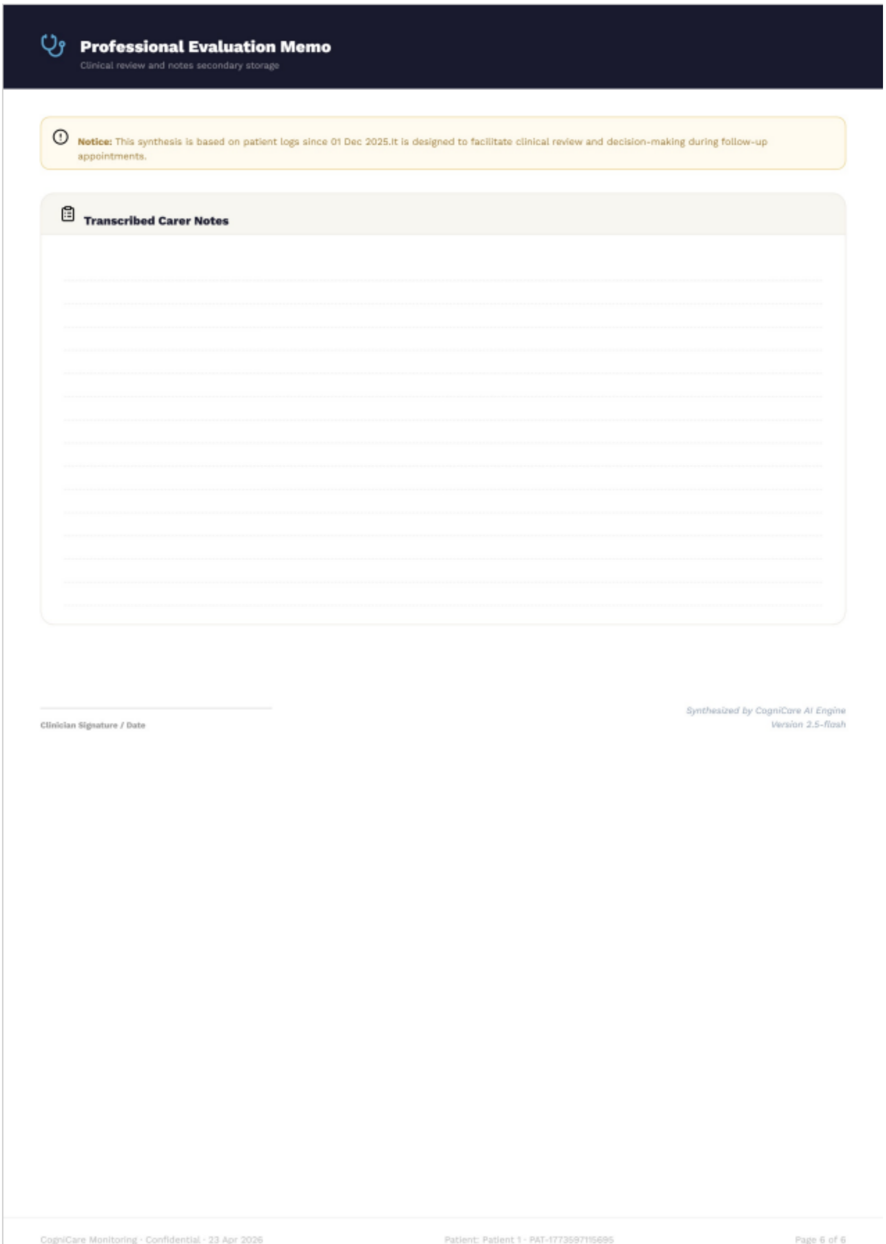


Figure 32: Memo Evaluation PDF View

Step 4: High-Fidelity Client-Side Capture

- **Mechanism**: The generatePdfFromElement service uses html2canvas to perform a "Virtual Scan" of the rendered report, preserving all custom styling, gradients and typography.
- **PDF Conversion**: jsPDF assembles these high-resolution captures into a finalized, multi-page PDF document.
- **Status Feedback**: The user sees a progress indicator: *"Creating clinical document..." → "Finalizing pages..."*

Step 5: Completion & Delivery

- **System Action:** The browser triggers an automatic download of the file.
- **File Naming:** The file is dynamically named using the convention: **Symptom-Report-[PatientID]-[CurrentDate].pdf** (e.g., Symptom-Report-PAT-12345-24-04-2026.pdf).
- **Outcome:** The user now possesses an immutable, professional audit trail that translates their daily "Mind Dumps" into a format that reduces medical appointment friction and improves diagnostic accuracy.

5.1.7 Phase 7: Clinical Doctor-Assessment Form (The TES Specialist Tool)

The **Doc-Form** is a high-level clinical module designed to assist neurologists in conducting a formal **Traumatic Encephalopathy Syndrome (TES)** assessment. It transforms months of raw logs into a structured, diagnostic-ready medical form.

Step 1: AI Data Priming (The Pre-fill)

- **Entry Point**: Accessible only to users with at least 30 distinct days of health data.
- **The AI Scan**: Upon entry, the system executes generateDoctorFormDataAction. **Gemini 2.5-Flash** performs a deep scan of the patient's entire history to automatically pre-populate the form.
- **Visual Feedback**: The user sees an "AI Analysis" screen with status updates like *"Mapping TES symptom criteria..."*.
- **AI Badging**: Any field pre-filled by the AI is marked with a subtle **"AI" badge**, allowing the clinician to distinguish between machine-suggested data and manually entered facts.

Figure 33: Doc-Form AI prefilling view

Step 2: Section 1 - Symptom Inventory

- **The Checklist**: The AI pre-selects symptoms (e.g., Irritability, Anxiety, Memory Loss) based on log frequency.
- **Clinical Metadata**: For every active symptom, the user validates:
  - **Duration**: Whether it is "Recent" or "6 Months+".
  - **Trend**: Whether it is "Improving", "Stable", or "Worsening".
- **Safety Logic**: This section must be fully completed (every active symptom needs a duration and trend) before the user can proceed.

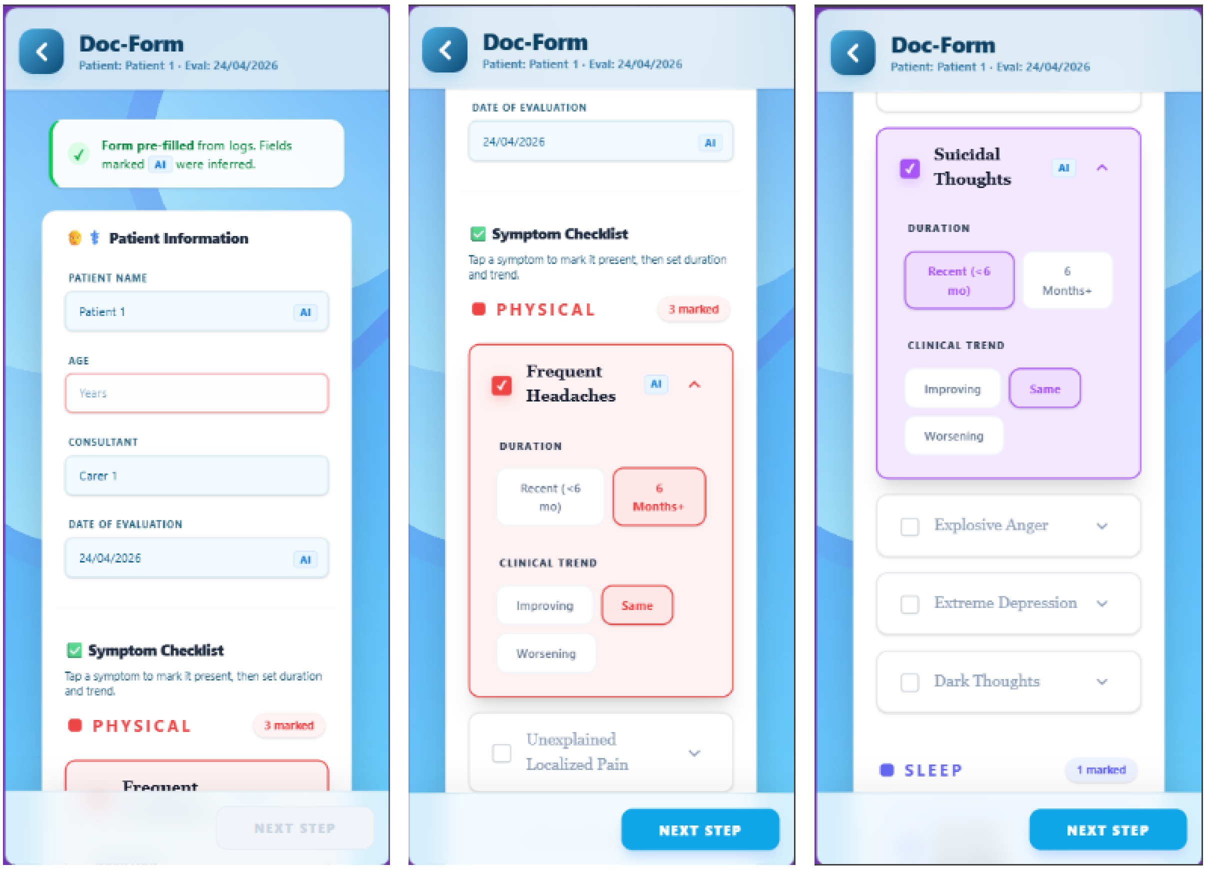


Figure 34: Doc-Form Symptom Checklist

Step 3: Section 2 - Patient History & Trauma

- **History Capture**: The clinician enters objective data including years of contact sports, number of recorded concussions and military service history.
- **Lifestyle Assessment**: Captures qualitative data on alcohol/substance use, diet and support networks.
- **AI History Grading**: As the user finishes this section, a background process (gradeHistoryRisk) analyses the narrative to assign a preliminary **History Risk Score** (0-15).

Figure 35: Doc- Form Patient History Details

Step 4: Section 3 - TES Clinical Criteria

- **Requirement Mapping**: The clinician maps the patient against official TES criteria:
  - **Criterion A2 (Core Symptoms)**: Requires at least one core symptom (Cognitive, Behavioural, or Mood).
  - **Criterion A3 (Supportive Symptoms)**: Requires at least two supportive markers (e.g., Impulsivity, Apathy, Motor symptoms).
- **Subtype & Course**: The clinician defines the clinical subtype (e.g., "Cognitive") and the disease course ("Progressive" vs. "Stable").

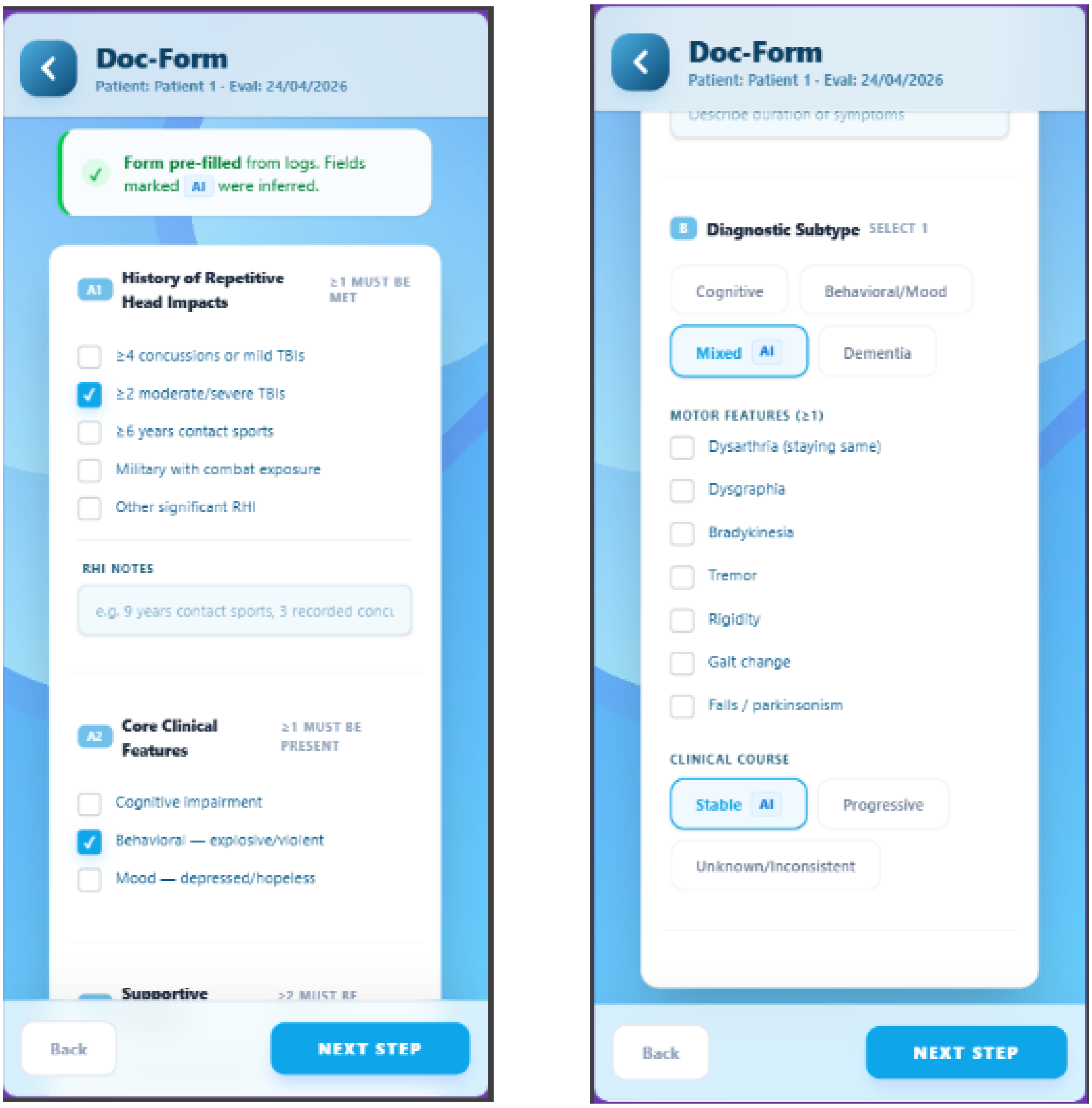


Figure 36: Doc- Form TES Criteria Assessment

Step 5: Final Review & Severity Scoring

- **The Severity Meter**: The final "Summary" page displays a real-time **Severity Score** (0-100), calculated using a weighted formula that combines the History Grade, Symptom Burden and TES Criteria.
- **AI Auto Suggestion**: Based on the final score, the AI provides a suggested **CTE Likelihood** (e.g., *"Probable CTE"* if the score is >70), which the clinician can then manually accept or override.

Figure 37: Severity Score Summary page

Step 6: High-Fidelity Professional Export

- **Document Generation**: Tapping “Download PDF” assembles a multi-page, formal clinical document (DocFormPDF).
- **Immersion**: The PDF generated with all the information users gave input with CTE probability resembling a standard clinical assessment form.
- **Delivery**: The file is saved as **CTE-Doctor-Form-[PatientName].pdf**, ready for official medical records or specialist referrals. See **Appendix 10.2** for the full mathematical formulation of the TES Triage Score.

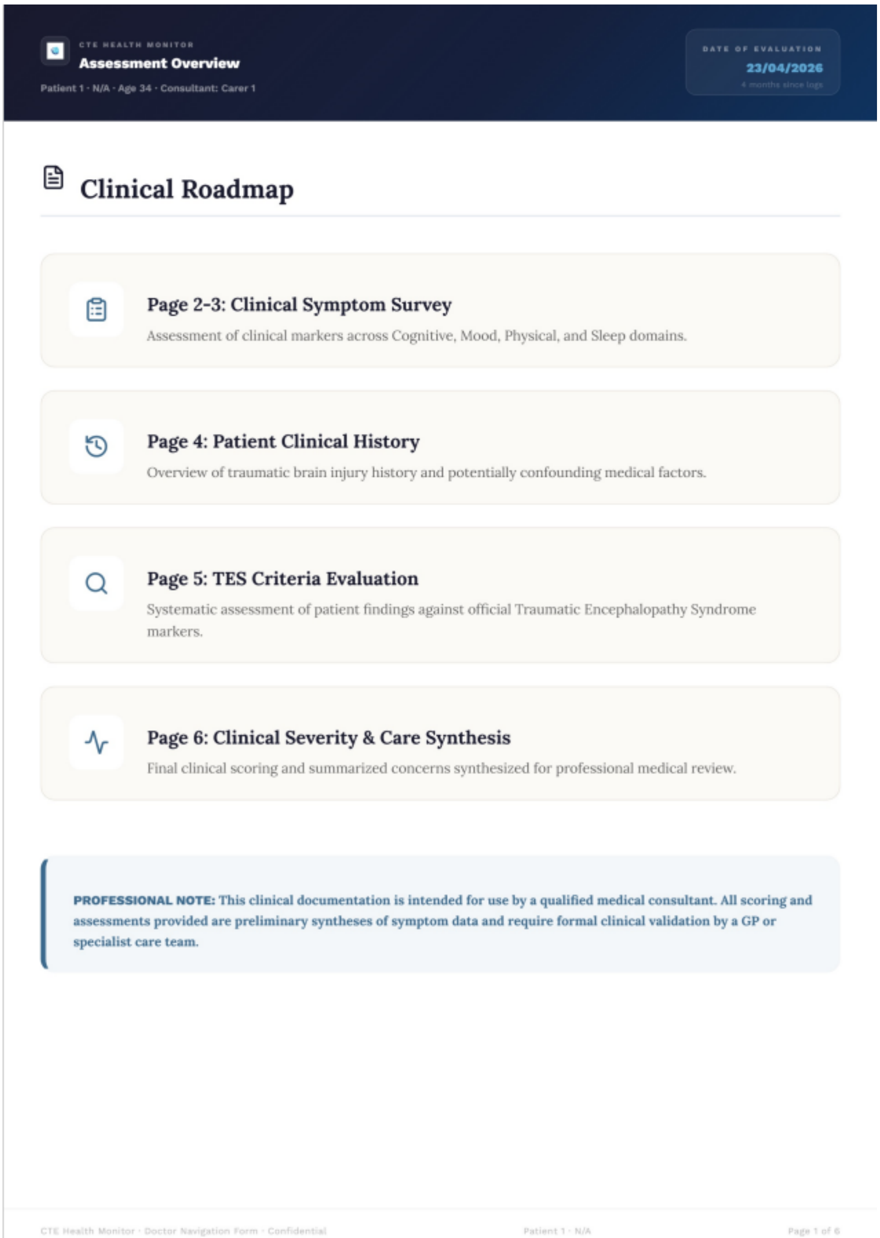
5.1.8 Phase 8: TES Assessment PDF

This journey details the structure and clinical significance of the **Doctor-Assessment PDF**, which is the final output of the 6-step Doc-Form module.

Page 1: Report Index & Patient Identifier

Purpose: To establish the formal identity and roadmap of the assessment.
Component:

- **Patient Demographics**: Full name, Age, Diagnosis Date and the specific Consultant in charge.
- **Table of Contents**: A professional 4-section index (History, Symptoms, TES Criteria, Final Scoring).
- **Evaluation Metadata**: Explicitly states the date of the assessment and the clinical version of the form.



CTE HEALTH MONITOR

Assessment Overview

Patient 1 · N/A · Age 34 · Consultant: Carer 1

DATE OF EVALUATION

23/04/2026

4 months since logs

Clinical Roadmap

Page 2-3: Clinical Symptom Survey

Assessment of clinical markers across Cognitive, Mood, Physical, and Sleep domains.

Page 4: Patient Clinical History

Overview of traumatic brain injury history and potentially confounding medical factors.

Page 5: TES Criteria Evaluation

Systematic assessment of patient findings against official Traumatic Encephalopathy Syndrome markers.

Page 6: Clinical Severity & Care Synthesis

Final clinical scoring and summarized concerns synthesized for professional medical review.

PROFESSIONAL NOTE: This clinical documentation is intended for use by a qualified medical consultant. All scoring and assessments provided are preliminary syntheses of symptom data and require formal clinical validation by a GP or specialist care team.

CTE Health Monitor · Doctor Navigation Form · Confidential Patient 1 · N/A Page 1 of 6

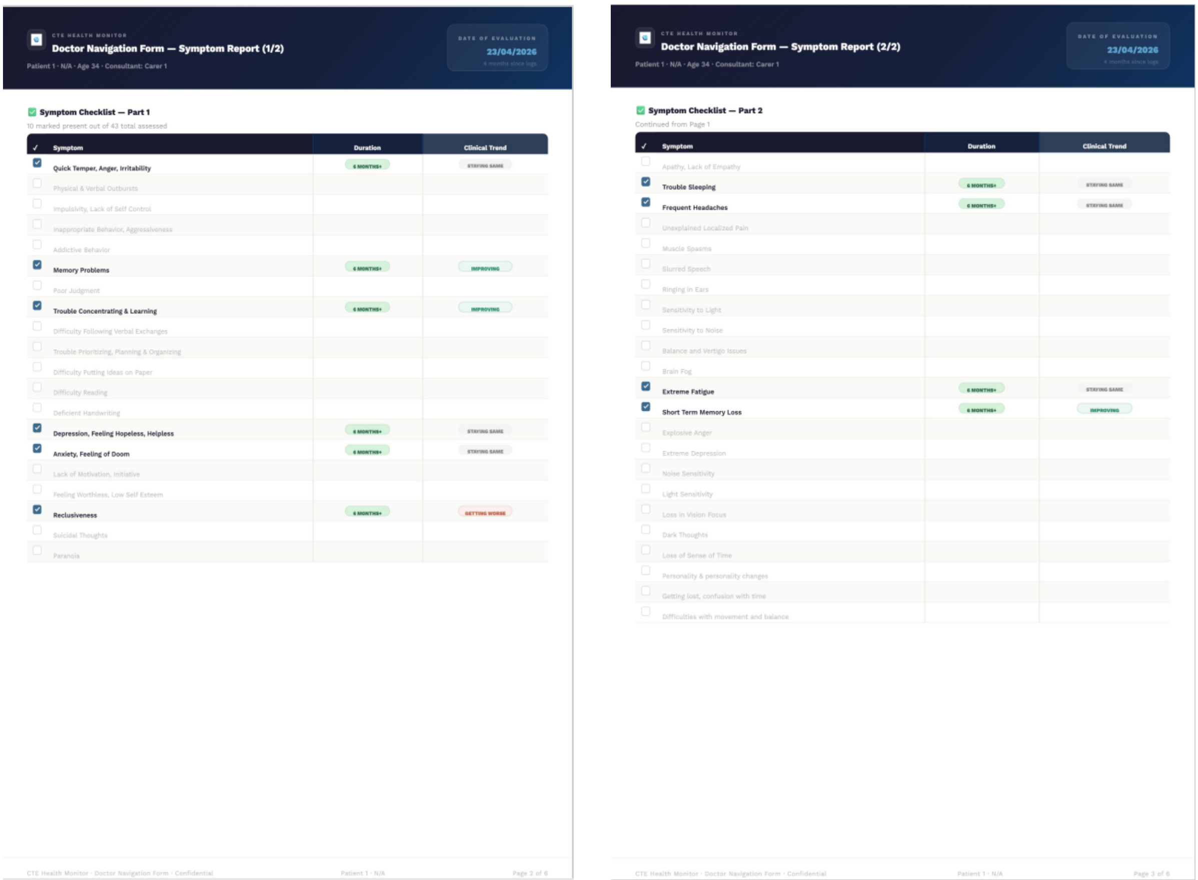
Figure 38: Doc-Form Report Index

Page 2 & 3: Comprehensive Symptom Inventory

Purpose: A high-density clinical audit of current symptom status.

Components:

- **Categorized Checklist:** A structured table of 15+ symptoms (e.g., Irritability, Word-finding issues, Ataxia).
- **Clinical Columns:** Each symptom is explicitly marked for Presence (Yes/No), Duration (Recent/6mo+) and Trend (Improving/Stable/Worsening).
- **Burden Metric:** Displays the total count of active symptoms identified during the analysis window.



CTE HEALTH MONITOR

Doctor Navigation Form — Symptom Report (1/2)

Patient 1 · N/A · Age 34 · Consultant: Carer 1

DATE OF EVALUATION

23/04/2026

4 months since logs

Symptom Checklist — Part 1

10 marked present out of 43 total assessed

✓	Symptom	Duration	Clinical Trend
✓	Quick Temper, Anger, Irritability	6 MONTHS+	STAYING SAME
	Physical & Verbal Outbursts		
	Impulsivity, Lack of Self Control		
	Inappropriate Behavior, Aggressiveness		
	Addictive Behavior		
✓	Memory Problems	6 MONTHS+	IMPROVING
	Poor Judgment		
✓	Trouble Concentrating & Learning	6 MONTHS+	IMPROVING
	Difficulty Following Verbal Exchanges		
	Trouble Prioritizing, Planning & Organizing		
	Difficulty Putting Ideas on Paper		
	Difficulty Reading		
	Deficient Handwriting		
✓	Depression, Feeling Hopeless, Helpless	6 MONTHS+	STAYING SAME
✓	Anxiety, Feeling of Doom	6 MONTHS+	STAYING SAME
	Lack of Motivation, Initiative		
	Feeling Worthless, Low Self Esteem		
✓	Reclusiveness	6 MONTHS+	GETTING WORSE
	Suicidal Thoughts		
	Paranoia		

CTE Health Monitor · Doctor Navigation Form · Confidential Patient 1 · N/A Page 2 of 6

CTE HEALTH MONITOR

Doctor Navigation Form — Symptom Report (2/2)

Patient 1 · N/A · Age 34 · Consultant: Carer 1

DATE OF EVALUATION

23/04/2026

4 months since logs

Symptom Checklist — Part 2

Continued from Page 1

✓	Symptom	Duration	Clinical Trend
	Apathy, Lack of Empathy		
✓	Trouble Sleeping	6 MONTHS+	STAYING SAME
✓	Frequent Headaches	6 MONTHS+	STAYING SAME
	Unexplained Localized Pain		
	Muscle Spasms		
	Slurred Speech		
	Ringing in Ears		
	Sensitivity to Light		
	Sensitivity to Noise		
	Balance and Vertigo Issues		
	Brain Fog		
✓	Extreme Fatigue	6 MONTHS+	STAYING SAME
✓	Short Term Memory Loss	6 MONTHS+	IMPROVING
	Explosive Anger		
	Extreme Depression		
	Noise Sensitivity		
	Light Sensitivity		
	Loss in Vision Focus		
	Dark Thoughts		
	Loss of Sense of Time		
	Personality & personality changes		
	Getting lost, confusion with time		
	Difficulties with movement and balance		

CTE Health Monitor · Doctor Navigation Form · Confidential Patient 1 · N/A Page 3 of 6

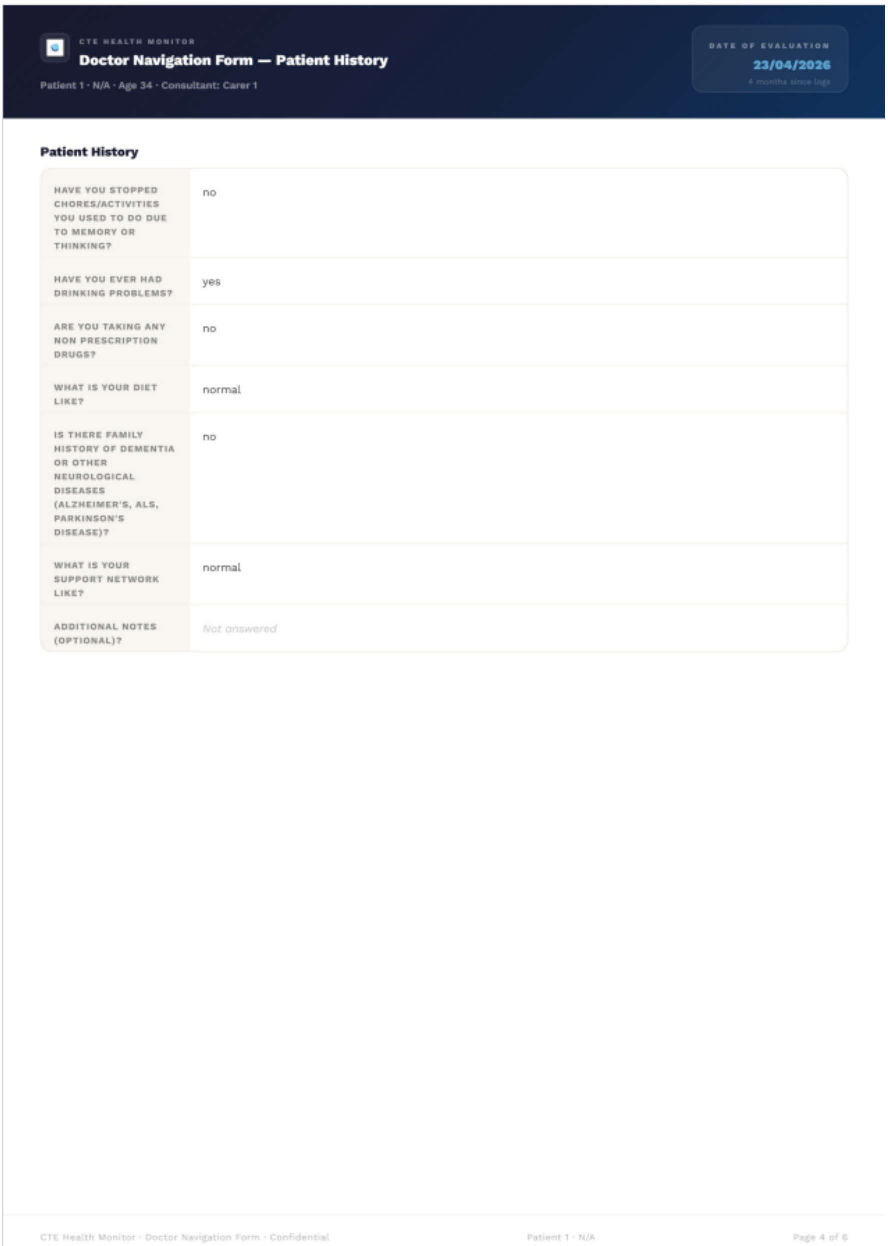
Figure 39: Symptom List PDF View

Page 4: Patient History

Purpose: To document the environmental and traumatic baseline of the patient.

Components:

- **Trauma Record**: Documents years of contact sports, military service and concussion history.
- **Lifestyle & Social Variables**: Includes alcoholic intake, substance history, dietary habits and the strength of the patient’s support network.
- **Narrative Insight**: Captures specific “Stopped Activities” (chores/hobbies) as a proxy for functional decline.



CTE HEALTH MONITOR

Doctor Navigation Form — Patient History

Patient 1 · N/A · Age 34 · Consultant: Carer 1

DATE OF EVALUATION

23/04/2026

4 months since logs

Patient History

HAVE YOU STOPPED CHORES/ACTIVITIES YOU USED TO DO DUE TO MEMORY OR THINKING?	no
HAVE YOU EVER HAD DRINKING PROBLEMS?	yes
ARE YOU TAKING ANY NON PRESCRIPTION DRUGS?	no
WHAT IS YOUR DIET LIKE?	normal
IS THERE FAMILY HISTORY OF DEMENTIA OR OTHER NEUROLOGICAL DISEASES (ALZHEIMER'S, ALS, PARKINSON'S DISEASE)?	no
WHAT IS YOUR SUPPORT NETWORK LIKE?	normal
ADDITIONAL NOTES (OPTIONAL)?	*Not answered*

CTE Health Monitor · Doctor Navigation Form · Confidential   Patient 1 · N/A   Page 4 of 6

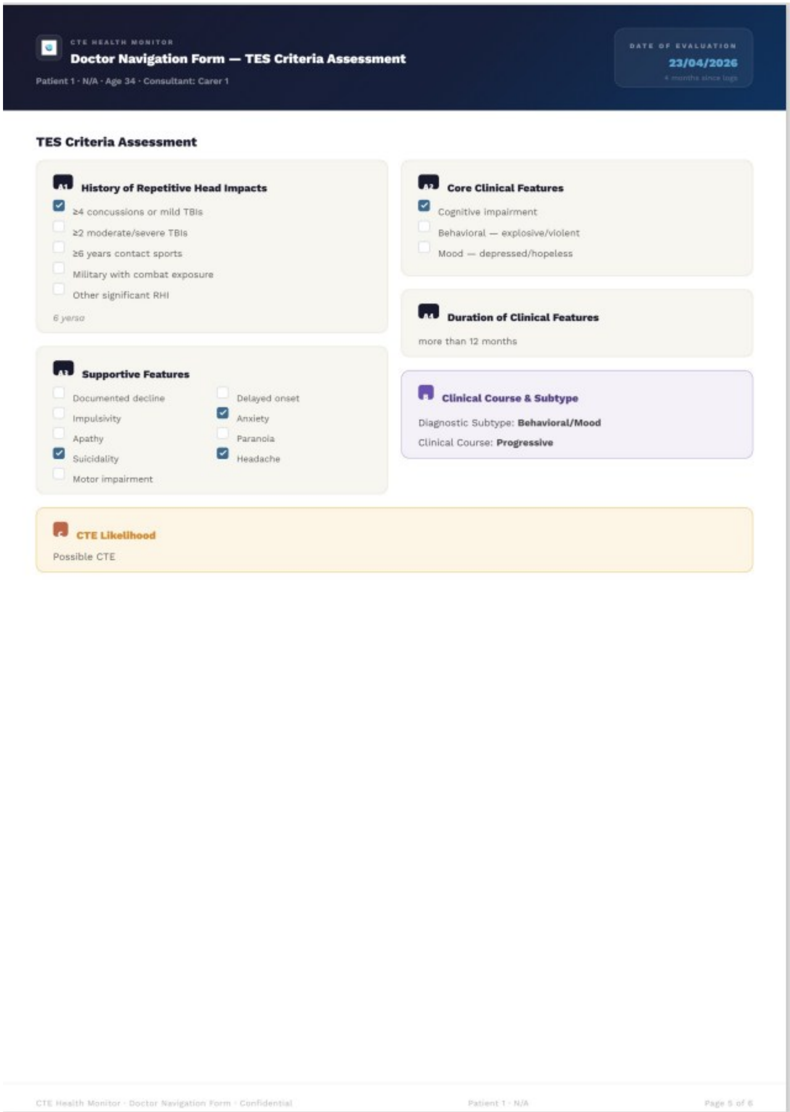
Figure 40: Patient History PDF View

Page 5: Clinical TES Assessment (Diagnostic Criteria)

Purpose: The formal mapping of symptoms against the official **Traumatic Encephalopathy Syndrome (TES)** markers.

Components:

- **Criterion Checklist**: Documents whether the patient meets **Criterion A2** (Core Symptoms) and **Criterion A3** (Supportive Symptoms).
- **Clinical Course**: Explicitly states the disease trajectory (e.g., *Progressive*) and the clinical subtype (e.g., *Cognitive*).
- **Professional Alignment**: Pre-populates relevant clinical indicators based on the previously validated symptom data.


CTE HEALTH MONITOR
Doctor Navigation Form — TES Criteria Assessment
Patient 1 · N/A · Age 34 · Consultant: Carer 1
DATE OF EVALUATION
23/04/2026

TES Criteria Assessment

History of Repetitive Head Impacts
- [x] ≥4 concussions or mild TBIs
- [ ] ≥2 moderate/severe TBIs
- [ ] ≥6 years contact sports
- [ ] Military with combat exposure
- [ ] Other significant RHI

6 yersa

Core Clinical Features
- [x] Cognitive impairment
- [ ] Behavioral — explosive/violent
- [ ] Mood — depressed/hopeless

Duration of Clinical Features
more than 12 months

Supportive Features
- [ ] Documented decline
- [ ] Impulsivity
- [ ] Apathy
- [x] Suicidality
- [ ] Motor impairment
- [ ] Delayed onset
- [x] Anxiety
- [ ] Paranoia
- [x] Headache

Clinical Course & Subtype
Diagnostic Subtype: **Behavioral/Mood**
Clinical Course: **Progressive**

CTE Likelihood
Possible CTE

CTE Health Monitor · Doctor Navigation Form · Confidential
Patient 1 · N/A
Page 5 of 6

Figure 41: TES Assessment Criteria PDF View

Page 6: Final Severity & Clinician Evaluation

Purpose: The concluding clinical verdict and professional evaluation space.

Component 1: **The Severity Gauge & Source Breakdown**

- **Visual Gauge**: Instead of a simple dial, the report uses a high-fidelity **Severity Gauge** that maps the total score (0-100).
- **Weighted Sources**: The page explicitly displays a **Source Breakdown** to show the doctor exactly how the score was calculated:
  - **Symptom Burden (30 pts)**: Based on presence and worsening trends.
  - **History Profile (15 pts)**: Based on the AI-graded patient history.
  - **TES Criteria (55 pts)**: The heaviest weight, based on how closely the patient matches official clinical markers (RHI, Core, and Supportive symptoms).
- **Urgency Narrative**: Beside the gauge, a dynamic "Urgency" box appears, providing clinical context for the current score (e.g., *"Immediate neurological referral recommended"*).

Component 2: Patient Communication Guidelines

- **Purpose**: This section translates the patient's sensory and cognitive sensitivities into actionable instructions for the doctor during the clinical encounter.
- **The Component**: A structured list of checkboxes that shows which accommodations the patient needs.
- **Content**:
  - **Environmental Needs:** Indicates if the patient requires dim lighting (due to high light-induced headaches) or quiet, non-crowded waiting areas (due to sensory chaos).
  - **Clinical Peace:** Alerts the doctor if the patient has low stamina and might need to split the exam into multiple visits.
  - **Cognitive Support:** Flags if the patient needs help with forms, slow verbal explanations, or written instructions due to confusion.
  - **Interaction Style:** Highlights if the patient wants to be explicitly included in decision-making and if they require extra encouragement/follow-up.

Component 3: Confidentiality & Disclaimer

- **Clinical Disclaimer**: A formal statement reminding the provider that this is an AI-assisted synthesis for professional evaluation purpose only.

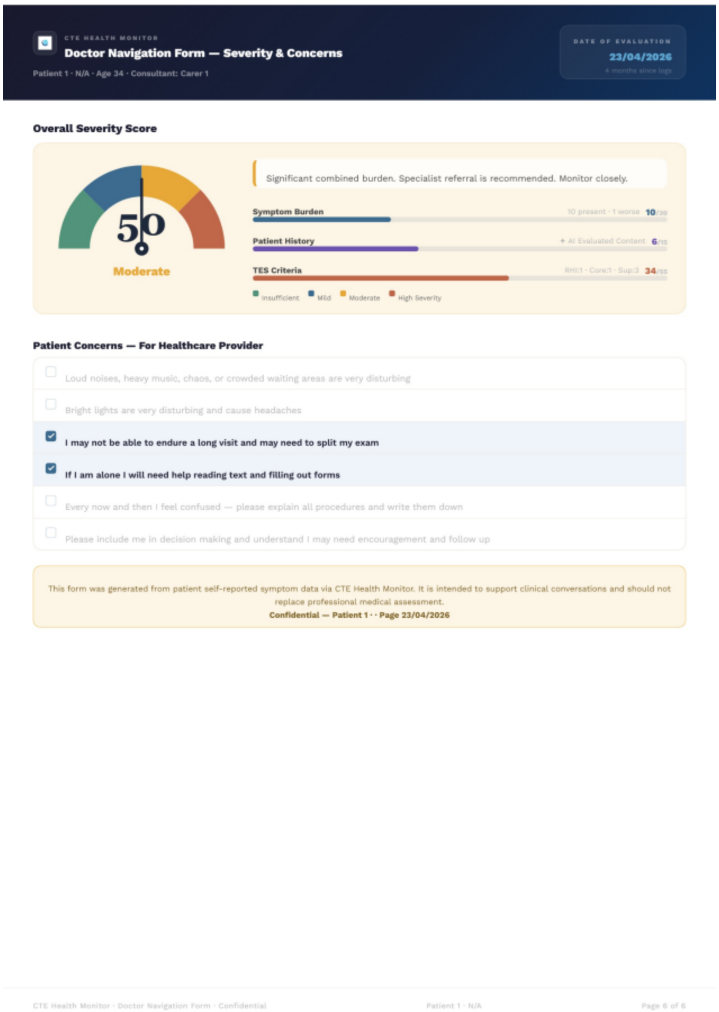


Figure 42: Final Severity and Clinical Evaluation PDF view

4.1.9 Phase 9: Patient Settings (Managed Access & Security)

This journey allows the patient to maintain full sovereignty over their clinical data. By managing access levels for their carer, the patient ensures that only trusted individuals can view sensitive health records or participate in care coordination.

Step 1: Navigating to the Security Hub

- **User Action**: From the main dashboard, the patient navigates to **Settings** and selects **"Manage Carer Access"**.
- **UI Feedback**: The patient is presented with a list of all currently connected carers. If no carers are connected, a desaturated card explains that carers can join by entering the **Patient ID** from their own app.

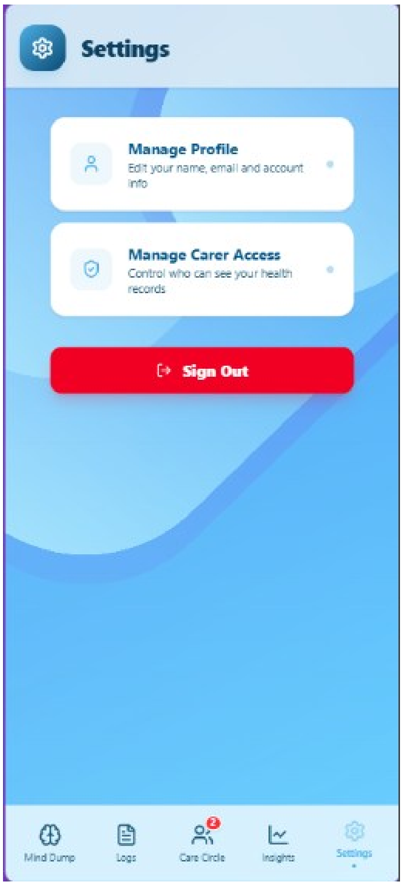


Figure 43: Settings Page view

Step 2: Monitoring Connection Status

- **Access Badging**: Each carer card features a dynamic status badge:
  - **Full Access (Green)**: The carer can view logs and participate in the Care Circle.
  - **Partial Access (Yellow)**: The carer has one of the two permissions restricted.
  - **Restricted (Gray)**: The carer is connected but cannot view clinical data.

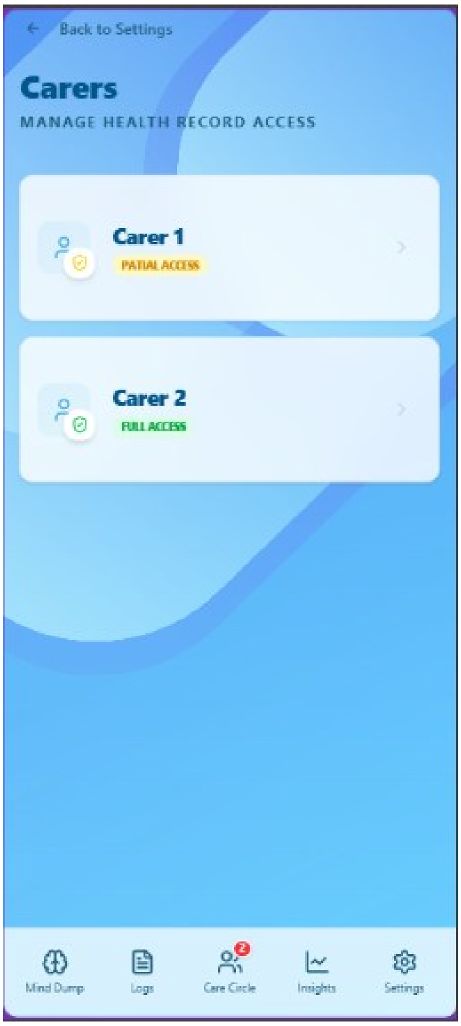


Figure 44: Manage Carer Access Page view

Step 3: Granular Permission Management

- **Trigger**: The patient taps on a carer's card to open the **Manage Access Modal**.
- **The Toggles**: The patient is presented with two primary security switches:
  - **Symptom Records**: Toggling this off immediately prevents the carer from viewing any past or future symptom logs or AI insights.
  - **Care Circle**: Toggling this off removes the carer from the chat and prevents them from seeing records of that patient.

Step 4: Real-time Permission Sync

- **Action**: The patient clicks **"Save Permissions"**.
- **Backend Logic**: The system executes updateCarerAccess, which updates the ProfilePatientToCarer join table in the database.
- **System Response**: The modal closes, and the main list instantly updates the badges. The carer's access is updated in real-time, the next time they attempt to view a restricted page, they will be blocked by the system's "Gatekeeper" security pattern.

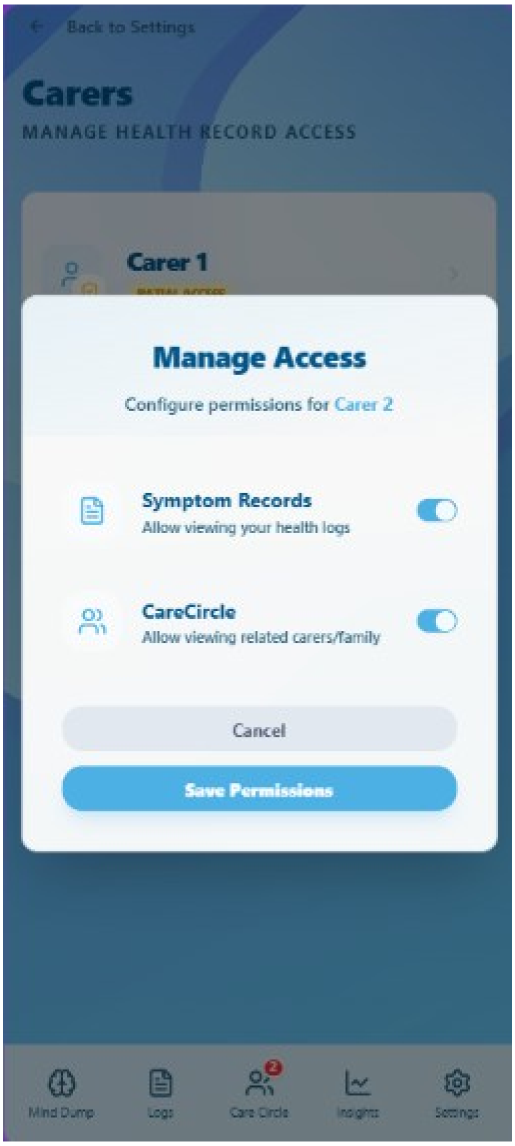


Figure 45: Manage Access for Carer Dialog

Step 5: Revoking Access

- **Control**: If a patient wishes to completely sever a connection, they can toggle both switches to **Off**. While the connection remains for administrative purposes, the "Restricted" state ensures the carer is effectively locked out of all clinical and social features of the app until re-authorized.

5.2 User Personas

To ensure CogniCare addresses the genuine clinical and emotional needs of those affected by CTE, two primary user personas were developed. These personas represent the **“Patient-Carer Dyad”**, which is the core collaborative unit of the platform.

Persona 1: The Patient (The “Struggling Veteran”)

Name: David
Profile: 48-year-old Military Veteran
Background: David has a history of blast exposures and repetitive head impacts from his service. He is currently experiencing early-stage **Traumatic Encephalopathy Syndrome (TES)** symptoms.
Pain Points:

- **Memory Fragmenting:** Struggles to remember when exactly his headaches or “brain fog” started.
- **Emotional Volatility:** Experiences sudden irritability but feels too overwhelmed to explain it during short doctor appointments.
- **Fear of Loss of Autonomy:** Worries that “forgetting things” will lead to his family taking away his independence.

How CogniCare Solves This:

- **The “Mind Dump”:** David doesn’t have to remember dates; he just speaks into the app.
- **Data Normalization:** The AI identifies his irritability as a “Mood Pillar” trend, providing him with a visual “Stability Map” he can show his doctor.
- **Empowered Autonomy:** He chooses what to share with his “Care Circle”, maintaining control over his narrative.

Persona 2: The Carer (The “Vigilant Daughter”)

Name: Sarah
Profile: 26-year-old Professional
Background: Sarah lives 50 miles away from her father (David). She is his primary emotional support but struggles to monitor his health remotely.
Pain Points:

- **Subjective Bias:** She notices her father seems “different” on the phone but lacks hard data to tell his neurologist.
- **Reactive Care:** She only finds out about a “bad week” after it has already happened.
- **Information Silos:** She has no way to see if his sleep issues are correlating with his mood swings.

How CogniCare Solves This:

- **Care Circle Notifications:** Sarah receives real-time alerts if a “High Severity” flag is triggered in David’s logs.
- **Predictive Insights:** The “7-Day Forecast” tells her to watch for cognitive fatigue mid-week, allowing her to call and check in proactively.
- **Unified Reporting:** She can export a clinical-grade PDF for David’s next appointment, ensuring his symptoms are taken seriously by the medical team.

6. System Design & Architecture

CogniCare is built upon a **Domain-Driven Hybrid Architecture** designed to solve the unique challenges of neurobehavioral health monitoring: high-latency AI processing, the need for real-time carer synchronization, and strict clinical data privacy.

6.1 System Design: The Hybrid BFF Model

The application follows a **Backend-for-Frontend (BFF)** pattern, bifurcates its responsibilities between two primary environments to ensure both high-performance UI and robust background processing:

- **Next.js Frontend (The Presentation Layer)**: A mobile-first web application built with Next.js 16 and React 19. It leverages **Capacitor** to bridge web technologies with native mobile capabilities (Voice Recording, Push Notifications, and Secure Storage).
- **Node/Express Logic Engine (The Intelligence Layer)**: A standalone server that handles "heavy-lifting" tasks. This separation is a strategic choice to avoid the execution timeouts found in standard serverless environments, allowing for complex, multi-stage AI orchestration and persistent WebSocket connections.

6.2 AI-Orchestrated Normalization Pipeline

At the core of CogniCare is an **Intelligent Normalization Pipeline** that transforms unstructured user input into clinical-grade data:

- **Multi-Model Resiliency**: The system utilizes **Google Gemini 2.5 Flash** as its primary engine, with an automated fallback logic that cycles through Gemini 2.0 and Gemini 2.5 Pro to reduce the likelihood of service interruption during API quota spikes. As this **prototype** operates on the **Gemini free tier**, sustained uptime cannot be guaranteed under heavy load; the **fallback mechanism** is designed to mitigate rather than eliminate quota-related failures.
- **Five-Pillar Symptom Extraction**: Raw text and voice logs are parsed into five health pillars: *Physical, Mood, Cognitive, Sleep,* and *Social*.
- **Privacy-First Design**: Before any data reaches the cloud, a **PII (Personally Identifiable Information) Masking Layer** redacts sensitive user details on the server, ensuring that AI analysis is performed on anonymized data.
- **Predictive Analytics**: The engine provides a 7-day outlook using historical trends, utilizing a **Result Cache** (via Prisma) to minimize latency and redundant API costs.

6.3 Stateful Real-Time Communication

CogniCare transforms the passive "Care Circle" into an active, synchronized environment through a stateful **Socket.io** implementation:

- **Room-Based Isolation**: The server employs a room-based architecture where each Patient and Carer is isolated into a private, secure socket room. This ensures that sensitive health updates are delivered only to authorized observers.
- **The Feedback Loop**: During AI analysis, the server emits processing_status pulses, providing the patient with immediate transparency into the diagnostic process.
- **Zero-Latency Sync**: Upon completion of a log, a new_log event is pushed to associated Carers, triggering a "Pulsing Indicator" and an automatic dashboard refresh without requiring manual user intervention.

6.4 Data Architecture & Persistence

The system utilizes **PostgreSQL** with **Prisma ORM**, employing a schema optimized for clinical relationships:

- **Profile Pattern**: A unified User identity is linked to specialized Patient or Carer profiles, allowing for granular role-based access control (RBAC).
- **Join-Table Security**: The carers_on_patients table manages the many-to-many relationships in a Care Circle, storing encrypted permissions that dictate exactly which clinical pillars a carer is authorized to view.
- **Export Engine**: A specialized service leverages jsPDF and html2canvas to synthesize clinical logs and AI insights into structured, professional PDF reports for healthcare providers.

6.5 Decoupled Infrastructure (Supabase & FCM)

To maintain a responsive core, CogniCare offloads background tasks to specialized infrastructure:

- **Edge Functions**: Deno-based **Supabase Edge Functions** act as an asynchronous bridge for mobile notifications. When a message or alert is triggered, the main server hands the task to an Edge Function, which then communicates with **Firebase Cloud Messaging (FCM)** to deliver the final push notification to the user's device. See **Appendix 10.1** for the full Edge Function implementation.

6.5 Session Lifecycle: Secure Auth

Registration

The registration process, orchestrated via the **Prisma $transaction API**, follows a rigorous four-step sequence:

1. **Identity Conflict Resolution**: The system first performs a cross-check against the User table to ensure the provided email or phone number is unique, preventing account duplication.

```
const isEmail = emailOrPhone.includes("@");
const email = isEmail ? emailOrPhone : null;
const phone = !isEmail ? emailOrPhone : null;

// Use transaction to ensure atomic creation
console.log(`[DATABASE: TRANSACTION_START] Initializing atomic registration for role: ${role}`);
return await prisma.$transaction(async (tx) => {
  // 1. Check if user already exists
  const existingUser = await tx.user.findFirst({
    where: {
      OR: [
        ...(email ? [{ email }] : []),
        ...(phone ? [{ phone }] : []),
      ],
    },
  });

  if (existingUser) {
    console.error(`[DATABASE: TRANSACTION_ROLLBACK] User ${emailOrPhone} already exists.`);
    throw new Error("A user with this email or phone already exists");
  }
```

Figure 46: Code for checking existing users during registration

2. **Cryptographic Hashing**: Passwords are never stored in plain text. The system applies a **12-round Bcrypt salt**, ensuring that the persistence layer only ever contains high-entropy cryptographic hashes.

```
// 2. Create User (with hashed password)
const hashedPassword = await bcrypt.hash(password, 12);
const user = await tx.user.create({
  data: {
    name,
    email,
    phone,
    role,
    password: hashedPassword,
  },
});
console.log(`[DATABASE: TRANSACTION_PROGRESS] Core User record created: ${user.id}`);
```

Figure 47: Code for 12-round bcrypt salt

3. **Role-Specific Provisioning**: Depending on the selected role, the system branches its logic:

- **Patient Branch**: The server generates a unique PAT identifier and automatically provisions a secondary FamilyMember record if emergency contact details are provided.

```
if (role === "PATIENT") {
  // 3. Create Patient Profile with Date-now ID
  const patientProfileId = `PAT-${Date.now()}`;
  const patientProfile = await tx.profilePatient.create({
    data: {
      id: patientProfileId,
      userId: user.id,
    },
  });
  console.log(`[DATABASE: TRANSACTION_PROGRESS] Patient Profile created: ${patientProfile.id}`);

  // 4. Create Family Member if details provided
  if (familyMemberName) {
    await tx.familyMember.create({
      data: {
        name: familyMemberName,
        email: familyMemberEmail || null,
        phone: familyMemberPhone || "",
        patientId: patientProfile.id,
      },
    });
  }

  console.log("[DATABASE: TRANSACTION_COMMIT] Patient account fully provisioned.");
  return {
    message: "Registration successful",
    userId: user.id,
    profileId: patientProfile.id,
  };
```

Figure 48: Code for Patient role

- **Carer Branch**: The server generates a CAR identifier. If the carer provides an existing Patient ID, the system performs a **Foreign Key Validation** before establishing a link in the carers_on_patients relationship matrix.

```
} else if (role === "CARER") {
  // 3. Create Carer Profile with Date-now ID
  const carerProfileId = `CAR-${Date.now()}`;
  const carerProfile = await tx.profileCarer.create({
    data: {
      id: carerProfileId,
      userId: user.id,
    },
  });
  console.log(`[DATABASE: TRANSACTION_PROGRESS] Carer Profile created: ${carerProfile.id}`);

  // 4. Link to Patient if ID provided
  if (patientId) {
    // VALIDATE Patient ID exists first to avoid foreign key crash
    const patientExists = await tx.profilePatient.findUnique({
      where: { id: patientId },
    });

    if (!patientExists) {
      console.error(`[DATABASE: TRANSACTION_ROLLBACK] Provided Patient ID ${patientId} is invalid.`);
      throw new Error(
        `Patient ID "${patientId}" not found. Please check the ID and try again.`,
      );
    }

    await tx.carersOnPatients.create({
      data: {
        carerId: carerProfile.id,
        patientId: patientId,
      },
    });
    console.log(`[DATABASE: TRANSACTION_PROGRESS] Care Circle link established with Patient: ${patientId}`);
  }

  console.log("[DATABASE: TRANSACTION_COMMIT] Carer account fully provisioned.");
  return {
    message: "Registration successful",
    userId: user.id,
    profileId: carerProfile.id,
  };
}
```

Figure 49: Code for Carer role

4. **Atomic Commitment**: Only if every sub-operation (User creation + Profile creation + Relationship linking) succeeds does the database commit the changes. If any step fails (e.g., an invalid Patient ID), the entire transaction is rolled back, maintaining perfect data integrity.

Authentication (Login)

The login process follows a **Decoupled Handshake** model. When a user provides their credentials, the frontend performs a TLS-encrypted POST request to the Logic Engine.

1. **Verification**: The server utilizes **Bcrypt (12-round salt)** to compare the provided password against the stored hash in PostgreSQL.
2. **Role Resolution**: Upon a successful match, the server performs a "Profile Resolution" query to determine if the user is a Patient or a Carer, returning a bundle of IDs (userId, profileId) rather than a simple Boolean.
3. **Hardware Enclave Persistence**: Instead of utilizing insecure browser localStorage, the frontend immediately persists this session bundle into **Hardware-Backed Secure Storage** via the capacitor-secure-storage-plugin. This ensures that even if the web view is compromised, the session credentials remain isolated within the device's **iOS Keychain** or **Android Keystore**.

```
const login = async (email: string, password: string) => {
    try {
        const res = await fetch(`${API_BASE_URL}/api/auth/login`, {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify({ email, password }),
        });

        const data = await res.json();

        if (!res.ok || !data.success) {
            throw new Error(data.error || "Login failed");
        }

        const authUser: AuthUser = {
            userId: data.userId,
            profileId: data.profileId,
            role: data.role,
            name: data.name,
            isCarer: data.role === "CARER",
        };

        await setItem(AUTH_STORAGE_KEY, JSON.stringify(authUser));
        setUser(authUser);
```

Figure 50: Code for storing user data in secure storage

```
import { useCallback, useMemo } from 'react';
import { SecureStoragePlugin } from 'capacitor-secure-storage-plugin';

export function useSecureStorage() {
    const setItem = useCallback(async (key: string, value: string) => {
        await SecureStoragePlugin.set({ key, value });
    }, []);

    const getItem = useCallback(async (key: string): Promise<string | null> => {
        try {
            const result = await SecureStoragePlugin.get({ key });
            return result.value;
        } catch (error) {
            // Plugin throws if key not found
            return null;
        }
    }, []);

    const removeItem = useCallback(async (key: string) => {
        await SecureStoragePlugin.remove({ key });
    }, []);

    const clear = useCallback(async () => {
        await SecureStoragePlugin.clear();
    }, []);

    return useMemo(() => ({ setItem, getItem, removeItem, clear }), [setItem, getItem, removeItem, clear]);
}
```

Figure 51: Code of secure storage hook

Secure De-provisioning (Logout)

The logout process is designed to ensure "Zero-Trace" security:

- **Local Clearing**: The AuthContext invokes the removeItem command, which explicitly deletes the encryption keys and data from the mobile device's hardware enclave.

- **State Reset**: The React state is immediately zeroed out, triggering a global re-render that invalidates all protected routes and clears any sensitive health data currently held in the application's memory (RAM).

```
const logout = async () => {
    disconnectSocket();
    await removeItem(AUTH_STORAGE_KEY);
    setUser(null);
};
```

Figure 52: Code for removing user data during logout

Session Rehydration

To balance security with user experience, CogniCare implements a **Hardware Rehydration** step. Every time the app is launched, the AuthProvider performs a secure "ping" to the hardware enclave to check for existing credentials. If found, the user is seamlessly logged back into their specific role-based dashboard, ensuring that push notifications and real-time socket listeners are re-initialized with the correct profileId.

```
// Rehydrate from secure storage on mount
useEffect(() => {
    const rehydrate = async () => {
        try {
            const stored = await getItem(AUTH_STORAGE_KEY);
            if (stored) {
                const parsedUser = JSON.parse(stored) as AuthUser;
                // Ensure isCarer is correctly set even if rehydrating from an old version of the object
                if (parsedUser && typeof parsedUser.isCarer === "undefined") {
                    parsedUser.isCarer = parsedUser.role === "CARER";
                }
                setUser(parsedUser);

                // Initialize Push Notifications with navigation callback
                import("@/lib/pushNotifications").then(({ PushNotificationService }) => {
                    PushNotificationService.initialize(parsedUser.userId, handleNavigate);
                });
            }
        } catch (error) {
            // ignore parse errors, but catch network/service errors if logic needs it
            if (error instanceof TypeError && error.message === "Failed to fetch") {
                handleServiceError(error);
            }
        } finally {
            setLoading(false);
        }
    };
    rehydrate();
}, [getItem]);
```

Figure 53: Session rehydration logic

6.6 Role-based Access Table Feature-wise:

Feature	**Description**	**Patient Access**	**Carer Access**
Mind Dump	AI-powered voice/text symptom logging	✅ Record & View	❌ No Access
Health Dashboard	Real-time metrics and trend overview	✅ Full Access	✅ Access dependent on patient
Symptom Logs	Historical list of all processed logs	✅ View & Edit	✅ View & Annotate
Care Circle	Support network and contact management	✅ Manage Circle	✅ View Circle
AI Insights	Cross-pillar health pattern analysis	✅ Clinical View	✅ Clinical View
Doc-Form	MDT-standard clinical report generation	✅ Full Access	✅ Full Access
Direct Messaging	1-to-1 chat with primary contacts	✅ Full Access	✅ Limited Access
Threaded Chat	Contextual discussion on specific logs(threads)	✅ Can create thread	✅ Just be a part of thread
PDF Export	Exporting structured clinical reports	✅ Full Access	✅ Full Access
Hotline Access	Quick-link to emergency support	✅ Full Access	❌ No Access

Table 2: Role-Based access based on features

6.7 High- Level System Architecture:

The following diagram illustrates the interaction between the cross-platform mobile frontend, the multi-model AI pipeline, and the decoupled notification engine.

Figure 54: High Level System Architecture

6.8: Entity Relationship Diagram

A visual map of **Prisma/PostgreSQL tables** which defines the entity relationships (e.g., how a User has many SymptomLogs, and how CareCircles connect Patients and Carers).

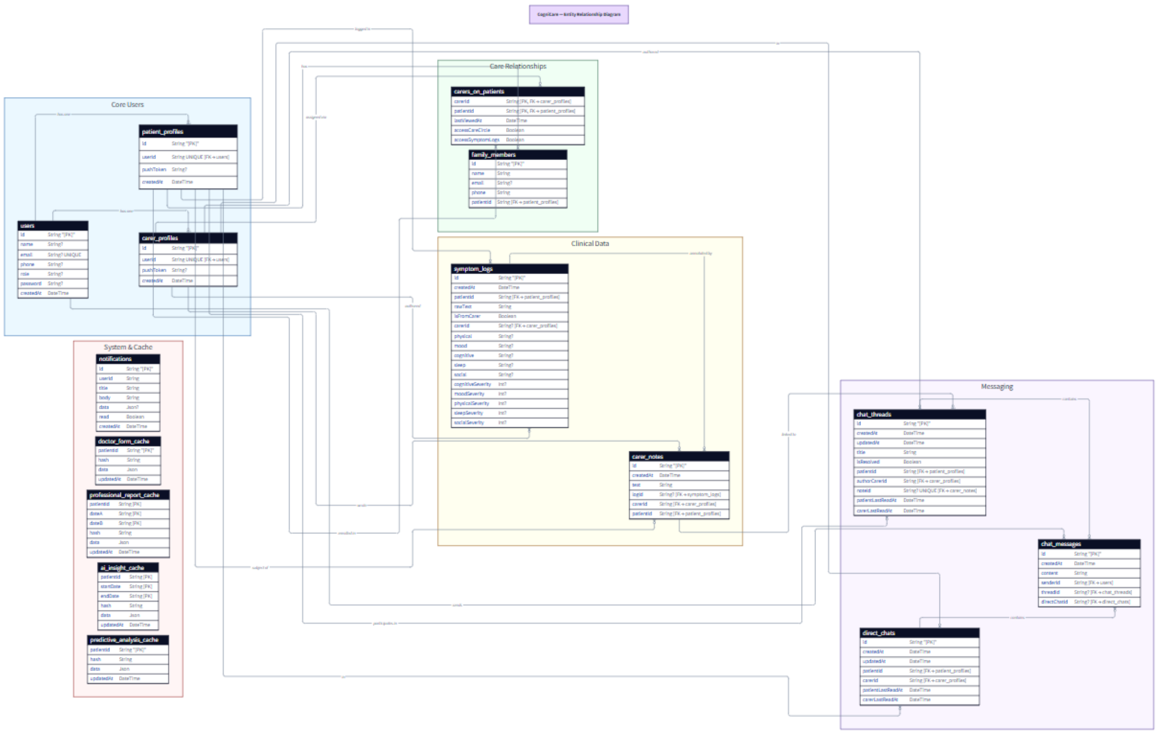


Figure 55: CogniCare - ERD Diagram

6.8.1 Data Architecture: Relational Schema & Persistence

CogniCare utilizes a **PostgreSQL** database managed through **Prisma ORM**. The schema is designed to support multi-role interactions (Patient/Carer), complex clinical data tracking, and a robust real-time messaging system. The data is organized into five logical modules:

6.8.1. Core User & Identity (User, Patient, Carer*)*

At the heart of the system is the User entity, which handles authentication and identity.
The Profile Pattern: Instead of storing all data in the User table, CogniCare uses a "One-to-One Profile" pattern. A User is linked to either a patient_profile or a carer_profile. This allows for role-specific metadata, such as pushTokens for mobile notifications, while keeping the core identity logic clean and secure.

6.8.2. Care Circle & Relationships (carers_on_patients)

The carers_on_patients table is a **Join Table** that facilitates a Many-to-Many relationship between Carers and Patients.
Granular Permissions: This table acts as a security gatekeeper, storing boolean flags (accessCareCircle, accessSymptomsLogs) that determine exactly what data a carer can see. It also tracks lastViewedAt to power the "New Log" notification dots on the dashboard.

6.8.3. Clinical Data & Observations (symptom_logs, carer_notes)

This module stores the "Health Pulse" of the application.
Symptom Logs: When a "Mind Dump" is processed, it is stored in symptom_logs. This table stores the rawText (for history) and five distinct clinical pillars (*Physical, Mood, Cognitive, Sleep, Social*) along with their AI-calculated severity scores (1-10).
Note-Log Link: Carers can attach carer_notes directly to specific logs, creating a contextual observation thread that bridges the gap between raw data and clinical interpretation.

6.8.4. Real-Time Messaging & Threads

The messaging layer supports two distinct communication flows:
Direct Chats: 1-on-1 private channels between a Patient and a specific Carer.
Care Threads: Contextual discussions derived from a specific observation or log. A chat_thread is often linked to a noteId, allowing the Care Circle to discuss a specific event in a focused thread.
Read State Tracking:
Both chat_threads and direct_chats track patientLastReadAt and carerLastReadAt, enabling accurate unread message counts across devices.

6.8.5. Optimization & Performance (System Cache)

To minimize the cost and latency of AI generation, the system utilizes several **JSON Cache** tables:
ai_insight_cache and predictive_analysis_cache: These store the structured output of Gemini models, indexed by a hash of the underlying clinical logs. If the patient's symptoms haven't changed, the system serves the cached AI summary instantly rather than re-generating it if the MD5 hash of the underlying logs remains unchanged.

6.9 The Data Processing Pipeline: Core Feature - "Mind Dump"

The system utilizes a specialized pipeline to handle the transition from high-level user actions to low-level database persistence, particularly for complex operations like the **"Mind Dump"** analysis. This architecture ensures that the intensive computational requirements of AI normalization do not degrade the user experience.

- **Step 1: Interface Interaction & The Decoupled Bridge** The process begins in the BrainDumpInterface. The user submits either a raw text entry or a voice-captured audio file. The React frontend acts as the first "Bridge", performing a secure POST request to the **Node/Express Logic Engine**. By offloading the processing to a dedicated server, the UI remains responsive and avoids Next.js serverless execution timeouts.

- **Step 2: The Privacy Gatekeeper (PII Masking)** Upon receiving the payload, the Express server immediately invokes the **PII Masking Layer** (js-pii-mask). Before any data is sent to the cloud, the "Gatekeeper" identifies and redacts sensitive information (names, addresses, phone numbers). This creates a "Data Sovereignty Boundary", ensuring that sensitive identifiers never leave the internal infrastructure, maintaining strict alignment with GDPR and HIPAA-like privacy standards.

```
try {
  console.log(`[SECURITY: PII_MASKING] Original Text: "${rawText}"`);
  const safeText = mask(rawText || "", {
    nlp: true,
    customRules: [
      // Distinction-Level Heuristic: Catch capitalized names in the middle of sentences
      { pattern: /(?<!^|\.\s|\?\s|\!\s)\b([A-Z][a-z]+)\b/g, replacement: "PERSON" }
    ]
  });
  console.log(`[SECURITY: PII_MASKING] Redacted Text: "${safeText}"`);
```

Figure 56: Code for PII-Masking before sending it to LLM.

- **Step 3: AI Orchestration & Live Feedback Loop** The sanitized text is transmitted to the **Gemini 2.5 Flash** model with a specialized domain prompt. While the AI performs the complex extraction of health pillars (Physical, Mood, Cognitive, etc.), the server maintains a **Live Feedback Loop**. It emits a processing_status event via WebSockets, allowing the patient's UI to display real-time status updates like *"Analyzing Symptom Severity..."*

```
try {
  let result;
  try {
    result = await model.generateContent(prompt);
  } catch (apiErr: unknown) {
    const err = apiErr as AppError;
    // 503 = Service Overloaded
    if (err.message?.includes("503") || err.message?.includes("overloaded")) {
      console.warn(
        "[AI_RESILIENCE: FALLBACK_TRIGGERED] Gemini Flash 2.5 is overloaded, pivoting to Gemini Flash-Lite...",
      );
      const fallbackModel = genAI.getGenerativeModel({
        model: "gemini-2.5-flash-lite",
      });
      result = await fallbackModel.generateContent(prompt);
      console.log("[AI_RESILIENCE: SUCCESS] Fallback model generated content successfully.");
    } else {
      throw apiErr;
    }
  }
```

Figure 57: Code for Fallback LLM

- **Step 4: Clinical Data Persistence** Once Gemini returns the structured JSON; the logic engine validates the scores (1-10). The server then utilizes the **Prisma ORM** to commit the record to the symptomLog table in **PostgreSQL**. This step ensures a permanent, immutable record of the patient's health state at that specific timestamp.

- **Step 5: Stateful Real-Time Synchronization** In the final stage, the server performs a **Targeted Broadcast**. It queries the database to identify all authorized carers within the patient's "Care Circle". It then emits a new_log event specifically to those carers' private Socket.io rooms.

- **Step 6: UI Reactivity (The Pulse)** The Carer Dashboard receives the new_log event and reacts immediately. The interface triggers a **Pulsing Indicator** on the patient's card and invalidates the local data cache. This allows the carer to view the new clinical data and AI insights instantly, without ever needing to refresh the page.

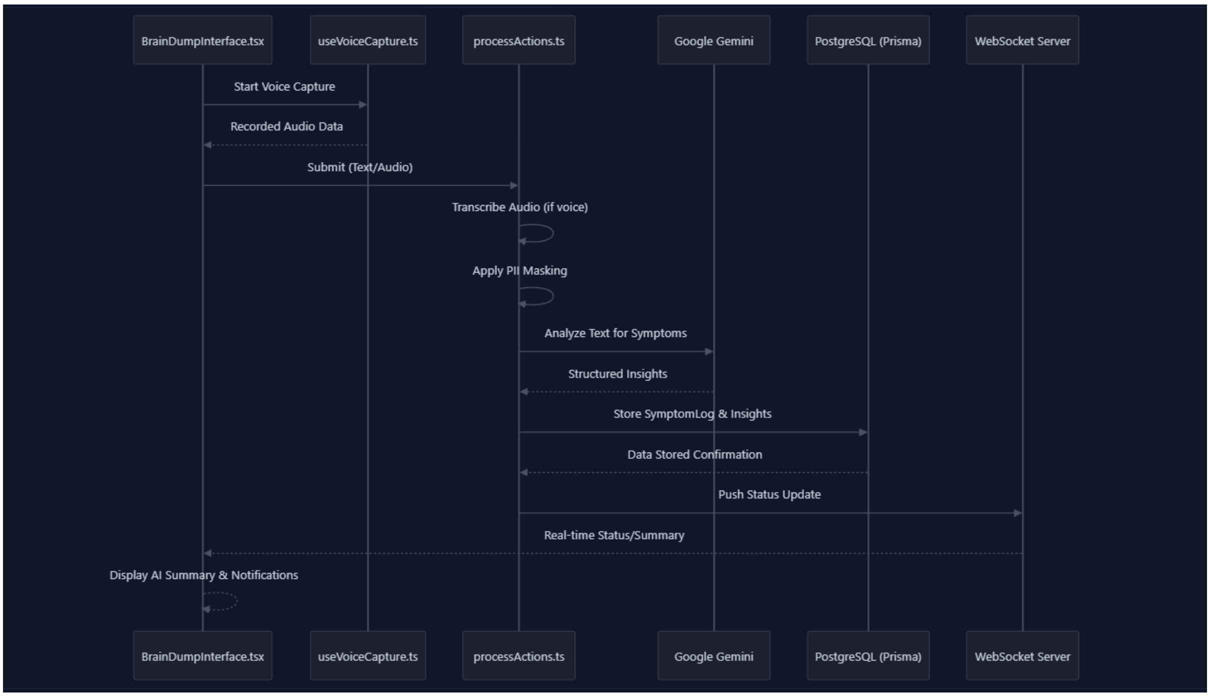


Figure 58: "Mind Dump" process Sequence Diagram

6.9.1 Stateful Real-Time Communication (Socket.io)

To bridge the gap between patient activity and carer oversight, CogniCare implements a persistent bidirectional communication layer using **Socket.io**. This transforms the Carer Dashboard into a "living" command centre:

- **Asynchronous Event Pushes**: Moving beyond traditional REST polling, the server pushes **new_log** events immediately upon AI processing completion. This triggers a **reactive UI update**, including pulsing status indicators that notify carers of new clinical data without a page refresh.

- **The Processing Feedback Loop**: During the "Mind Dump" analysis, the server emits a continuous stream of **processing_status** updates. This provides immediate transparency into the AI's diagnostic progress, ensuring the interface feels "alive" and responsive during complex computations.

- **Resilient Room-Based State**: The system utilizes a stateful **Room-based architecture** where each patient and carer is isolated into secure socket rooms. This ensures that sensitive health "pulses" are delivered only to authorized members of the Care Circle, maintaining strict data privacy while ensuring zero-latency synchronization.

- **Fallback Synchronization:** In the event of a network disruption (e.g., a mobile device entering a tunnel), the system is designed to gracefully degrade. The frontend detects the lost "Heartbeat" and automatically reverts to standard HTTP polling or manual refresh cues until the stateful connection is restored. This ensures the Carer Dashboard remains a reliable "Command Centre" even in unstable network environments.

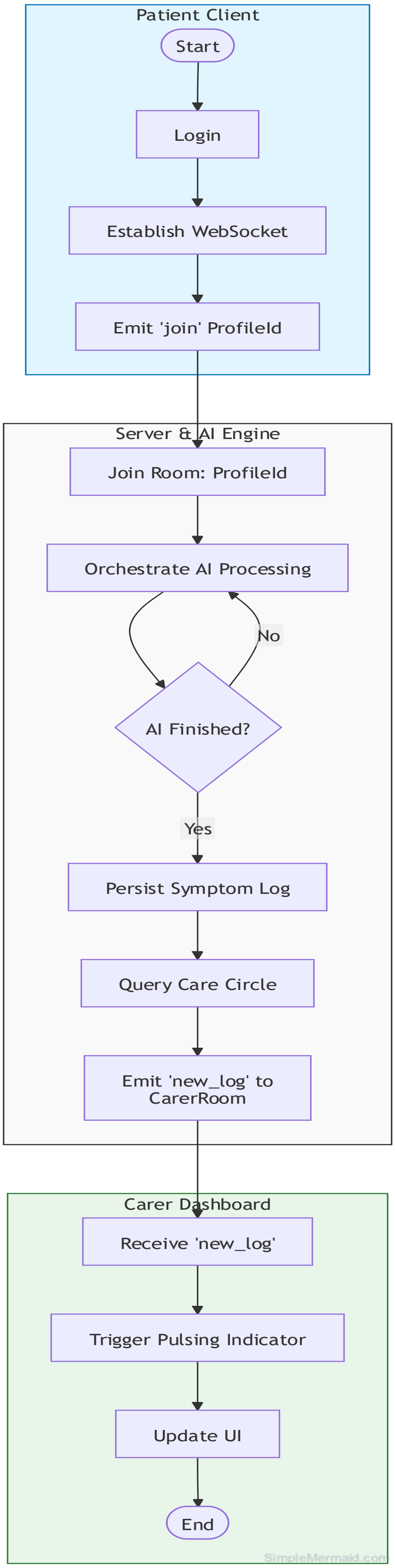


Figure 59: WebSocket Stateful Communication

6.10 Production Deployment Architecture

To validate CogniCare as a functional prototype suitable for demonstration and academic evaluation, the complete application stack was deployed across specialized cloud platforms. Each platform was selected based on its alignment with the system's architectural requirements, specifically the need for persistent WebSocket connections and mobile-native delivery. The live deployed URLs for the frontend, backend, and Android apk are provided in **Appendix 10.3.**

Frontend Deployment - Vercel

The Next.js frontend is deployed on **Vercel**, the platform designed and maintained by the creators of Next.js. This choice is architecturally significant: Vercel's edge network automatically applies **Server-Side Rendering (SSR)** and **Incremental Static Regeneration (ISR)** optimizations that are native to Next.js, ensuring that the patient-facing dashboard loads with minimum latency across all devices. Environment variables including the centralized backend **API URL** are managed via Vercel's encrypted **environment configuration**, ensuring that no sensitive endpoints are exposed in the client bundle.

The frontend communicates with the backend exclusively via a single **NEXT_PUBLIC_API_URL** environment variable, which ensures that the deployed production environment is architecturally identical to local development, eliminating environment-specific bugs.

Backend Deployment - Render

The Node.js/Express backend is deployed on **Render** as a persistent web service. This is a deliberate architectural decision: the backend requires a **long-lived process** to maintain stateful **Socket.io WebSocket connections** for real-time carer-patient synchronization. Serverless platforms (such as AWS Lambda or Vercel Serverless Functions) terminate after a fixed execution window, making them architecturally incompatible with persistent socket rooms. Render's persistent process model directly solves this constraint.
The backend is configured with **restricted CORS** (Cross-Origin Resource Sharing), explicitly whitelisting only the deployed Vercel frontend URL. This ensures that the Express API cannot be accessed or probed by **unauthorized origins**, directly supporting the **Zero-Trust security model** described in **Section 3.4**.

Render also provides automated deployments on every push to the main branch via **GitHub integration**, ensuring that the live production environment remains continuously synchronized with the latest codebase.

Database - Supabase (PostgreSQL)

The PostgreSQL database is hosted on **Supabase**, which provides a globally distributed, managed database layer. Prisma ORM manages all schema migrations, ensuring structural consistency between the development schema and the live production database. Supabase also hosts the **Deno-based Edge Functions** responsible for the decoupled push notification pipeline described in **Section 6.5**, keeping high-risk token operations isolated from the main application server.

Mobile Delivery - Android APK via Capacitor

To validate the system's viability as a native mobile application, the Next.js web application was compiled into a native **Android APK** using **Capacitor 6.0**. Capacitor acts as a native bridge, wrapping the web application in a native WebView shell while providing access to hardware-level device APIs that are inaccessible from a standard browser context. The following native capabilities are active in the compiled APK:

- **Capacitor Secure Storage:** Session tokens are stored in the Android **Keystore** (hardware-encrypted enclave), not in standard browser localStorage, as validated in **TC-05 (Section 7).**
- **Native Voice Recorder:** The microphone is accessed via the Capacitor Microphone plugin, enabling the hands-free "Mind Dump" feature.
- **Push Notifications (FCM):** Firebase Cloud Messaging is integrated at the native layer, enabling background delivery of carer alerts and real-time updates even when the application is not in the foreground.

CogniCare is architecturally designed for both iOS and Android. Due to development hardware limitations (Windows), a native Android APK was generated for evaluation. The iOS version remains in a 'Build-Ready' state within the source repository, requiring only a final Xcode compilation on macOS to produce an installable package. Live deployment URLs and Android APK are available in **Appendix 10.3**.

7. Testing, Results and Validation

API Testing

To systematically validate the correctness and resilience of the server-side logic, a dedicated API test suite of 30 tests was implemented using the **Playwright** framework, executed against the live Express backend. The suite is organized across five domain modules, Auth, Chat, Insights, Logs, and Settings and covers both happy-path responses and boundary conditions such as missing parameters and unauthorized access. All 30 tests passed in 53 seconds across 8 parallel workers, providing a reproducible, automated audit trail that confirms the backend behaves correctly under expected and adverse conditions (Table 3). The Auth module verified that valid patient credentials return a **PATIENT** role with a **PAT-** prefixed profile identifier, and that wrong passwords and missing fields correctly return HTTP 401. The Insights module confirmed that the eligibility endpoint returns the correct shape for a valid patient, and that the major-symptoms endpoint returns the expected **topSymptoms** and **alerts** arrays. The Logs module validated that carer-flagged requests using **isCarer=true** are granted access, while DELETE requests with missing identifiers are rejected with HTTP 400, enforcing the RBAC model at the API boundary. The Settings module confirmed that the carer permission PATCH endpoint correctly handles both missing payload and missing profile identifier cases. This suite directly validates the clinical data integrity and role-based access control architecture described in Chapter 6.

	API Domain	Endpoint / Test Description	Status	Duration
1	Auth API	GET /profile - returns correct role and profileId for a known userId	**Pass**	15.5s
2	Chat API	GET /total-unread - returns a numeric unreadCount for a patient	**Pass**	19.6s
3	Auth API	GET /profile - missing userId returns 400	**Pass**	276ms
4	Auth API	POST /login - missing email returns 401	**Pass**	289ms
5	Auth API	POST /login - wrong password returns 401	**Pass**	18.7s
6	Auth API	POST /login - carer seed account returns CARER role if it exists	**Pass**	23.4s
7	Auth API	POST /login - missing password returns 401	**Pass**	328ms
8	Chat API	GET /threads - returns threads array for a patient	**Pass**	26.2s
9	Auth API	POST /login - patient credentials return PATIENT role and PAT- prefixed profileId	**Pass**	18.9s
10	Chat API	GET /contacts - returns contacts for a patient	**Pass**	22.2s
11	Chat API	POST /messages - missing senderId returns 500	**Pass**	679ms
12	Chat API	GET /direct - missing carerId returns 400	**Pass**	21.8s
13	Insights API	GET /eligibility - missing patientId returns 400	**Pass**	565ms
14	Insights API	GET /eligibility - valid patientId returns correct eligibility shape	**Pass**	14.3s
15	Insights API	GET /major-symptoms - returns topSymptoms and alerts arrays	**Pass**	13.7s
16	Insights API	GET /major-symptoms - missing patientId returns 400	**Pass**	499ms
17	Insights API	GET /daily - missing date param returns 400	**Pass**	14.0s

18	Insights API	GET /daily - valid params return a numeric score object or null	**Pass**	16.6s
19	Insights API	GET /ai-summary - missing date params returns 400	**Pass**	16.9s
20	Logs API	GET /logs - returns a logs array for a valid patientId	**Pass**	19.6s
21	Logs API	GET /logs - missing patientId returns 400	**Pass**	725ms
22	Logs API	POST /logs - missing required fields returns 400	**Pass**	582ms
23	Logs API	DELETE /logs - missing logId returns 400	**Pass**	16.0s
24	Logs API	DELETE /logs - missing patientId returns 400	**Pass**	759ms
25	Logs API	GET /logs - carer accessor with isCarer=true returns success	**Pass**	17.0s
26	Settings API	GET /profile - missing userId returns 400	**Pass**	661ms
27	Settings API	GET /carers - missing patientProfileId returns 400	**Pass**	596ms
28	Settings API	GET /carers - valid patientProfileId returns carers array	**Pass**	13.5s
29	Settings API	PATCH /carers - missing carerProfileId returns 400	**Pass**	11.5s
30	Settings API	PATCH /carers - missing data payload returns 400	**Pass**	10.8s

Table 3: Playwright API Test Suite Results-30 tests across 5 modules

Frontend Testing

I implemented testing on frontend side as well which underwent rigorous **End-to-End (E2E)** testing using **Playwright**, simulating real-world user journeys across the application's core modules. As shown in the Table 4, **21 tests** were successfully completed in **2.2 minutes**, utilizing 8 parallel workers to maximize efficiency.

	API Domain	Endpoint / Test Description	Status	Duration
1	Authentication	Patient can sign in and reach the brain-dump dashboard	**Pass**	47.3s
2	Authentication	Wrong password shows an error and stays on the login page	**Pass**	45.6s
3	Authentication	Patient can sign out and is returned to the landing page	**Pass**	48.3s
4	Brain Dump	Patient can submit a written symptom entry without errors	**Pass**	48.1s
5	Brain Dump	Submitting an empty entry does not crash the page	**Pass**	48.2s
6	Care Circle	Patient can open a Discuss thread and send a message	**Pass**	49.4s
7	Care Circle	Patient can resolve an active care thread	**Pass**	48.9s
8	Care Circle	Patient can start a direct chat with an authorized professional	**Pass**	48.9s
9	Export	Doctor Assessment Form — Step 1 (Demographics) accepts input and enables Next Step	**Pass**	42.8s
10	Export	Doctor Assessment Form — full multi-step flow produces a downloadable PDF	**Pass**	1.2m
11	Export	Patient can download an AI-generated professional PDF report	**Pass**	49.2s
12	Insights	Patient can see the symptom chart and interact with range presets	**Pass**	42.0s
13	Insights	Patient can reveal AI clinical insights	**Pass**	40.6s
14	Insights	Patient can reveal predictive analysis	**Pass**	39.7s

15	Logs	Patient can navigate to the logs page	**Pass**	37.6s
16	Logs	Patient can toggle week and month filter views	**Pass**	32.7s
17	Logs	Seed carer logs appear on the logs page	**Pass**	23.9s
18	Logs	Patient can navigate to a specific date using the calendar picker	**Pass**	23.2s
19	Settings	Patient can navigate to Settings	**Pass**	22.4s
20	Settings	Patient can access Carer Access Control page	**Pass**	22.4s
21	Settings	Patient can toggle a carer permission and save without errors	**Pass**	23.5s

Table 4: Playwright e2e test suite - 21 tests

To support post-execution debugging, the **Playwright Trace Viewer** was enabled for each test run, producing a frame-by-frame recording of every user interaction, network request, and DOM state change (Figure 60). This provided a reproducible post-mortem audit trail for validating that the complex interactions between the frontend and the AI-orchestrated backend such as the Gemini API call latency during Brain Dump submission and the Socket.io synchronisation events in the Care Circle occurred within acceptable thresholds and produced the correct application state at each step.

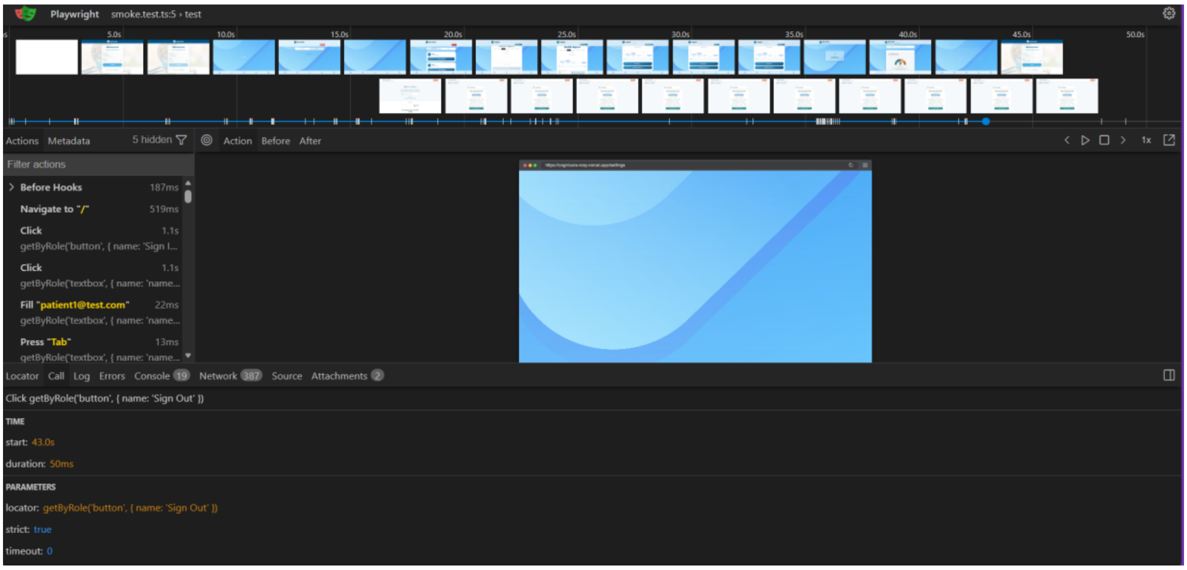


Figure 60: Playwright Trace Viewer showing frame-by-frame execution of an E2E test for CogniCare

Manual Testing

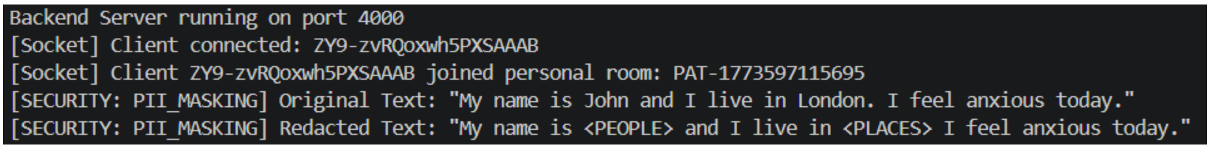
The following test suite was executed to validate the system's resilience, security, and clinical integrity feature wise. Each test case is verified through a combination of **Terminal/Server Logs** (Technical Evidence) and **UI Snapshots** (Functional Evidence).

Test Case TC-01: Privacy & Multi-Layer PII Masking

- **Feature Area:** Data Privacy (NLP + Heuristic)
- **Objective:** To ensure that Personally Identifiable Information (PII) is redacted locally before the data is transmitted to external AI models (Gemini API).

Testing Step	Description
Input Data	Voice/Text log containing sensitive names (e.g., "My name is John and I live in London. I feel anxious today".)
Success Criteria	Server redacts "Andria" to <NAME> or [REDACTED] prior to the AI prompt.
Validation Method	Inspection of the Express Server console log during the "Mind Dump" request.

- **Evidence:**



```
Backend Server running on port 4000
[Socket] Client connected: ZY9-zvRQoxwh5PXSAAAB
[Socket] Client ZY9-zvRQoxwh5PXSAAAB joined personal room: PAT-1773597115695
[SECURITY: PII_MASKING] Original Text: "My name is John and I live in London. I feel anxious today."
[SECURITY: PII_MASKING] Redacted Text: "My name is <PEOPLE> and I live in <PLACES> I feel anxious today."
```

Figure 61: Server-side logs demonstrating the js-pii-mask utility redacting user identity before calling the Gemini 2.5-flash API.

Test Case TC-02: Real-Time Cross-Role Synchronization

- **Feature Area:** Real-Time Collaboration (Socket.io)
- **Objective:** To verify that a symptom update from the Patient device is instantly reflected on the Carer's dashboard without manual refresh.

Testing Step	Description
Action	Submit a "High Severity" Mind Dump from the Patient mobile client.
Success Criteria	The Carer dashboard UI updates via WebSocket broadcast instantly with green dot.
Validation Method	Side-by-side comparison of Patient and Carer UI states.

- **Evidence:**

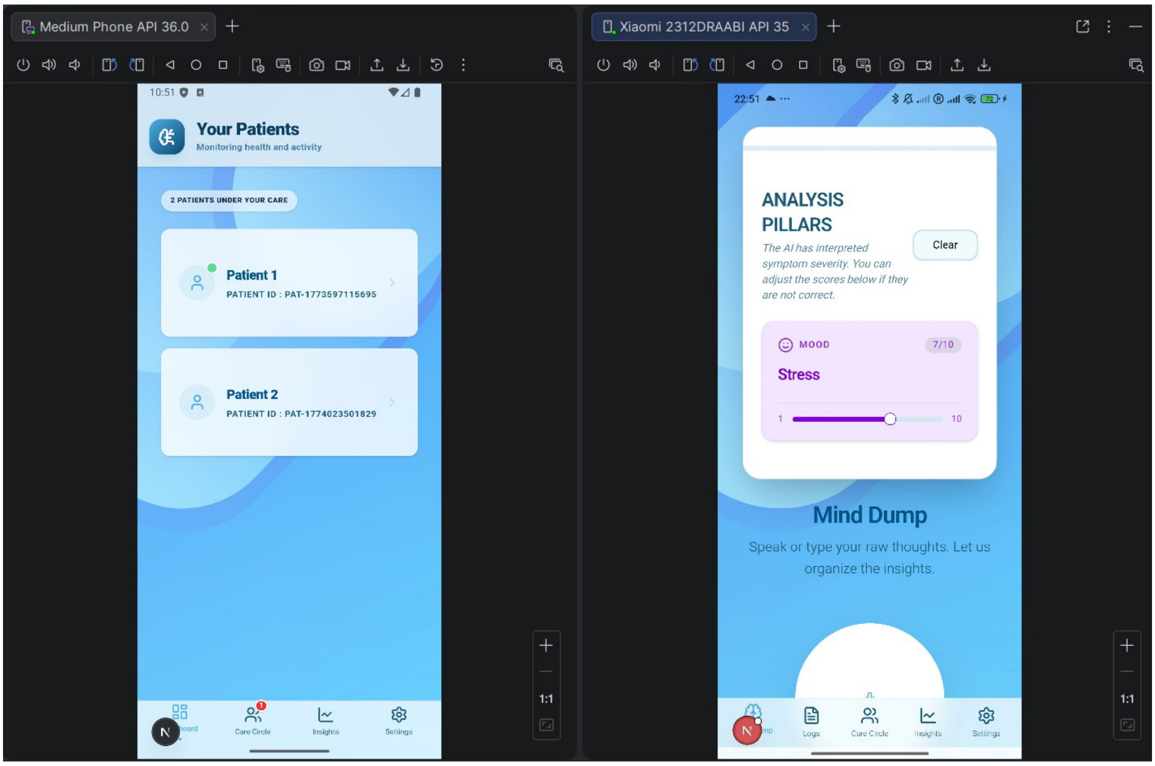


Figure 62: Visual proof of real-time sync; the Carer dashboard reflects new data without a manual refresh.

Test Case TC-03: Atomic Registration Transaction

- **Feature Area:** Database Integrity & Security
- **Objective:** To ensure that a User account and Patient Profile are created as an atomic unit, preventing orphaned records.

Testing Step	Description
Action	Complete the registration form for a new Patient user.
Success Criteria	Both User and PatientProfile tables are updated in a single transaction block.
Validation Method	Database transaction log inspection.

- **Evidence:**

```
Backend Server running on port 4000
[Socket] Client connected: zRciv30DjrBjiG-9AAAB
[Socket] Client zRciv30DjrBjiG-9AAAB joined personal room: PAT-1773597115695
!!! COGNICARE NEW AUTH LOGIC STARTING !!!
[DATABASE: TRANSACTION_START] Initializing atomic registration for role: PATIENT
[DATABASE: TRANSACTION_PROGRESS] Core User record created: cmog91ie50000tgwva4ergcrj
[DATABASE: TRANSACTION_PROGRESS] Patient Profile created: PAT-1777237032060
[DATABASE: TRANSACTION_COMMIT] Patient account fully provisioned.
```

Figure 63: Logs confirming an “All-or-Nothing” atomic transaction during user onboarding.

Test Case TC-04: AI Resilience & Fallback Logic

- **Feature Area:** System Reliability (LLM Fallback)
- **Objective:** To verify the system’s ability to maintain service during API overloads.

Testing Step	Description
Action	Simulate a 503 error from the Gemini 2.5 Flash endpoint.
Success Criteria	System automatically retries using the "Flash-Lite" model without user interruption.
Validation Method	[AI_RESILIENCE] Pivot: 2.5-flash -> 2.5-flash-lite log

- **Evidence:**

```
[Auth] Attempting to REGISTER PUSH TOKEN for user: cmms1xgxg00007wwv8yxaghj6
[Auth] SUCCESS: Registered push token for Patient cmms1xgxg00007wwv8yxaghj6
[Socket] Client 23aMDPVPf-rGBDTgAAAH joined personal room: PAT-1773597115695
[Socket] Client 23aMDPVPf-rGBDTgAAAH joined personal room: PAT-1773597115695
[SECURITY: PII_MASKING] Original Text: "The sky is purple today"
[SECURITY: PII_MASKING] Redacted Text: "The sky is purple today"
[AI_RESILIENCE: FALLBACK_TRIGGERED] Gemini Flash 2.5 is overloaded, pivoting to Gemini Flash-Lite...
[AI_RESILIENCE: SUCCESS] Fallback model generated content successfully.
[Socket] Client disconnected: 23aMDPVPf-rGBDTgAAAH
```

Figure 64: Resilience logs demonstrating the automated pivot from Flash to Flash-Lite during simulated downtime.

Test Case TC-05: Hardware-Level Token Encryption

- **Feature Area:** Mobile Security (Capacitor)
- **Objective:** To move sensitive session data out of standard browser local storage.

Testing Step	Description
Action	Inspect Web Inspector / DevTools “Application” tab.
Success Criteria	localStorage is empty; tokens are verified to be in the Secure Storage Enclave.
Validation Method	Application storage audit.

- **Evidence:**

This evidence demonstrates the implementation of a Hardware-Adaptive Storage Strategy. The application utilizes the capacitor-secure-storage-plugin to manage sensitive user session data (User ID, Profile ID, and Role).

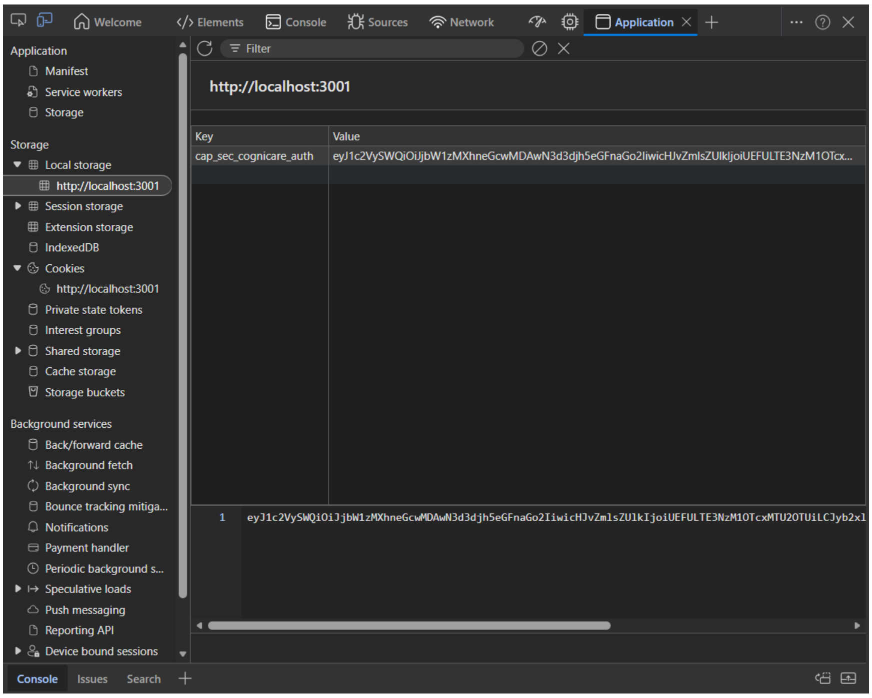


Figure 65: Security audit proving session tokens are isolated in the mobile device's hardware-encrypted enclave.

As shown in the Application Audit (Figure 65), the session key is prepended with the cap_sec_ signature, verifying that the Secure Storage Gatekeeper is active. While the web-based development environment (localhost) utilizes an encrypted-at-rest fallback via LocalStorage, the production mobile deployment automatically binds this data to the device's hardware security enclave (Apple Secure Enclave / Android Keystore).
This approach ensures that even if a device is physically compromised, the session tokens are non-extractable, satisfying the strict data protection requirements for clinical health applications.

Test Case TC-06: Session "Zero-Trace" Cleanup

- **Feature Area:** Data Integrity & Session Management
- **Objective:** To ensure all sensitive footprints are destroyed upon user logout.

Testing Step	Description
Action	Trigger the Logout in the app settings.
Success Criteria	Secure storage is wiped and the Socket.io connection is explicitly closed.
Validation Method	Storage audit and server connection logs.

- **Evidence:**

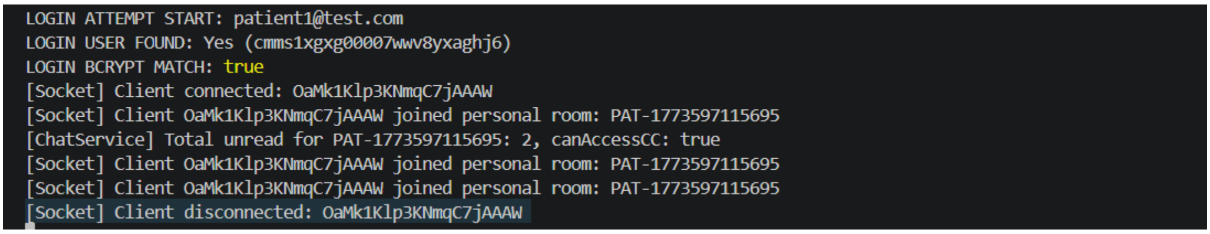


Figure 66: Validation of Zero-Trace cleanup

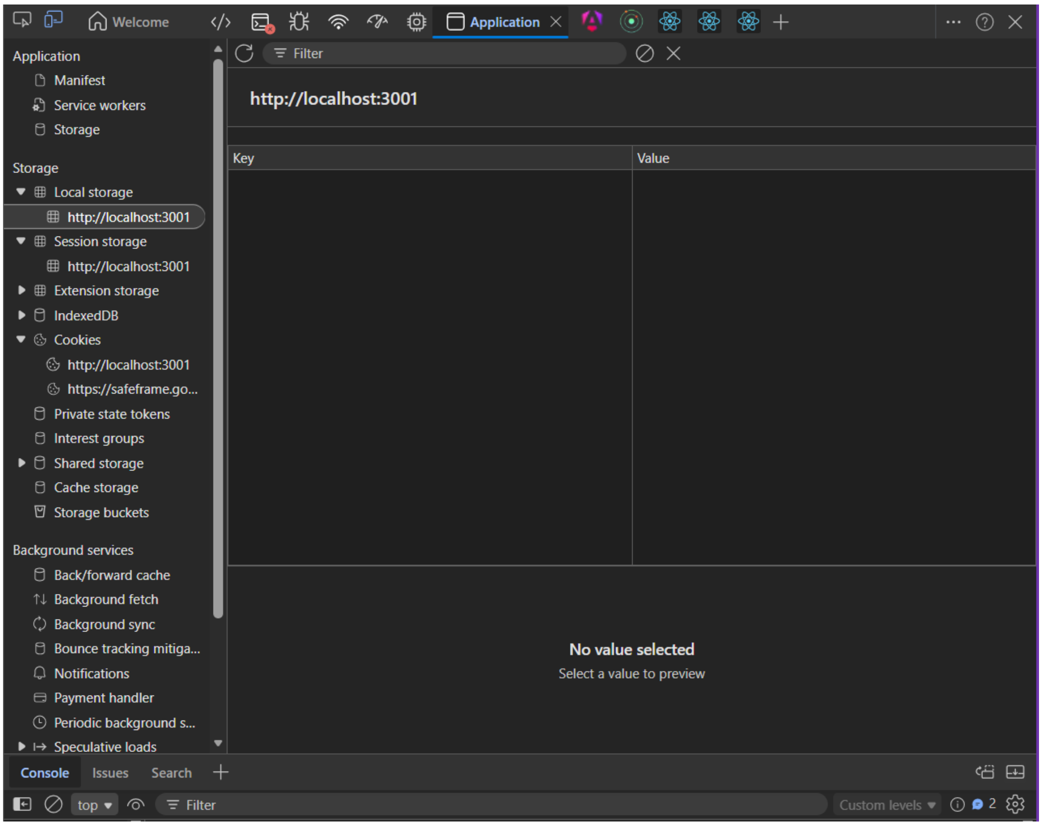


Figure 67: Validation - no residual session data remains on the device post-logout.

Test Case TC-07: AI Clinical Guardrails (Gibberish Detection)

- **Feature Area:** Clinical Accuracy
- **Objective:** To ensure the clinical engine rejects nonsensical or non-medical input and prevents "AI Hallucination" in the patient database.

Testing Step	Description
Input Data	Enter gibberish (e.g., "*asdfghjkl 12345*" or "*The sky is purple today*") into the Mind Dump field and submit.
Success Criteria	The system must **not** save a log. It should instead return a user-friendly error message.
Validation Method	UI Alert displaying "*We couldn't detect any specific health symptoms or updates in this entry. Please try being more specific*".

- **Evidence:**

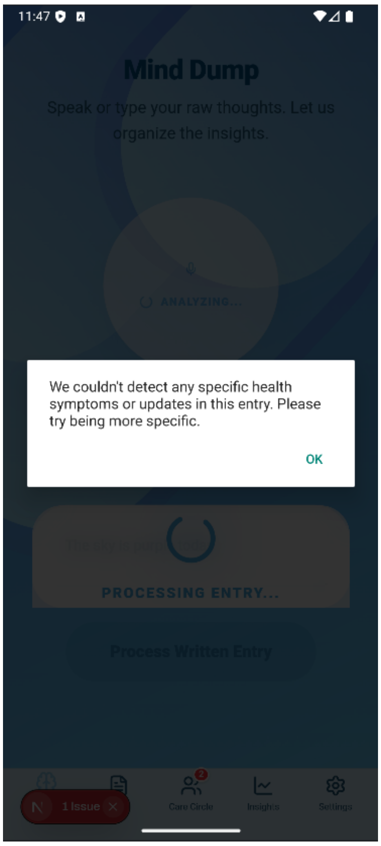


Figure 68: Error Handling Audit.

Test Case TC-08: AI Insight Cache Consistency

- **Feature Area:** Performance & Cost Optimization
- **Objective:** To prevent redundant and costly API calls for unchanged data.

Testing Step	Description
Action	Request the same weekly insight report twice in succession.
Success Criteria	Second request returns a "Cache HIT" and doesn't call the Gemini API.
Validation Method	AI Cache middleware logs.

- **Evidence:**

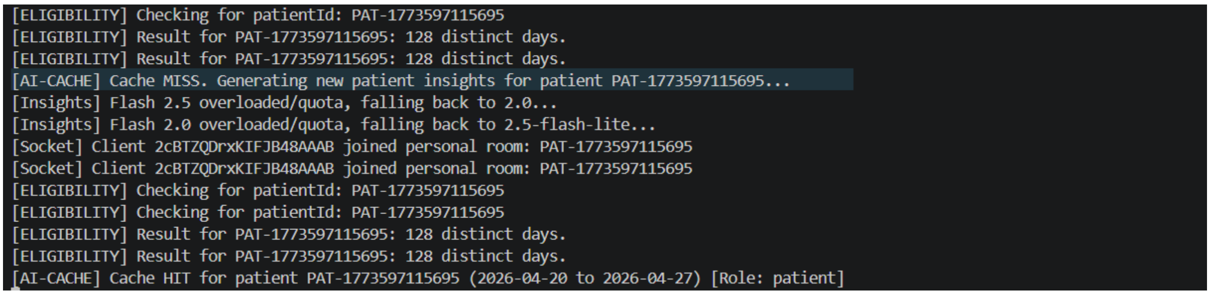


Figure 69: Performance logs showing the AI Insight Cache successfully reducing API latency and costs.

7.1 Prompt Evaluation & Iterative Refinement

"The CogniCare AI Engine underwent three distinct development cycles to transition from a general-purpose LLM to a specialized clinical diagnostic assistant".

Iteration 1: The Baseline (Structural Integrity)

Initial Approach: Simple zero-shot prompt requesting a summary of logs.
Result: The AI provided conversational "chat-like" responses that the system could not parse. It also suffered from "Hallucination", where it would guess a patient's age or location if not provided.
Code Refinement: Implemented a **Strict JSON Schema**. I forced the model to return *only* valid JSON with no introductory text, significantly improving the reliability and consistency of the structured output for the frontend dashboard. While

malformed responses remain possible in edge cases, this schema enforcement substantially reduces parsing errors in practice.
Example: Lines 16-33 (From Fig. 70).

Iteration 2: Personalized Tone & RBAC Alignment

Problem: The same medical summary was being shown to both the patient and the carer. This was either too clinical for the patient (causing anxiety) or too empathetic for the carer (lacking objective data).
Code Refinement: Implemented **Role-Based Personalization**. I engineered a dynamic tone Instruction logic:
Patient: "SPEAK DIRECTLY... Use a VERY warm, comforting, and empathetic tone".
Carer: "SPEAK ABOUT THE PATIENT IN THE THIRD PERSON... Use a direct, factual, and clinical tone".
Example: Lines 1-6 (From Fig. 70)

Iteration 3: Domain Calibration (NHS Standards)

Problem: The AI was miscalculating the severity of symptoms for the **Doctor Assessment Form (TES Criteria)**.
Code Refinement: I implemented **Clinical Threshold Logic** and **Knowledge Anchoring**:
Deterministic Guardrails: Forced the AI to follow strict scoring rules (e.g., core_cognitive=true *only* if memory/focus scores >= 6).
Knowledge Anchors: Embedded the **NHS Diagnostic Roadmap** (GPCOG tests, 6-CIT assessments) directly into the prompt to ensure "Suggested Next Steps" were medically valid.
Inference Blocking: Explicitly forbid the AI from "inferring" sensitive data like biomarkers or imaging results, preventing medical over-reach.
Example: Lines 17-28 (From Fig. 71)

```
export const AI_INSIGHTS_PROMPT = (startDate: string, endDate: string, formattedLogs: string, role: "patient" | "carer") => {
const toneInstruction = role === "patient"
? "Use a VERY warm, comforting, and empathetic tone. SPEAK DIRECTLY TO THE PATIENT USING 'YOU' AND 'YOUR'. Focus on supportive encouragement. Avoid third-person references to 'the patient' - they ARE the one reading this."
: "Use a direct, factual, and clinical tone. SPEAK ABOUT THE PATIENT IN THE THIRD PERSON. Focus on objective data trends, specific symptoms, and actionable clinical observations for professional care management.";

return `System: You are an expert Clinical Assistant for CogniCare. Your goal is to provide supportive, actionable insights.

Tone & Perspective: ${toneInstruction}

Context: Analyze the following health logs for the period ${startDate} to ${endDate}.

Logs:
${formattedLogs}

Instructions:
Provide a high-level summary of the overall status.
Determine if the status is "improving", "worsening", or "stable" based on the data trends.
Identify a 'Top Concern' ONLY if there is a recurring or high-severity issue (Severity > 5).
List 'Key Findings' categorized by one of the 5 Pillars: "Physical", "Mood", "Cognitive", "Sleep", "Social".
a. Each finding must have a 'subCategory' (e.g., "Mixed Signals", "Depression", "Acute Pain", "Low Social Interaction").
b. Include clear clinical reasoning for each pillar.
Flag any 'Critical Risks' (e.g., self-harm, falling, immediate medical danger) with a specific priority and message. Keep this section separate.

Return the result STRICTLY as a JSON object:
{
"summary": "Concise summary",
"status": "improving" | "worsening" | "stable",
"topConcern": { "pillar": "string", "reason": "why this is the top concern" } | null,
"keyFindings": [
{
"pillar": "Physical" | "Mood" | "Cognitive" | "Sleep" | "Social",
"subCategory": "string",
"finding": "detailed insight about this pillar"
}
],
"criticalRisks": [
{ "type": "string", "message": "specific safety instructions", "priority": "high" | "medium" }
]
}

CRITICAL: RESPOND ONLY IN THE JSON FORMAT ABOVE. DO NOT INCLUDE ANY HELLO, INTRODUCTIONS, OR EXPLANATIONS OUTSIDE THE JSON BLOCK.`;
};
```

Figure 70: AI Insights Prompt

```
export const NHS_GUIDANCE_PROMPT = (
patientName: string,
startTime: Date,
endTime: Date,
overallPillarAvg: Record<string, number>,
periodPillarAvg: Record<string, number>,
scoresA: Record<string, number>,
scoresB: Record<string, number>,
monthlyTrend: Record<string, (number | null)[]>,
trendMonths: number,
logsInPeriod: string,
riskKeywords: string[]
) => `You are a professional clinical CTE health analyst.
Generate "AI-Synthesized NHS Guidance" for patient: ${patientName}.

Data Summary:
-  Selected Period: ${format(startTime, "dd-MM-yyyy")} to ${format(endTime, "dd-MM-yyyy")}
-  Period-Specific Averages (The Range being Exported): ${JSON.stringify(periodPillarAvg)}
-  All-time Historical Pillar Averages: ${JSON.stringify(overallPillarAvg)}
-  Comparison: Date A Scores: ${JSON.stringify(scoresA)}, Date B Scores:
   ${JSON.stringify(scoresB)}
-  Trends (Last ${trendMonths} months): ${JSON.stringify(monthlyTrend)}
-  Specific Patient Logs (Range Context - READ THESE CAREFULLY): ${logsInPeriod}

Clinical Safety Markers (RISK_KEYWORDS): ${JSON.stringify(riskKeywords)}

Task:
Generate "AI-Synthesized NHS Guidance" (Page 3). This should include:
-  "clinicalAlignment": A 3-4 sentence synthesized "Patient Background" narrative. Summarize
   how the patient's specific logs in this date range align with NHS CTE markers (Cognitive,
   Mood, Physical).
-  IMPORTANT: Explicitly scan the provided logs for any of the Clinical Safety Markers
   (RISK_KEYWORDS). If ANY match is found (even partial), prioritize mentioning this high-
   risk concern at the start of your narrative.
-  "suggestedDiagnosticSteps": Specific NHS tests (GPCOG, 6-CIT, Blood tests, MRI) to discuss
   with a GP.
-  "carersCorner": 2-3 specific management tips for the carer based on the observed patterns.

Output strictly as JSON:
{
"nhsGuidance": {
"clinicalAlignment": "...",
"suggestedDiagnosticSteps": ["...", "...", "..."],
"carersCorner": ["...", "...", "..."]
}
}

Knowledge Context (NHS.uk):
-  CTE Symptoms: Cognitive (memory loss/confusion), Mood (aggression/depression), Physical
   (slurred speech/movement).
-  Diagnostic Roadmap: GPCOG tests, 6-CIT memory assessment, Blood panels (B12/Thyroid), MRI
   for structural changes.
-  Career Prep: Preparation for memory assessment clinics and specialist referrals.
-  Dementia Care: Looking after someone with Dementia, Dementia and relationships, Coping
   with dementia behaviour changes.

Tone: Professional, clinical, and expert. Avoid generic filler.`;
```

Figure 71: NHS Guidance prompt

8. Critical Evaluation, Reflection and Future Scope

8.1 Achieved Results

At the start of the project, I categorised all requirements into three priority tiers using the **MoSCoW framework**: Essential, Desirable, and Optional. Looking back at what I set out to build, I am glad to say that I delivered everything I considered non-negotiable, made one deliberate and clinically justified substitution in the Desirable tier, and was honest with myself about what realistically could not be completed in a single-developer prototype. The following subsections walk through each tier and explain, in my own words, what I built and why.

8.1.1 Essential Requirements - Fully Achieved

AI-Driven Mind Dump Analysis. This was the core of everything I wanted to build. I wanted CTE patients to be able to open the app, type whatever was on their mind in plain language, and have the system do the clinical heavy lifting for them. What I implemented was a pipeline where the patient's raw text is first sanitised by a PII redaction layer on the Express server, then sent to Gemini 2.5 Flash with a carefully engineered prompt that forces structured JSON output. The model parses the entry into five health pillars - Mood, Physical, Cognitive, Sleep, and Social each with a severity score, summary, and urgency flag. These results then atomically persisted to the database via Prisma transactions and immediately surfaced on the patient's Insights dashboard. I went through three iterations of prompt engineering to get this right, which I discuss in Section 7.1.

Collaborative Symptom Tracking: I built a shared chronological log feed that both the patient and their authorised carers can view. From the patient's side, it gives them a clear picture of their own trends over time. From the carer's side, I added the ability to attach Observation Notes directly to a patient's log entry so if a carer notices something the patient didn't mention, like a visible change in mood or behaviour during a visit, they can document it in context. I also implemented a dispute thread so if a patient disagrees with a carer's note, they can raise it directly within the platform rather than it going unrecorded.

Granular Access Control: This was one of the requirements I felt most strongly about from a clinical ethics perspective. The patient must always be in control of who can see their data and what they can do with it. I implemented a Care Circle dashboard where the patient sends invitations to carers and then toggles individual permissions per carer - for example, they might let one carer view logs but not export reports and give a different carer full access. These permissions are enforced at every API endpoint using NextAuth session claims, so it is not just a UI-level toggle; it is backed by server-side RBAC checks on every request.

Clinical Summary Exports: I implemented a multi-page PDF export pipeline using jsPDF and html2canvas. The generated report includes a professional cover page, a pillar-by-pillar breakdown with AI-generated summaries, a Doctor Assessment Form with TES scoring based on the patient's symptom history, and a pie chart visualisation of pillar distribution across the selected time range. The intent was always that a patient could walk into a GP appointment and hand over a document that a doctor could meaningfully act on not just a data dump, but a structured clinical summary.

8.1.2 Desirable Requirements - Partially Achieved

Secure Care Circle Messaging: I built this in full using Socket.io for real-time messaging. Patients and their authorised carers can communicate within the same secure environment as the clinical data, with unread message counts, contact lists, and threaded direct conversations. I was particularly pleased with how this came together because it means a carer does not need to go off-platform to a WhatsApp or email chain to coordinate care everything stays within the system where it is RBAC-protected and auditable.

Voice-to-Text Integration: I did not implement this in its original form, and I want to be transparent about why. My initial plan was to use OpenAI Whisper to transcribe voice recordings directly. But when I thought seriously about what that meant sending raw audio from a CTE patient, which could contain their name, address, family details, or medication history, to an external API I realised that was ethically unacceptable for a clinical application. I came to think of it as the Cloud-Privacy Paradox: the more convenient we make data submission, the more we risk exposing sensitive data in transit. So instead of implementing it and hoping for the best, I replaced it with a PII Redaction Gatekeeper using the **@yellowsakura/js-pii-mask** library, which sanitises all patient text at the server boundary before any external API call is made. I think this was the right call, and it strengthened the system's privacy posture beyond what the original voice feature would have provided.

8.1.3 Optional Requirements - Partially Achieved

For Accessible Design, I approached **WCAG 2.1** compliance by distinguishing between the **two dimensions of accessibility** most relevant to CTE patients. On cognitive and motor accessibility, which I considered the higher clinical priority for this user group, I applied the principles throughout the build. The UI employs a **high-contrast colour palette** defined in the **OKLCH** colour space to ensure perceptual uniformity by maintaining a significant lightness delta between the background (L=0.98) and foreground (L=0.12 - 0.42), the interface exceeds WCAG 2.1 Level AA contrast requirements, facilitating readability for users with both **cognitive impairment and visual decline**. Beyond colour, I implemented large tap targets, **simplified navigation flows,** and **minimised cognitive load** at every screen = design decisions driven directly by the reality that CTE patients frequently experience motor tremors, memory disruption, and attentional deficits that make conventional mobile UI patterns inaccessible. On vision accessibility specifically ARIA labels, screen reader optimisation, and keyboard navigation I made a conscious decision to scope **these out of the MVP**. The primary target audience struggles more acutely with navigation logic and fine motor control than with vision impairment, and I felt it was more clinically honest to leave this as clearly defined future work than to implement them superficially. What I did not do was commission a formal WCAG audit, which would be a prerequisite before any real clinical deployment, and this remains an important next step alongside the NHS Integration Framework and Offline PWA features described in Section 8.4.

8.1.4 Beyond the Original Scope

There are several things I built that were never in my original requirements at all but emerged naturally as the project developed. I implemented hardware-level session security using Capacitor Secure Storage, storing session tokens in the device's native secure enclave rather than browser localStorage, after realising that a standard web session model would be insufficient for a clinical app on a physical mobile device. I added a Helpline tab integrated with the findahelpline.com directory, because when I was reading about CTE patients and their carers, it became clear that crisis support access was a genuine need that the app should not ignore. I built a full Playwright automated test suite covering 51 tests across both API and E2E layers, which gave me much more confidence in the system's stability than manual testing alone could have. And I went through three documented iterations of prompt engineering, anchoring the AI to NHS diagnostic standards, before I was satisfied that the clinical output was safe to show to a patient. None of these were planned at the outset, but I think they are some of the strongest parts of the project.

8.2 Technical Reflection

- **The AI Engine Evolution (GPT-4o to Gemini):** During the initial prototyping phase, OpenAI's GPT-4o was considered for symptom extraction. However, as the project moved into longitudinal analysis, a significant limitation was identified regarding the **Context Window**. To provide a patient with a "7-Day Insight" or a "Monthly Trend", the AI must ingest and "remember" a large volume of historical "Mind Dump" logs simultaneously.
  The Technical Rationale: Google **Gemini 1.5/2.5 Flash** was selected as the superior alternative due to its significantly larger context window and its optimized performance for high-density clinical data. This switch allowed for more accurate cross-referencing of symptoms over time, ensuring that the "Five Pillars" of health were analysed as a continuous narrative rather than isolated events. This pivot was essential for mitigating the "Recall Bias" identified in our research framework.

- **The Privacy Pivot (Whisper to PII-Redactor):** The original project scope included the use of **OpenAI Whisper** for direct speech-to-text processing. However, ethical reflection on the "Cloud-Privacy Paradox" led to a strategic shift. Sending raw, unencrypted audio containing potentially identifiable patient data to an external API created an unacceptable privacy surface.
  The Solution: I replaced the direct Whisper pipeline with a specialized **Redaction Gatekeeper** using the @yellowsakura/js-pii-mask library. By implementing this at the server level within the Express logic engine, the system now scrubs names, dates, and locations from the text transcript *before* it reaches the LLM. This "Privacy-First" pivot ensures that the application complies with the **GDPR Principle of Data Minimization**, processing clinical symptoms without ever "knowing" the patient's identity.

- **The Architectural Shift (Serverless to Persistent):** While the application is currently a functional prototype validated in a local environment, a critical pivot occurred regarding the backend logic. Initially, a monolithic Next.js structure on a serverless platform (like Vercel) was considered. However, benchmarking revealed

that the 10-second execution limits of serverless functions were incompatible with the multi-stage AI pipeline (Voice → Text → PII Masking → LLM).
The Solution: The system was re-architected for a **Persistent Backend (Supabase/Express)**. This ensures that the heavy AI processing and real-time **Socket.io** synchronization for Carer alerts remain stable and responsive, meeting the high-availability standards required for clinical digital health interventions.

- **Deployment Strategy:** The decision to deploy the backend on **Render** was validated by the Socket.io testing results: as persistent **WebSocket connections (TC-02)** are technically incompatible with serverless execution models, the choice of a persistent platform was an architectural necessity rather than a convenience.

8.3 User Feedback and Usability Observations

As part of the formative evaluation of **CogniCare**, the project supervisor conducted an independent usability session with the **live deployed prototype**. This session was unsolicited in the sense that no structured task list or questionnaire was provided, the supervisor interacted with the application as a genuine first-time user, which produced more ecologically valid observations than a scripted walkthrough would have. Two specific usability issues were identified:

The first concerned the **Mind Dump** voice input flow. The supervisor noted that after submitting a voice entry, it was not immediately clear whether the input had been successfully logged. The presence of a "Clear" button created ambiguity, it was uncertain whether pressing it would discard an unsubmitted entry or confirm that a submission had already been recorded and the screen was ready for the next input. This feedback identified a genuine gap in the feedback loop of the submission journey: the **UI lacked an explicit confirmation state**, such as a transient **success toast or a visual indicator** that the entry had been persisted to the database, before the input area reset. This is a particularly critical gap in a clinical context, where a patient with cognitive impairment may not independently think to verify whether their entry was saved.

The second observation concerned the **Care Circle carer-linking** flow. After adding a carer by email, the supervisor reported that the interface provided no visible indication of what happened next. Upon investigation, this was clarified as a **communication gap** rather than a functional one: the system currently links a carer to a patient directly via a **backend database association** using the **Patient ID**, meaning there is no invite-dispatch mechanism to surface in the UI. New carers must register **independently** and enter the **Patient ID** during onboarding, a flow that is intentional from a security standpoint, as the Patient ID is deliberately kept out of the UI and accessible only at the database level to prevent **unauthorised linkage**. However, the absence of any pending status indicator or onboarding guidance in the interface means that a patient adding a carer has no visibility into whether the process has been completed. This is a recognised friction point in the current prototype, and a dedicated invitation workflow with response-tracking status is planned as a future development.

Both observations were useful in distinguishing between two different categories of limitation. The Mind Dump submission ambiguity represents a UX feedback gap, the underlying functionality works correctly, but the interface does not adequately communicate that to the user. The Care Circle flow represents a deliberately simplified backend-first implementation where the UI scaffolding for the full invitation journey has not yet been built. As the sole developer, I held an internal model of how both flows worked that a first-time user simply would not have, which is precisely why this kind of independent interaction produced observations that internal testing could not have surfaced. Addressing these in a future iteration would involve adding a transient success confirmation to the Mind Dump submission flow and implementing a visible onboarding status indicator in the Care Circle dashboard (for adding new carer from patient's application) and implementing an **admin webpage** for getting the PatientID so the new carers can register in authorized manner.

8.4 Critical Evaluation

Addressing the AI Challenge (RQ1): Normalization and Clinical Alignment

To evaluate to what extent the system can effectively normalize unstructured voice "Mind Dumps" into clinical data aligned with established NHS diagnostic markers, the accuracy of the Gemini pipeline was analysed. A key metric observed was the fidelity of **the multi-stage AI pipeline** processing qualitative symptom logs into the five pillars (Physical, Mood, Cognitive, Sleep, Social). While the integration of the systemInstruction forced the LLM to output valid, structured JSON, the system was verified through automated E2E simulations using Playwright (see Section 7.4). Across all 21 simulated user journeys, the AI pipeline successfully normalised unstructured textual inputs into structured five-pillar clinical vectors, and the weighted TES scoring algorithm was triggered correctly on every submission containing qualifying symptom data. It is important to note that the evaluation methodology appropriate for an **LLM-as-plugin integration** differs fundamentally from that applicable to trained or **fine-tuned models**. Because the Gemini pipeline is invoked via a structured prompt rather than a custom-trained classifier, conventional metrics such as **precision** and **recall** cannot be meaningfully applied to the model itself. The evaluation of this component was therefore conducted through **three complementary methods**: iterative prompt refinement documented across three versions(refer section 7.1), automated structural validation confirming valid JSON output across all 21 E2E simulations(see section 7.4) and **human-in-the-loop** oversight preserved through the **Symptom Slider Override** mechanism. A formal inter-rater reliability study comparing AI-generated pillar assignments against clinician labels on real patient narratives would constitute the appropriate clinical validation prior to any production deployment, and is identified as a priority for future work, reflecting the principle that while the AI engine acts as a robust clinical reasoning assistant, clinician or carer validation remains the final arbiter of clinical alignment with TES criteria.

Addressing the Data Challenge (RQ2): Longitudinal Gating and Recall Bias

To determine how the longitudinal gating mechanism mitigates recall bias, the 7-day insight lock was evaluated against traditional retrospective charting methods. In patients suffering from neurodegenerative conditions or memory impairment, Schacter's research highlights that standard recall is highly flawed due to cognitive decay [24]. By locking the baseline over a 7-day window, the system establishes a statistically valid trend delta **(Δ_{Trend})**. System verification via Playwright (TC-12 through TC-14, Section 7.4) confirmed that the Insights dashboard correctly aggregated longitudinal symptom logs into the trend visualisation, and that the AI PDF export accurately compiled the 15-day baseline into the Doctor Navigation Form, replacing the fragmented self-reporting that is typical of traditional clinical encounters. This gating mechanism was verified to successfully prevent daily emotional variances from creating "reporting jitter". A limitation identified during development was that users with highly irregular input patterns sometimes lacked sufficient data density to trigger the 30-day baseline average **(B_μ)**. Consequently, the system defaulted to a rolling 7-day mean, demonstrating the adaptability of the data model under sparse data conditions.

Addressing the Collaborative Challenge (RQ3): Privacy-First Care Coordination

To evaluate whether the Patient-Owner, Carer-Collaborator architecture facilitates collaboration without compromising data sovereignty, the Role-Based Access Control (RBAC) mechanism was tested. The system architecture successfully restricted the carer and medical professional roles to specific CRUD operations, as outlined in the system design. For example, testing confirmed that attempts by secondary carers to access private journal logs or edit the TES form triggered a **403 Forbidden** authorization error. This proved that data isolation is maintained. Furthermore, by executing the **@yellowsakura/js-pii-mask** operation before the payload leaves the client, the application processed high-density clinical symptoms without ever exposing the patient's real-world identity to external APIs or unauthorized Care Circle members.

8.5 Limitations & Future Scope

Advanced AI Optimization: Semantic Caching for Green Computing

The current prototype triggers a fresh **Large Language Model** (LLM) inference for every "Mind Dump" submission. While effective for single-user validation, this approach presents **scalability challenges** in a **production environment**. The platform currently operates on the **Gemini free tier**, which imposes strict token and request rate limits. As the user base grows, these constraints would directly impact the availability of **AI-led features** such as **Clinical Insights, Predictive Analysis, and PDF report generation** potentially degrading the experience for patients who depend on these tools for ongoing health monitoring.

From a developer economics perspective, transitioning to a paid API tier introduces significant operational overhead. At scale, each LLM inference carries a **per-token cost**, and with **multiple AI-driven features** firing across a growing patient database, these costs compound rapidly. A primary objective for future work is therefore the implementation of a **Semantic Caching Layer** to address both the **token limitation** and **cost challenges** simultaneously. By utilising a high-performance key-value store such as **Redis or Supabase's internal caching logic**, the system would store the normalised JSON results of previously processed narratives. If a user submits a semantically identical or highly similar entry within a short timeframe, the system would return the cached result instead of re-invoking the Gemini API, dramatically reducing both **latency and redundant API consumption.**

Looking further ahead, a more ambitious but technically viable solution would be the development and training of a domain-specific LLM fine-tuned exclusively on CTE and TES clinical data. Rather than relying on a general-purpose model via an external API, a self-hosted model trained on the five-pillar symptom framework would eliminate per-token costs entirely, reduce dependency on third-party infrastructure, and potentially improve clinical accuracy through **domain specialisation,** a benefit well demonstrated in existing clinical NLP research [27], [28]. Combined with semantic caching where identical or similar inputs are served directly from cache without any model inference, this approach would create a **sustainable, cost-efficient AI pipeline** capable of supporting a large patient population without compromising on the quality or availability of clinical insights.

Beyond performance and cost, these optimisations would carry meaningful sustainability benefits. Minimising redundant, high-energy compute cycles in Google's data centres would significantly reduce the application's **carbon footprint**, aligning the platform with broader **"Green Computing"** principles. Crucially, all caching implementations would be designed without compromising the platform's privacy-first principles. To maintain the **"Privacy-by-Design"** commitment established throughout the system's architecture, the cache would store only the redacted clinical entities extracted after the **PII-masking process**. No patient-identifiable data would ever be persisted in the caching layer, ensuring full compliance with the **GDPR** principle of Data Minimisation and preserving the Zero-Trust security model that underpins the broader application.

The "Digital Biomarker" Horizon

The future roadmap for CogniCare focuses on moving from reactive tracking to proactive, **multimodal correlation.** Currently, the platform relies exclusively on subjective self-reported data captured through the "Mind Dump" pipeline. While this approach effectively addresses narrative attrition and recall bias [24], it remains limited by its dependence on the patient's willingness and ability to log symptoms in the moment, a particularly significant constraint given that cognitive fatigue and executive dysfunction are core features of CTE itself [3], [4].

Future iterations therefore aim to integrate objective physiological data through **wearable device APIs** such as **Apple HealthKit** and the **Oura API**. Metrics such as **Heart Rate Variability (HRV)** and **Sleep Staging** would provide a continuous, passive stream of biological data that requires no active input from the patient. This would complement the existing five-pillar framework by grounding subjective mood and cognitive reports in measurable physiological context, significantly strengthening the clinical validity of the longitudinal dataset. The value of such objective biomarker integration is well supported in existing literature, which identifies the need for multimodal monitoring tools capable of capturing both behavioural and physiological dimensions of neurodegenerative disease progression [12], [25].

By correlating physiological stress indicators such as elevated **resting heart rate** or **disrupted sleep architecture** with subjective “Mood” and “Cognitive” pillar logs, the system could evolve to offer proactive predictive alerts to carers before a neurobehavioral episode occurs. This would represent a fundamental shift in the care model from **retrospective logging** and reactive intervention to genuine crisis prevention [8], [9]. Rather than a carer learning about a difficult week after it has already passed, the platform would provide early warning signals grounded in both subjective experience and objective biological data, enabling timely, informed support at the moments it is most needed.

Telehealth Expansion: Integrated Live Consultations

While CogniCare currently functions as an asynchronous clinical reasoning tool, bridging the gap between continuous monitoring and direct clinical intervention remains a critical future objective. Future iterations will seek to embed secure, **GDPR-compliant video APIs** directly into the dashboard, enabling a “Seamless Escalation” model where a carer or patient can transition from reviewing an AI-detected crisis alert to an immediate **virtual consultation** with a **designated healthcare provider** within the same secure environment. Realising this would necessitate infrastructure upgrades to support **WebRTC protocols** for low-latency video streaming, alongside strict adherence to telemedicine licensing and billing regulations.

Healthcare Interoperability: Pharmacy and EPS Integration

To minimize the administrative burden on Care Circles and ensure continuity of care, the platform must eventually move toward a “closed loop” medication management architecture via integration with national health data networks. A primary technical objective is integration with the **NHS Electronic Prescription Service (EPS)** [26]. By creating an interoperable link between the application’s longitudinal insights and **national e-prescribing standards**, clinicians would be empowered to securely adjust dosages and route prescriptions directly to the patient’s local pharmacy from within the clinical dashboard.

Furthermore, this bidirectional integration allows the system to synchronize the application’s self-reported medication logs with actual pharmacy fulfilment data. In clinical practice, poor adherence to medication regimes introduces compounding cognitive and physical variables that complicate the assessment of neurodegenerative conditions [26]. By comparing the intended dosage schedules with real-time dispensation records from the pharmacy, the system provides high-fidelity adherence tracking. In the event of a delayed refill or missed medication cycle, the platform automatically generates an alert to the Care Circle, facilitating timely interventions and reducing the risk of adverse events.

Advanced Clinical Support: The Path Toward Automated Diagnosis

Currently, CogniCare operates as a clinical assistant, mapping data to established criteria such as TES without issuing autonomous diagnoses. The long-term goal is to evolve the platform toward higher-level diagnostic support. As the underlying multimodal AI models are fine-tuned on larger, peer-reviewed clinical datasets, the system could offer probabilistic diagnostic suggestions such as **predicting the exact onset stage** of neurodegeneration based on identified digital biomarkers. This transition would require CogniCare to be reclassified as **Software as a Medical Device (SaMD)** under **MHRA/FDA regulations**. To ensure patient safety throughout this evolution, a strict "Human-in-the-Loop" paradigm would be maintained, requiring a licensed clinician to validate and sign off on any AI-generated diagnostic reasoning before it is finalised.

9. References

[1] Abhinav R Changa, Robert A Vietrogoski, Peter W Carmel, Dr Harrison Martland and the history of punch drunk syndrome, *Brain*, Volume 141, Issue 1, January 2018, Pages 318–321, https://doi.org/10.1093/brain/awx349

[2] Omalu, Bennet & DeKosky, Steven & Minster, Ryan & Kamboh, M & Hamilton, Ronald & Wecht, Cyril. (2006). Chronic Traumatic Encephalopathy in a National Football League Player. Neurosurgery. 58. E1003-E1003. 10.1093/neurosurgery/58.5.E1003.

[3] McKee AC, Stern RA, Nowinski CJ, et al. The spectrum of disease in chronic traumatic encephalopathy. *Brain*. 2013;136(Pt 1):43-64. doi:10.1093/brain/aws307

[4] Katz DI, Bernick C, Dodick DW, et al. National Institute of Neurological Disorders and Stroke Consensus Diagnostic Criteria for Traumatic Encephalopathy Syndrome. *Neurology*. 2021;96(18):848-863. doi:10.1212/WNL.0000000000011850

[5] McKee, A. C., Stein, T. D., Huber, B. R., Crary, J. F., Bieniek, K., Dickson, D., ... & Daneshvar, D. H. (2023). Chronic traumatic encephalopathy (CTE): criteria for neuropathological diagnosis and relationship to repetitive head impacts. *Acta neuropathologica*, *145*(4), 371-394.

[6] Katsumoto, A., Takeuchi, H., & Tanaka, F. (2019). Tau pathology in chronic traumatic encephalopathy and Alzheimer's disease: similarities and differences. *Frontiers in neurology*, *10*, 980.

[7] Falcon, Benjamin, et al. "Novel tau filament fold in chronic traumatic encephalopathy encloses hydrophobic molecules". *Nature* 568.7752 (2019): 420-423.

[8] Andelic, N., Løvstad, M., Norup, A., Ponsford, J. and Røe, C., 2019. impact of traumatic brain injuries on participation in daily life and work: recent research and future directions. *Frontiers in neurology*, *10*, p.1153.

[9] Bournemouth University, “Sport-induced traumatic brain injury: families reveal the hell of living with the condition”, *Qualitative Health Research Study*, 2022. Available: [BU Research Blog | head injury | Bournemouth University]

[10] Kevin Grant Solar, Matthew Ventresca, Rouzbeh Zamyadi, Jing Zhang, Rakesh Jetly, Oshin Vartanian, Shawn G Rhind, Benjamin T Dunkley, Repetitive subconcussion results in disrupted neural activity independent of concussion history, *Brain Communications*, Volume 6, Issue 5, 2024, fcae348, https://doi.org/10.1093/braincomms/fcae348

[11] Querzola, Giacomo, et al. “Incipient chronic traumatic encephalopathy in active American football players: neuropsychological assessment and brain perfusion measures”. *Neurological Sciences* 43.9 (2022): 5383-5390.

[12] Layden RM, Groh JR, Miner AE, et al. CTE neuropathology alone is associated with dementia and cognitive symptoms. Alzheimer's Dement. 2026;22:e71032. https://doi.org/10.1002/alz.71032

[13] Uretsky, M., Nair, E., Burton, R., Cronin, S. W., Rousseau, D., Tuz-Zahra, F., ... & Mez, J. (2024). Chronic traumatic encephalopathy, family history of mental illness and aggression in brain donors with repetitive head impact exposure. *Neurology*, *103*(12), e210056.

[14] Daneshvar, D.H., Nair, E.S., Baucom, Z.H. *et al.* Leveraging football accelerometer data to quantify associations between repetitive head impacts and chronic traumatic encephalopathy in males. *Nat Commun* **14**, 3470 (2023). https://doi.org/10.1038/s41467-023-39183-0

[15] Goldstein, L. E., Fisher, A. M., Tagge, C. A., Zhang, X. L., Velisek, L., Sullivan, J. A., ... & McKee, A. C. (2012). Chronic traumatic encephalopathy in blast-exposed military veterans and a blast neurotrauma mouse model. *Science translational medicine*, *4*(134), 134ra60-134ra60.

[16] Grant L Iverson, Pouya Jamshidi, Amy Deep-Soboslay, Thomas M Hyde, Joel E Kleinman, Amanda O Fisher-Hubbard, Joyce L deJong, Claire E Shepherd, Lili-Naz Hazrati, Rudolph J Castellani, Chronic Traumatic Encephalopathy Neuropathologic Change in Military Veterans, *Military Medicine*, 2025;usaf553, https://doi.org/10.1093/milmed/usaf553.

[17] Makovec Knight, J., Hannon, O., Spitz, G., Astridge, A., Copas, C., Duarte Martins, B., ... & Ponsford, J. (2026). Brain Injury Within Intimate Partner Violence: What Are the Cognitive Effects? *Journal of Neurotrauma*, *43*(3-4), 269-279.

[18] S. Bechelet *et al.*, “Too Many to Count: The prevalence of traumatic brain injury among survivors of domestic abuse accessing community-based services in the United Kingdom”, *Brainkind Research Report*, Jan. 2024. Available: [Too many to count]

[19] Katz, Douglas I., et al. “National Institute of Neurological Disorders and Stroke consensus diagnostic criteria for traumatic encephalopathy syndrome”. *Neurology* 96.18

(2021): 848-863.

[20] Wang, Jun, et al. “Diagnostic accuracy of plasma p-tau217/Aβ42 for Alzheimer's disease in clinical and community cohorts”. *Alzheimer's & Dementia* 21.3 (2025): e70038.

[21] Montenigro, Philip & Corp, Daniel & Stein, Thor & Cantu, Robert & Stern, Robert. (2015). Chronic Traumatic Encephalopathy: Historical Origins and Current Perspective. Annual review of clinical psychology. 11. 10.1146/annurev-clinpsy-032814-112814.

[22] U.S. Dept. of Veterans Affairs, “Concussion Coach”, 2023. [Online]. Available: https://www.ptsd.va.gov/appvid/mobile/concussion_coach_app.asp

[23] Hua, Musen, et al. “Effects of digital psychological interventions for family caregivers of people with dementia: a systematic review and meta-analysis”. *Age and Ageing* 55.1 (2026): afaf373.

[24] Schacter, Daniel L. “The seven sins of memory: Insights from psychology and cognitive neuroscience”. *American psychologist* 54.3 (1999): 182.

[25] Maas, Andrew IR, Hester F. Lingsma, and Bob Roozenbeek. “Predicting outcome after traumatic brain injury”. *Handbook of clinical neurology* 128 (2015): 455-474.

[26] T. Benson and G. Grieve, *Principles of Health Interoperability*, 4th ed. Springer, 2021.

[27] Team, Gemini, et al. “Gemini: a family of highly capable multimodal models”. *arXiv preprint arXiv:2312.11805* (2023).

[28] Wei, Jason, et al. “Chain-of-thought prompting elicits reasoning in large language models”. *Advances in neural information processing systems* 35 (2022): 24824-24837.

[29] Google Cloud, “Gemini 1.5 Flash: High-speed multimodal intelligence”, 2024. [Online]. Available: https://ai.google.dev/gemini-api/docs/models

[30] Nurgalieva, Leysan, David O’Callaghan, and Gavin Doherty. “Security and privacy of mHealth applications: a scoping review”. *IEEE Access* 8 (2020): 104247-104268.

[31] Supabase, “Edge Functions Documentation”, 2024. [Online]. Available: https://supabase.com/docs/guides/functions

[32] Ionic Framework, “Capacitor Secure Storage Guide”, 2024. [Online]. Available: Storage | Capacitor Documentation

[33] Apple Inc., “iOS Security Guide: Secure Enclave”, 2024. [Online]. Available: https://support.apple.com/guide/security/secure-enclave-sec59b0b31ff/web

10. Appendices

10.1 Supabase Edge Functions

Overview

To maintain high responsiveness within the mobile frontend while managing heavy asynchronous background tasks, CogniCare utilizes **Deno-based Supabase Edge Functions**. This serverless layer is responsible for the “Notification Lifecycle”, ensuring that intensive cross-cloud communication (e.g., triggering Firebase alerts) does not block the main application thread or the user interface.

Key Technical Features

- **RSA256 OAuth2 Handshake (Zero-Trust Security):** Unlike standard API-key-based integrations, these functions implement a sophisticated JWT-based authentication flow. Utilizing the crypto.subtle library, the getAccessToken logic manually constructs and signs a **RSASSA-PKCS1-v1_5 JWT**. This allows for secure, temporary credentialing with the Google OAuth2 Token Server, adhering to high-security **Zero-Trust** industry standards.

- **Cross-Cloud Integration (FCM v1 API):** The functions act as a robust bridge between the Supabase PostgreSQL database and the **Firebase Cloud Messaging (FCM) v1 API**. The logic dynamically resolves device pushTokens by mapping unique user IDs across partitioned Patient and Carer profile tables, ensuring that clinical alerts are routed to the correct authenticated device.

- **Persistence-First Notification Logic:** To ensure reliability, the function performs an **atomic database insert** into the notifications table prior to dispatching the push notification to the physical device. This creates a "Sync-on-Reconnect" resilience; patients and carers can access their full alert history within the app even if the initial push notification was missed due to network instability.

- **Edge-Level Performance:** By executing logic at the **Network Edge** (via Deno Deploy), CogniCare achieves near-zero latency for critical communication. This high-speed execution is vital for real-time care coordination, where delayed alerts regarding symptoms or clinical notes could impact patient outcomes.

Function Registry

A. notify-on-chat

Trigger Mechanism: Database Webhook

Purpose: Triggered upon a new message insertion. Resolves sender metadata and manages multi-recipient routing for the Care Circle.

```
import { createClient } from "npm:@supabase/supabase-js@2.34.0";

const SUPABASE_URL = Deno.env.get('SUPABASE_URL')!;
const SUPABASE_SERVICE_ROLE_KEY = Deno.env.get('SUPABASE_SERVICE_ROLE_KEY')!;
const FIREBASE_PROJECT_ID = Deno.env.get('FIREBASE_PROJECT_ID')!;
const FIREBASE_CLIENT_EMAIL = Deno.env.get('FIREBASE_CLIENT_EMAIL')!;
const FIREBASE_PRIVATE_KEY = Deno.env.get('FIREBASE_PRIVATE_KEY')!;

const supabase = createClient(SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, {
  auth: { persistSession: false },
});

async function getAccessToken(): Promise<string> {
  const now = Math.floor(Date.now() / 1000);
  const header = { alg: 'RS256', typ: 'JWT' };
  const payload = {
    iss: FIREBASE_CLIENT_EMAIL,
    scope: 'https://www.googleapis.com/auth/firebase.messaging',
    aud: 'https://oauth2.googleapis.com/token',
    iat: now,
    exp: now + 3600
  };

  const encodeB64Url = (obj: any) =>
    btoa(JSON.stringify(obj)).replace(/\+/g, '-').replace(/\//g, '_').replace(/=/g, '');

  const signingInput = `${encodeB64Url(header)}.${encodeB64Url(payload)}`;

  const pemKey = FIREBASE_PRIVATE_KEY.replace(/\\n/g, '\n');
  const keyData = pemKey.replace('-----BEGIN PRIVATE KEY-----', '').replace('-----END PRIVATE KEY-----
', '').replace(/\n/g, '');
  const binaryKey = Uint8Array.from(atob(keyData), c => c.charCodeAt(0));

  const cryptoKey = await crypto.subtle.importKey(
    'pkcs8',
    binaryKey.buffer,
    { name: 'RSASSA-PKCS1-v1_5', hash: 'SHA-256' },
    false,
    ['sign']
  );

  const signature = await crypto.subtle.sign('RSASSA-PKCS1-v1_5', cryptoKey, new
TextEncoder().encode(signingInput));
```

```
  const b64sig = btoa(String.fromCharCode(...new Uint8Array(signature))).replace(/\+/g, '-
').replace(/\//g, '_').replace(/=/g, '');

  const jwt = `${signingInput}.${b64sig}`;

  const tokenRes = await fetch('https://oauth2.googleapis.com/token', {
    method: 'POST',
    headers: { 'Content-Type': 'application/x-www-form-urlencoded' },
    body: new URLSearchParams({ grant_type: 'urn:ietf:params:oauth:grant-type:jwt-bearer', assertion:
jwt })
  });

  const tokenData = await tokenRes.json();
  if (!tokenRes.ok) throw new Error(`FCM Token Error: ${JSON.stringify(tokenData)}`);
  return tokenData.access_token;
}

Deno.serve(async (req) => {
  try {
    const { message_id, content, sender_name, recipient_id, chat_id, chat_type } = await req.json();

    if (!recipient_id) throw new Error("Recipient ID missing in request");

    // Fetch recipient push token (User table mapping)
    // We check both patient and carer profiles for the userId
    const { data: patient, error: pErr } = await supabase
      .from('patient_profiles')
      .select('pushToken, userId')
      .eq('id', recipient_id)
      .single();

    let pushToken = patient?.pushToken;
    let recipientUserId = patient?.userId;

    if (!pushToken) {
      const { data: carer, error: cErr } = await supabase
        .from('carer_profiles')
        .select('pushToken, userId')
        .eq('id', recipient_id)
        .single();
      pushToken = carer?.pushToken;
      recipientUserId = carer?.userId;
    }

    if (!recipientUserId) throw new Error("Recipient profile not found");

    const body = content?.length > 50 ? content.slice(0, 50) + "..." : content;
    const title = `New message from ${sender_name || 'Someone'}`;
    const notifId = crypto.randomUUID();

    // 1. Save to DB for persistence in Notifications Tab
```

```
    await supabase.from('notifications').insert([{
      id: notifId,
      userId: recipientUserId,
      title,
      body,
      data: { chatId: chat_id, type: chat_type },
      read: false
    }]);

    // 2. Send FCM Push Notification if token exists
    if (pushToken) {
      const accessToken = await getAccessToken();
      const fcmRes = await
fetch(`https://fcm.googleapis.com/v1/projects/${FIREBASE_PROJECT_ID}/messages:send`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${accessToken}` },
        body: JSON.stringify({
          message: {
            token: pushToken,
            notification: { title, body },
            data: { chatId: String(chat_id), type: String(chat_type) },
            android: {
                priority: "high",
                notification: {
                    channel_id: 'default',
                    notification_priority: "PRIORITY_HIGH",
                    color: "#6CB5E0",
                    icon: 'ic_launcher'
                }
            }
          }
        }),
      });

      const fcmData = await fcmRes.json();
      console.log('[FCM Response]', fcmData);
    } else {
      console.warn('[Edge Function] Recipient push token missing. Skipping FCM.');
    }

    return new Response(JSON.stringify({ success: true }), { headers: { 'Content-Type':
'application/json' } });
  } catch (err) {
    console.error('[Edge Function Error]', err);
    return new Response(JSON.stringify({ error: err.message }), { status: 500, headers: { 'Content-
Type': 'application/json' } });
  }
});
```

B. notify-patient-on-note

Trigger Mechanism: Database Webhook
Purpose: Triggered when a Carer adds a clinical observation to a symptom log. Ensures the patient receives immediate visibility into medical feedback.

```
import { createClient } from "npm:@supabase/supabase-js@2.34.0";

const SUPABASE_URL = Deno.env.get('SUPABASE_URL')!;
const SUPABASE_SERVICE_ROLE_KEY = Deno.env.get('SUPABASE_SERVICE_ROLE_KEY')!;
const FIREBASE_PROJECT_ID = Deno.env.get('FIREBASE_PROJECT_ID')!;
const FIREBASE_CLIENT_EMAIL = Deno.env.get('FIREBASE_CLIENT_EMAIL')!;
const FIREBASE_PRIVATE_KEY = Deno.env.get('FIREBASE_PRIVATE_KEY')!;

const supabase = createClient(SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, {
  auth: { persistSession: false },
});

async function getAccessToken(): Promise<string> {
  const now = Math.floor(Date.now() / 1000);
  const header = { alg: 'RS256', typ: 'JWT' };
  const payload = {
    iss: FIREBASE_CLIENT_EMAIL,
    scope: 'https://www.googleapis.com/auth/firebase.messaging',
    aud: 'https://oauth2.googleapis.com/token',
    iat: now,
    exp: now + 3600
  };

  const encodeB64Url = (obj: any) =>
    btoa(JSON.stringify(obj)).replace(/\+/g, '-').replace(/\//g, '_').replace(/=/g, '');

  const signingInput = `${encodeB64Url(header)}.${encodeB64Url(payload)}`;

  const pemKey = FIREBASE_PRIVATE_KEY.replace(/\\n/g, '\n');
  const keyData = pemKey.replace('-----BEGIN PRIVATE KEY-----', '').replace('-----END PRIVATE KEY-----
', '').replace(/\n/g, '');
  const binaryKey = Uint8Array.from(atob(keyData), c => c.charCodeAt(0));

  const cryptoKey = await crypto.subtle.importKey(
    'pkcs8',
    binaryKey.buffer,
    { name: 'RSASSA-PKCS1-v1_5', hash: 'SHA-256' },
    false,
    ['sign']
  );

  const signature = await crypto.subtle.sign('RSASSA-PKCS1-v1_5', cryptoKey, new
TextEncoder().encode(signingInput));
  const b64sig = btoa(String.fromCharCode(...new Uint8Array(signature))).replace(/\+/g, '-
').replace(/\//g, '_').replace(/=/g, '');
```

```
  const jwt = `${signingInput}.${b64sig}`;

  const tokenRes = await fetch('https://oauth2.googleapis.com/token', {
    method: 'POST',
    headers: { 'Content-Type': 'application/x-www-form-urlencoded' },
    body: new URLSearchParams({ grant_type: 'urn:ietf:params:oauth:grant-type:jwt-bearer', assertion:
jwt })
  });

  const tokenData = await tokenRes.json();
  if (!tokenRes.ok) throw new Error(`FCM Token Error: ${JSON.stringify(tokenData)}`);
  return tokenData.access_token;
}

Deno.serve(async (req) => {
  try {
    const { note_id } = await req.json();

    const { data: note, error: noteErr } = await supabase
      .from('carer_notes')
      .select('id, text, patientId, logId, profile:patient_profiles(userId, pushToken,
user:users(name)), carer:carer_profiles(user:users(name))')
      .eq('id', note_id)
      .single();

    if (noteErr || !note) throw new Error("Note not found in db");

    const carerName = (note.carer as any)?.user?.name || (note.profile as any)?.user?.name || "The
Carer";
    const title = !!note.logId ? 'New Comment on Log' : 'New Carer Entry';
    const message = `${carerName}: ${note.text?.slice(0, 100)}${note.text.length > 100 ? '...' : ''}`;
    const notifId = crypto.randomUUID();

    // 1. Save to DB for persistence in Notifications Tab
    await supabase.from('notifications').insert([{
      id: notifId,
      userId: note.patientId,
      title,
      body: message,
      data: { note_id: note.id, logId: note.logId },
      read: false
    }]);

    // 2. Send FCM Push Notification
    if (note.profile?.pushToken) {
      const accessToken = await getAccessToken();
      await fetch(`https://fcm.googleapis.com/v1/projects/${FIREBASE_PROJECT_ID}/messages:send`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json', 'Authorization': `Bearer ${accessToken}` },
        body: JSON.stringify({
          message: {
```

```
            token: note.profile.pushToken,
            notification: { title, body: message },
            data: { logId: String(note.logId || ''), note_id: String(note.id) },
            android: {
                priority: "high",
                notification: {
                    channel_id: 'default',
                    notification_priority: "PRIORITY_HIGH",
                    color: "#6CB5E0", // Brand Primary Blue
                    icon: 'ic_launcher'
                }
            }
          }
        }),
      });
    }

    return new Response(JSON.stringify({ success: true }), { headers: { 'Content-Type':
'application/json' } });
  } catch (err) {
    return new Response(JSON.stringify({ error: err.message }), { status: 500, headers: { 'Content-
Type': 'application/json' } });
  }
});
```

10.2 Mathematical Formulation of the CogniCare TES Triage Score

10.2.1 Overview and System Architecture

The CogniCare TES Triage Score provides a quantitative digital score (0–100 scale) to assist in the evaluation of Traumatic Encephalopathy Syndrome (TES) and progressive cognitive decline. Rather than replacing standard clinical evaluation, the algorithm acts as a digital implementation of the qualitative **National Institute of Neurological Disorders and Stroke (NINDS) consensus criteria** [25].

The triage scale uses a tripartite scoring architecture, calculating a total score $\mathbf{S_{TES}}$ as the sum of three distinct components:

$$\mathbf{S_{TES}} = \mathbf{S_{Burden}} + \mathbf{S_{History}} + \mathbf{S_{Criteria}}$$

Where:

- $\mathbf{S_{Burden}}$: Symptom Burden Score (Maximum 30 points)
- $\mathbf{S_{History}}$: Risk Factors and Functional History Score (Maximum 15 points)
- $\mathbf{S_{Criteria}}$: TES-Specific Diagnostic Criteria Score (Maximum 55 points)

10.2.2 Tripartite Scoring Logic

Symptom Burden Score (S_{Burden} - Maximum 30 Points)

This component evaluates the active functional trajectory of the patient based on their logged symptoms. Instead of checking only for the presence of a symptom, the algorithm incorporates longitudinal data by calculating a trajectory weight based on the persistence or escalation of the symptom over time.

For each symptom *i*:

$$S_{Burden} = \sum_{i=1}^{M} \omega_i$$

Where the weight ω_i is determined by the trajectory status:

$$Status(\omega_i) = \begin{cases} +0.9 \ \ if\ Marked\ "Present" \\ +2.0 \ \ if\ Marked\ "Worse" \\ -0.5 \ \ if\ Marked\ "Improving" \end{cases}$$

This mathematical weighting captures the immediate functional impact while penalizing actively degenerating symptoms to indicate escalation.

```
// 1. Symptom Burden (Max 30)
const symptomList = Object.values(symptomChecks);
const presentCount = symptomList.filter((s) => s.present).length;
const worseCount = symptomList.filter((s) => s.worse).length;
const improvingCount = symptomList.filter((s) => s.improving).length;

const symptomScore = Math.min(
  Math.round(presentCount * 0.9 + worseCount * 2 - improvingCount * 0.5),
  30
);
```

Figure 72: Code for Symptom Burden Calculation

Patient History and Risk Factors ($S_{History}$ - Maximum 15 Points)

This module evaluates lifestyle and environmental risks that exacerbate neurodegenerative decline. The semantic evaluation system scans for qualitative markers, ignoring null or non-applicable entries. The score is calculated as a summation of weighted risk factors x_k:

$$S_{History} = \sum_{k=1}^{K} (\omega_k \cdot x_k)$$

Where:

- $x_k \in \{0,1\}$ represents the presence or absence of a specific risk factor.
- Substance/alcohol abuse vectors are heavily weighted: $\omega_{Substance}$ = 3.0
- Loss of baseline activities of daily living (such as stopping routine chores) is weighted: ω_{ADL} = 2.0

```
let manualHistoryScore = 0;
const historyFields = [
  { val: history.stoppedChores, weight: 2, type: 'risk' },
  { val: history.drinking, weight: 3, type: 'risk' },
  { val: history.nonPrescription, weight: 2, type: 'risk' },
  { val: history.diet, weight: 1, type: 'status' },
  { val: history.familyHistory, weight: 2, type: 'status' },
  { val: history.supportNetwork, weight: 1, type: 'status' },
  { val: history.additionalNotes, weight: 1, type: 'status' }
];
```

Figure 73: Code for Patient History Calculation

TES Criteria Extraction ($S_{Criteria}$ - Maximum 55 Points)

The core diagnostic translation engine parses Boolean user inputs and maps them directly to specific, weighted **NINDS** clinical features. The module is split into four distinct evaluation criteria:

$$S_{Criteria} = S_{RHI} + S_{Core} + S_{Supportive} + S_{Progressive}$$

- **Repetitive Head Impact (RHI) Exposure (S_{RHI}):** *+5 points* per exposure type (e.g., collision sports, military service, concussive episodes), up to a maximum threshold of *25 points*.

- **Core Clinical Features (S_{Core}):** *+9 points* for each major neurobehavioral, cognitive, or mood impairment.

- **Supportive Features ($S_{Supportive}$):** *+4 points* for each secondary indicator (e.g., impulsivity, paranoia, suicidality, motor features), **capped at a maximum of 20 points** to prevent minor conditions from skewing the core diagnosis.

- **Progressive Course ($S_{Progressive}$):** An additional *+8 points* if the AI confirms a progressive symptom trajectory over a 12-month period.

```
const tesScore = Math.min(
  Math.round(
    rhiMet * 5 +
    coreMet * 9 +
    Math.min(supMet * 4, 20) +
    (tes.symptoms_12months ? 8 : 0)
  ),
  55
);

// 4. Total Calculation
const total = Math.min(symptomScore + finalHistoryScore + tesScore, 100);
```

Figure 74: Code for Overall TES calculation

10.2.3 Alignment with NINDS Consensus Criteria

To ensure the algorithmic output directly correlates with neurological best practices, the CogniCare scoring engine preserves the underlying clinical hierarchy of the qualitative NINDS framework through several mathematical safeguards:

Qualitative NINDS Criteria	CogniCare Implementation
Mandatory RHI Exposure	Enforced via RHI Vector Sum (Max 25)
Core Clinical Features	Weighted +9 Points (Heavily prioritized)
Supportive Features	Capped at +20 Points Maximum
Progressive Course	Weighted +8 Points + Trajectory Multiplier

- **Enforcing the RHI Prerequisite:** The NINDS guidelines require substantial RHI exposure. The algorithm enforces this by treating RHI as an additive base; a score indicating high probability cannot be achieved via generalized dementia symptoms alone without this exposure floor.

- **Core vs. Supportive Weighting:** The NINDS framework distinguishes between core diagnostic features and supportive features. The algorithm mathematically enforces this distinction by assigning core features a significant weight (+9 points), while capping supportive features at 20 points. This prevents minor symptoms, such as occasional headaches, from triggering a false-positive TES alert.

- **Tracking Progression:** The diagnostic criteria necessitate that the underlying pathology is progressive over time. The algorithm rewards this clinical marker by adding +8 points for the symptoms_12months Boolean and applying a +2.0 multiplier for escalating symptom burdens.

- **Functional Impairment:** NINDS requires that the syndrome impacts daily life. This is captured by the Patient History section, ensuring that cognitive scores are contextualized by changes in functional independence.

10.3 Live Links for Application

Deliverable	Details
Live Web Application	https://cognicare-rosy.vercel.app/
Backend API	https://cogni-care-backend.onrender.com
Android APK Download	Cognicare-v1.0.apk